Spatial Transcriptomics

Martin Hemberg

BST282

Why spatial transcriptomics?

- scRNA-seq dissociates tissue -> all cellular context is lost

Why spatial transcriptomics?

- scRNA-seq dissociates tissue -> all cellular context is lost
- Measure gene expression *in situ*

Technology Feature | Published: 06 January 2021

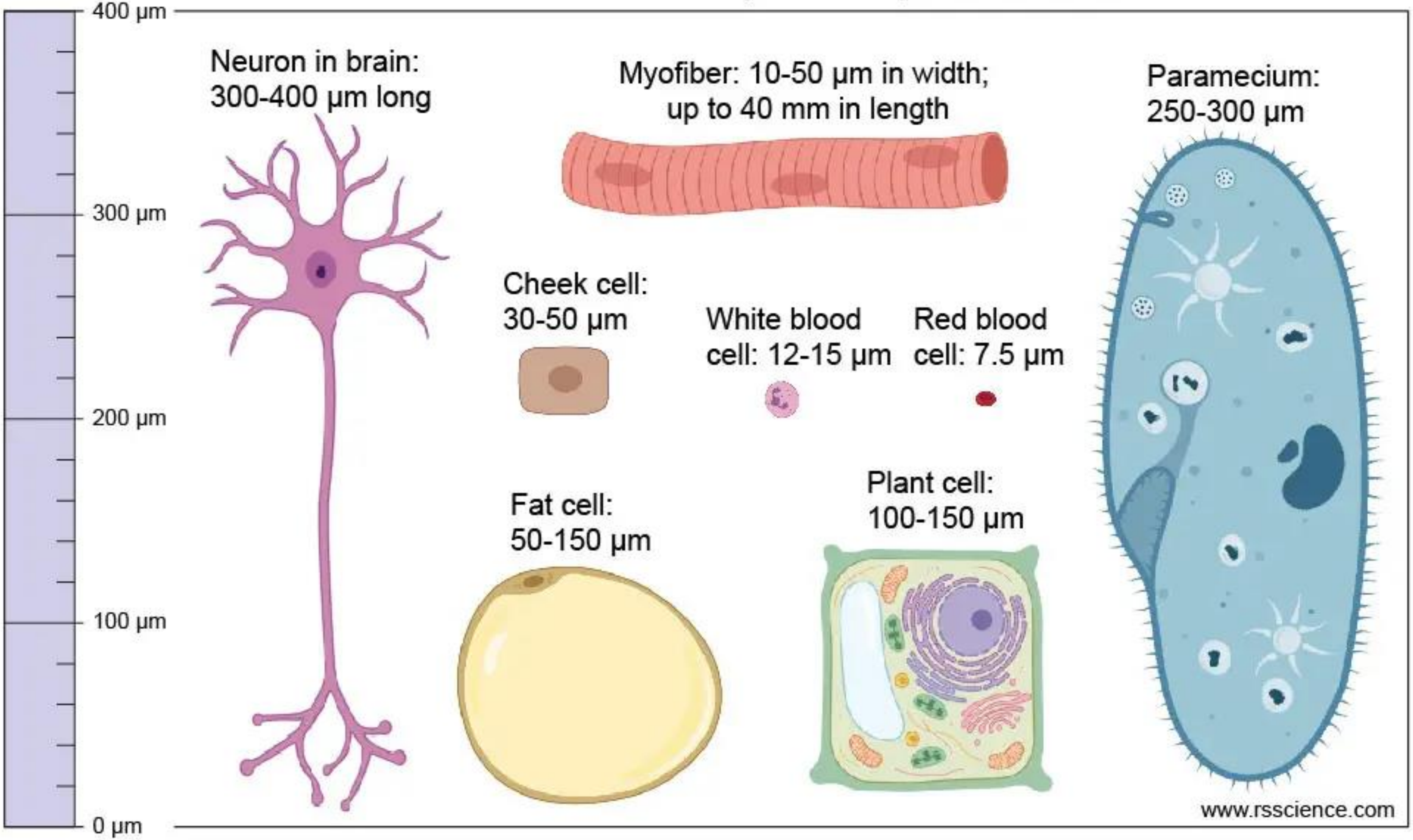
Method of the Year: spatially resolved transcriptomics

Vivien Marx

Nature Methods **18**, 9–14 (2021) | Cite this article

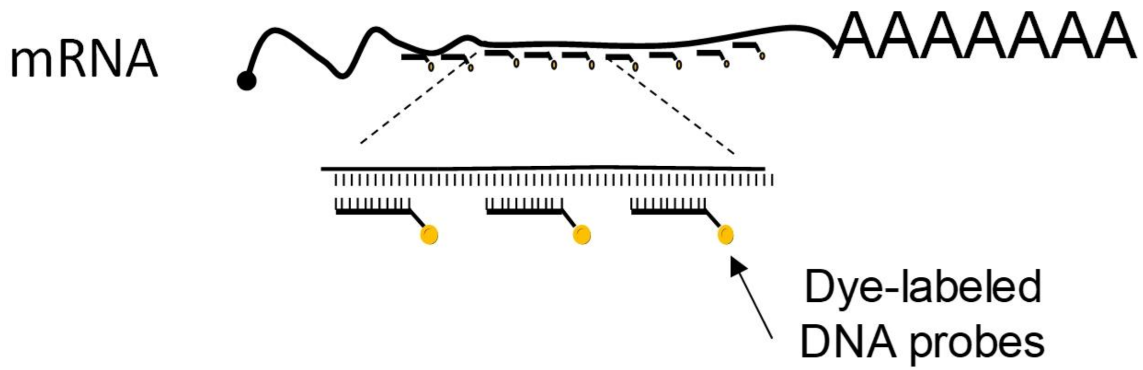
Technologies

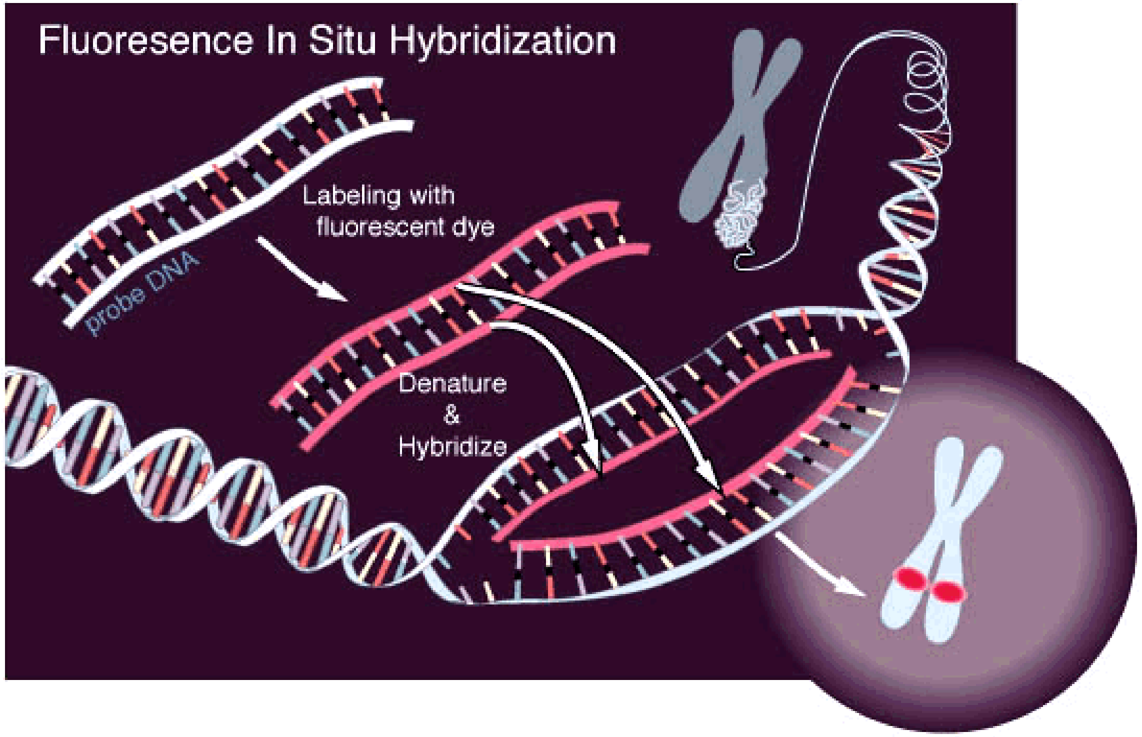
Cell Size and Shape Comparison



Fluorescence in situ hybridization (FISH)

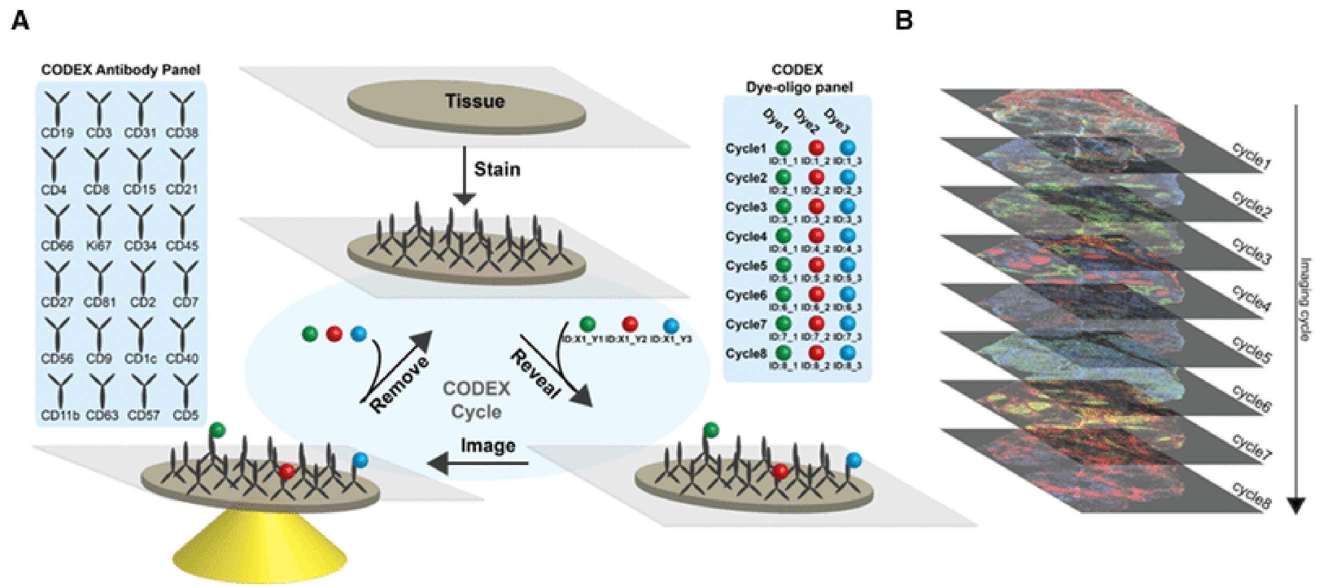
- Use fluorescent probes that bind specifically to a particular nucleotide
  - 20-50 nt target





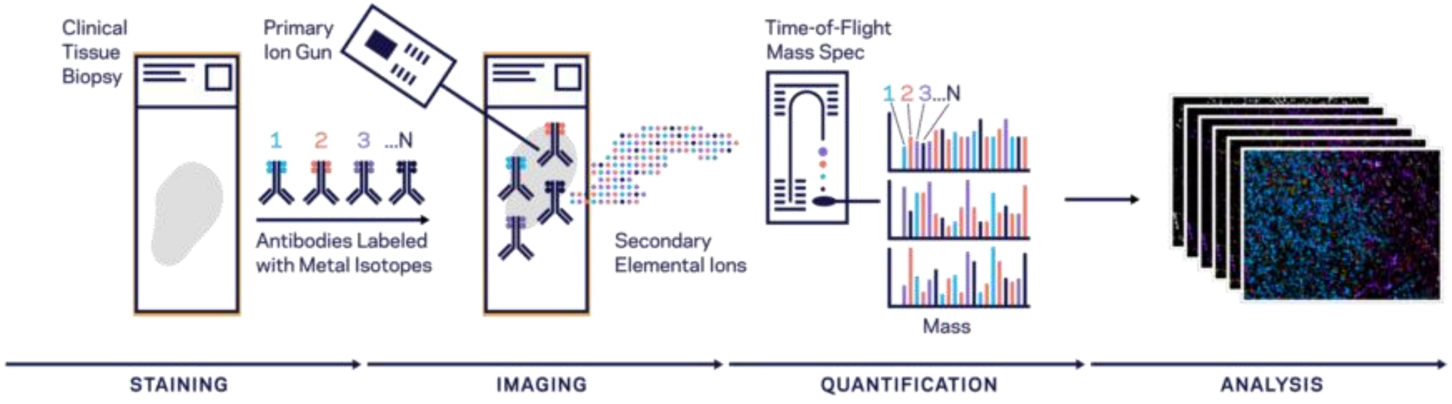
Co-detection by indexing (CODEX)

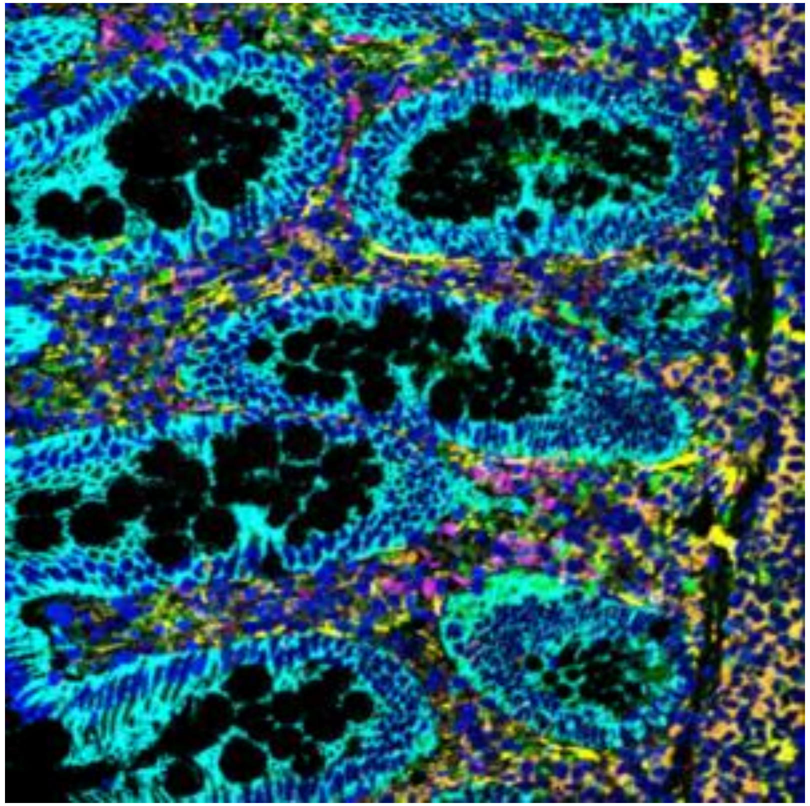
- 40+ Abs with conjugated oligonucleotide barcodes
- Barcodes read bye a dye-labeled reporter
  - High sensitivity
  - High throughput
  - Only surface proteins



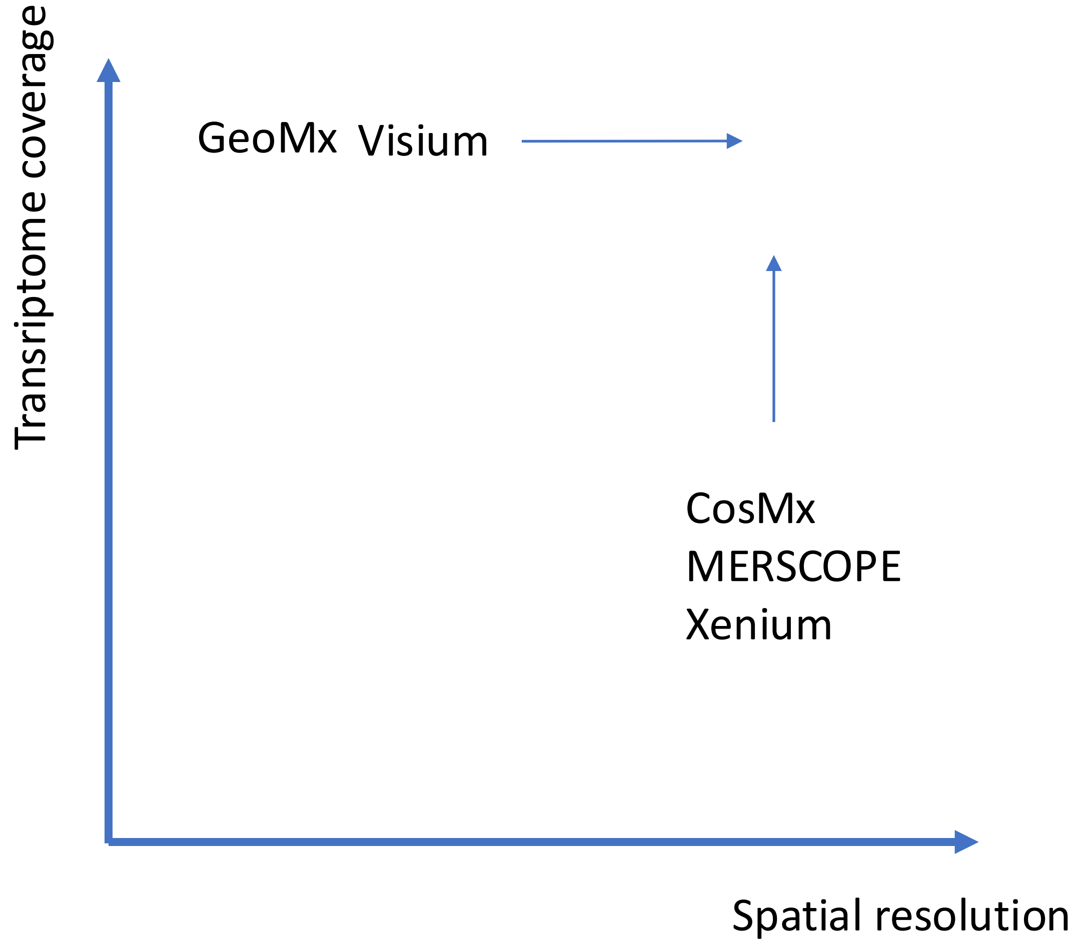
Mass spectrometry imaging

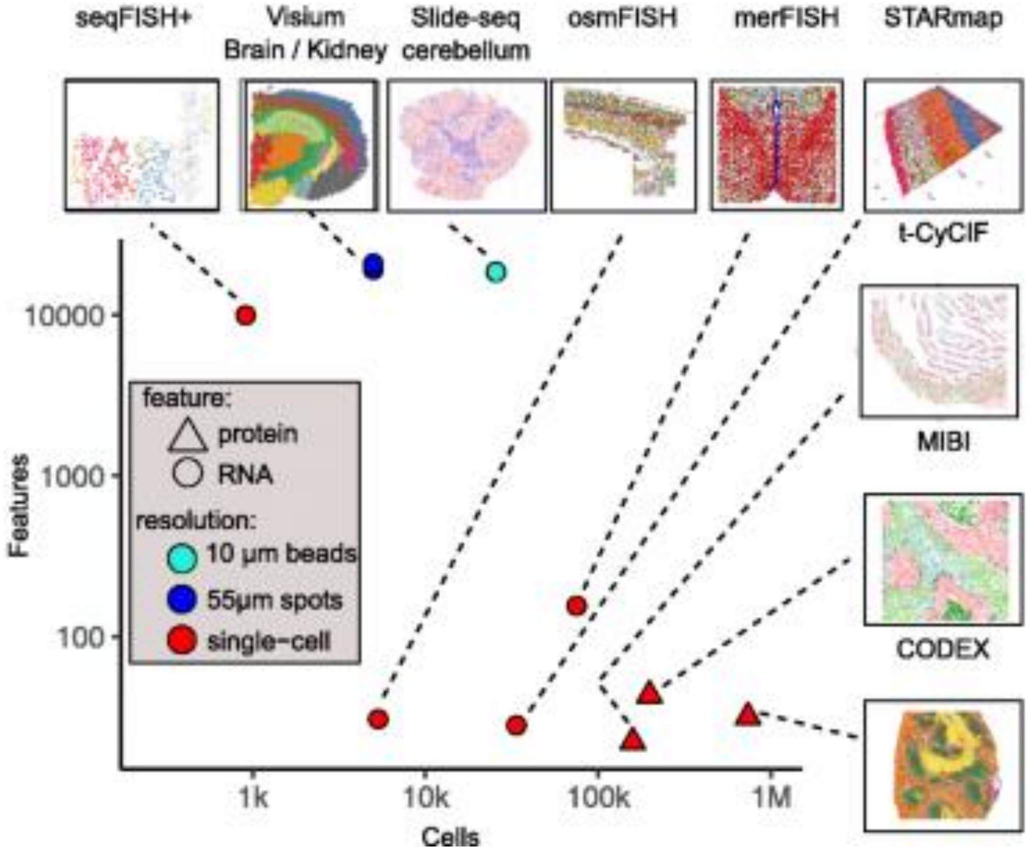
- Stain tissue with ~40 Abs, each tagged with metal isotope
- Use ion beam to dislodge metals, measure using mass spectrometer
  - Sub-cellular resolution
  - Very high sensitivity





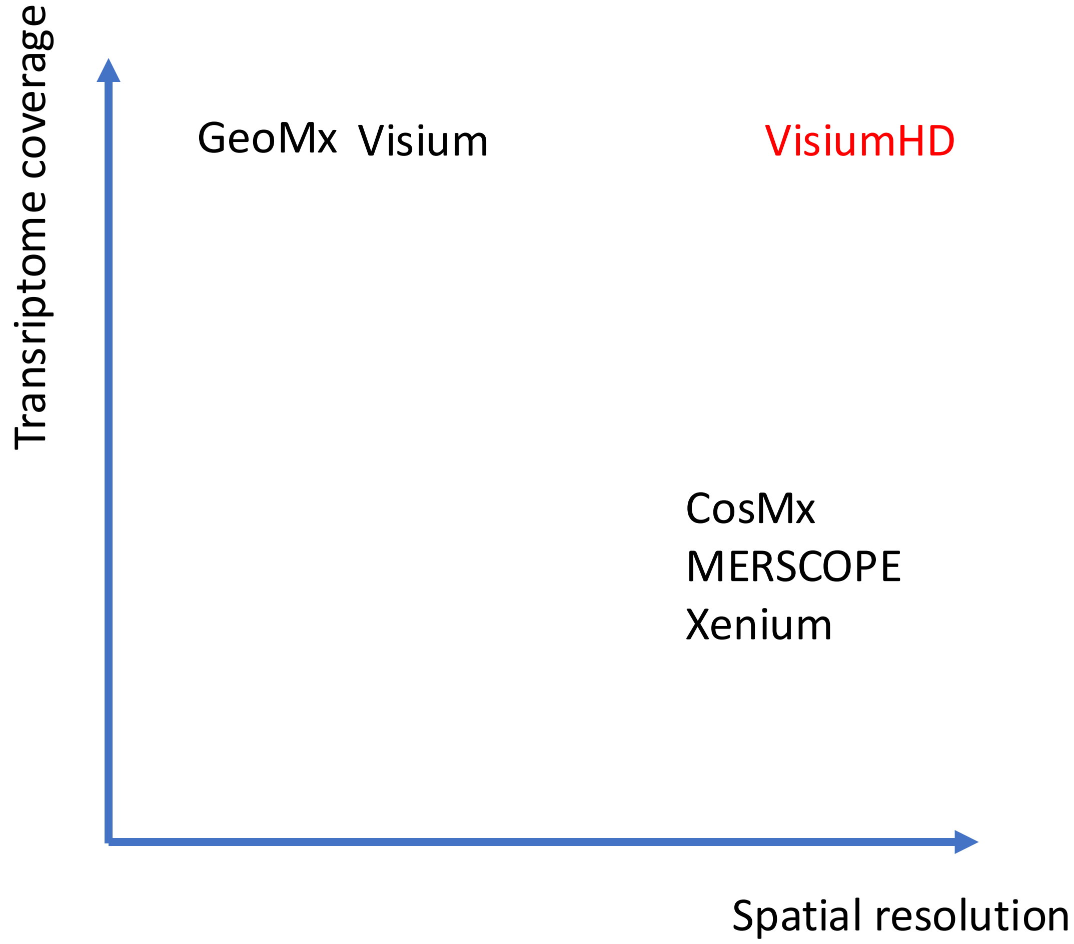
Technology overview

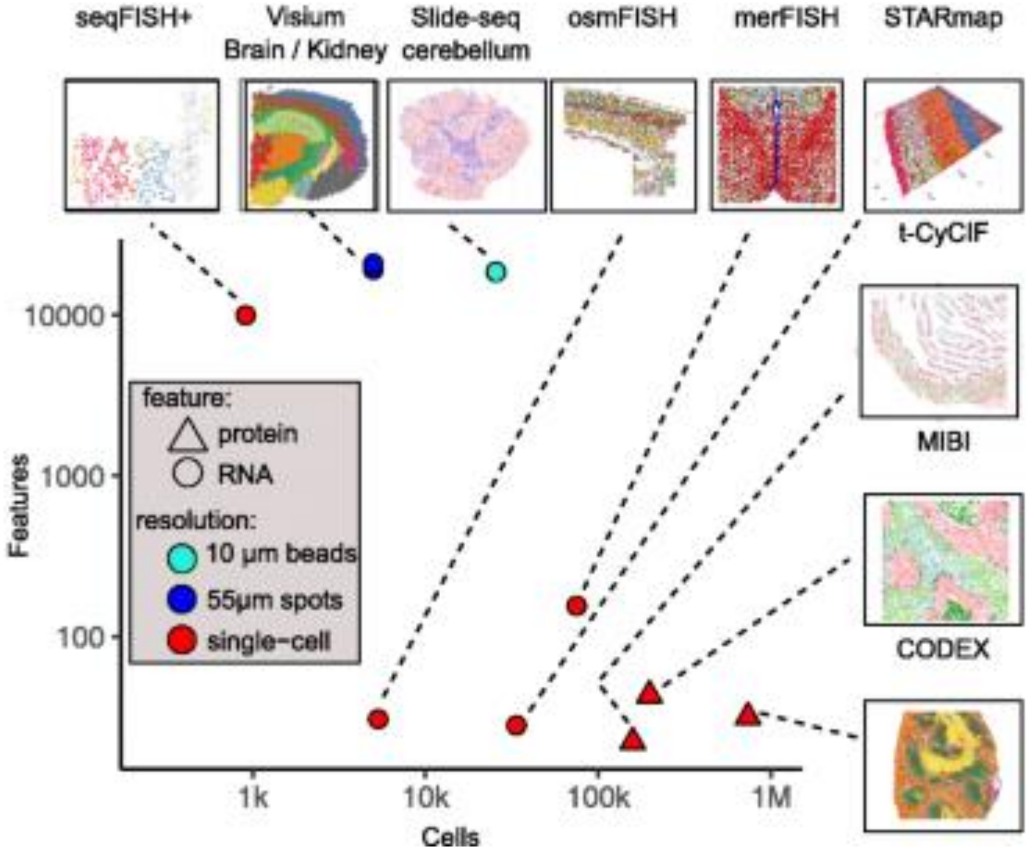




Dries et al, Genome Biology, 2021

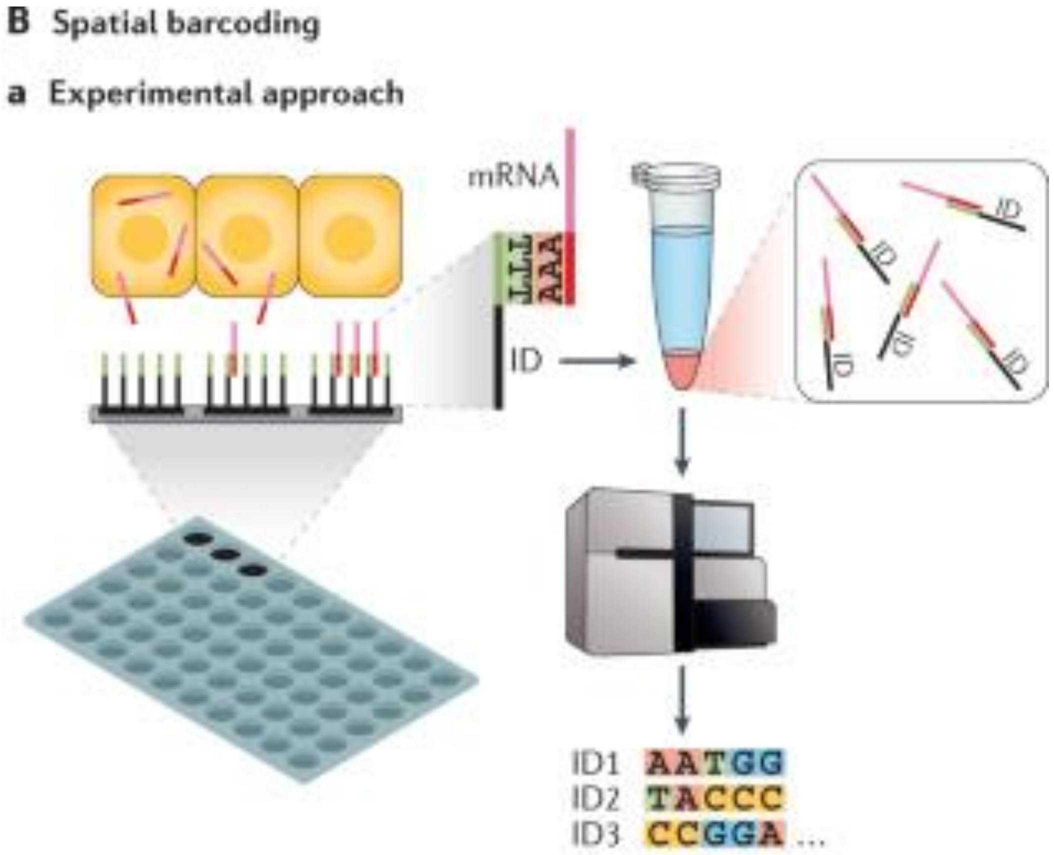
Technology overview

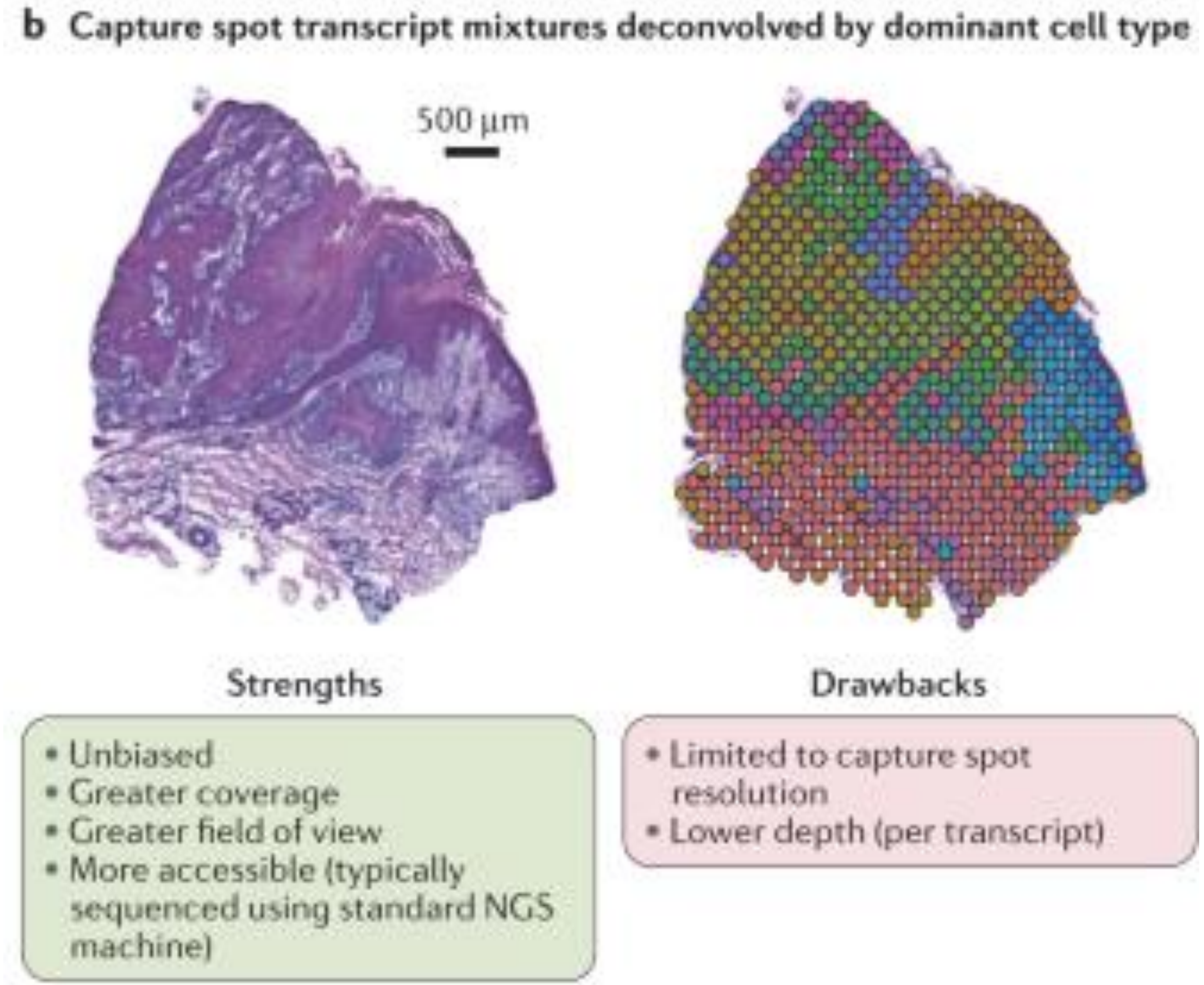




Dries et al, Genome Biology, 2021

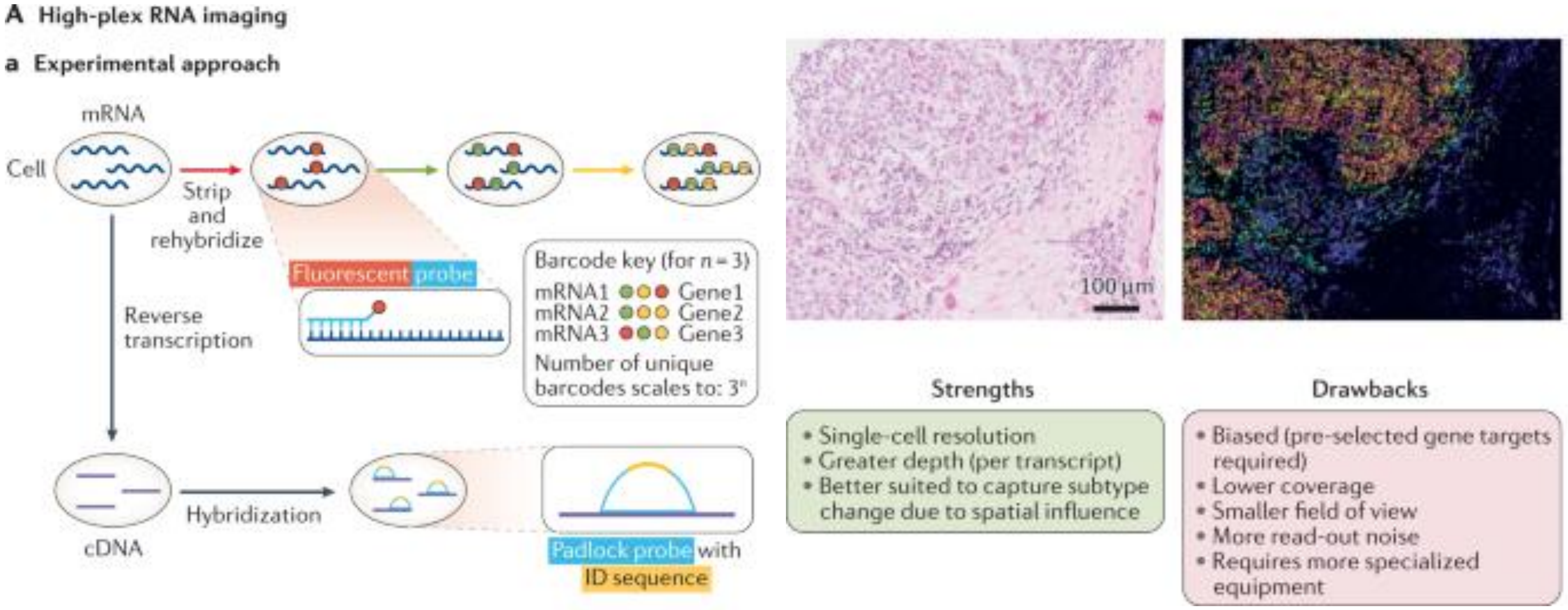
Sequencing based approaches





Longo et al, Nat Rev Gen, 2021

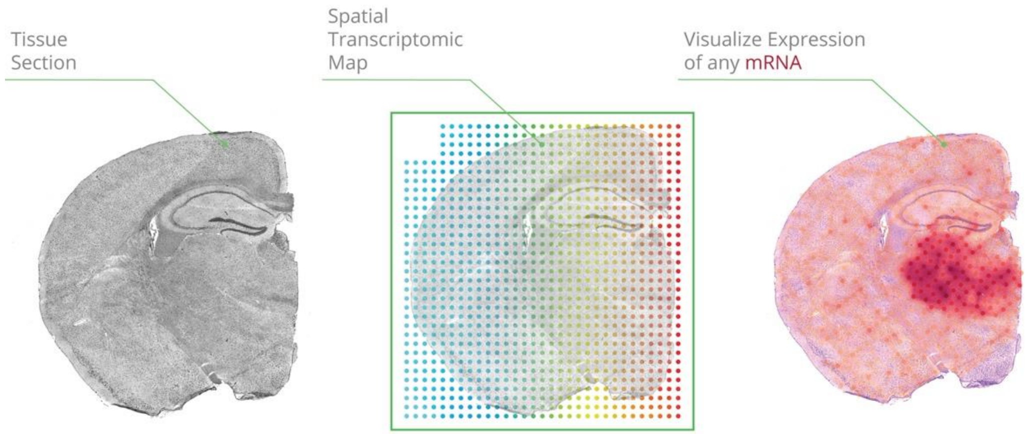
Probe based approaches

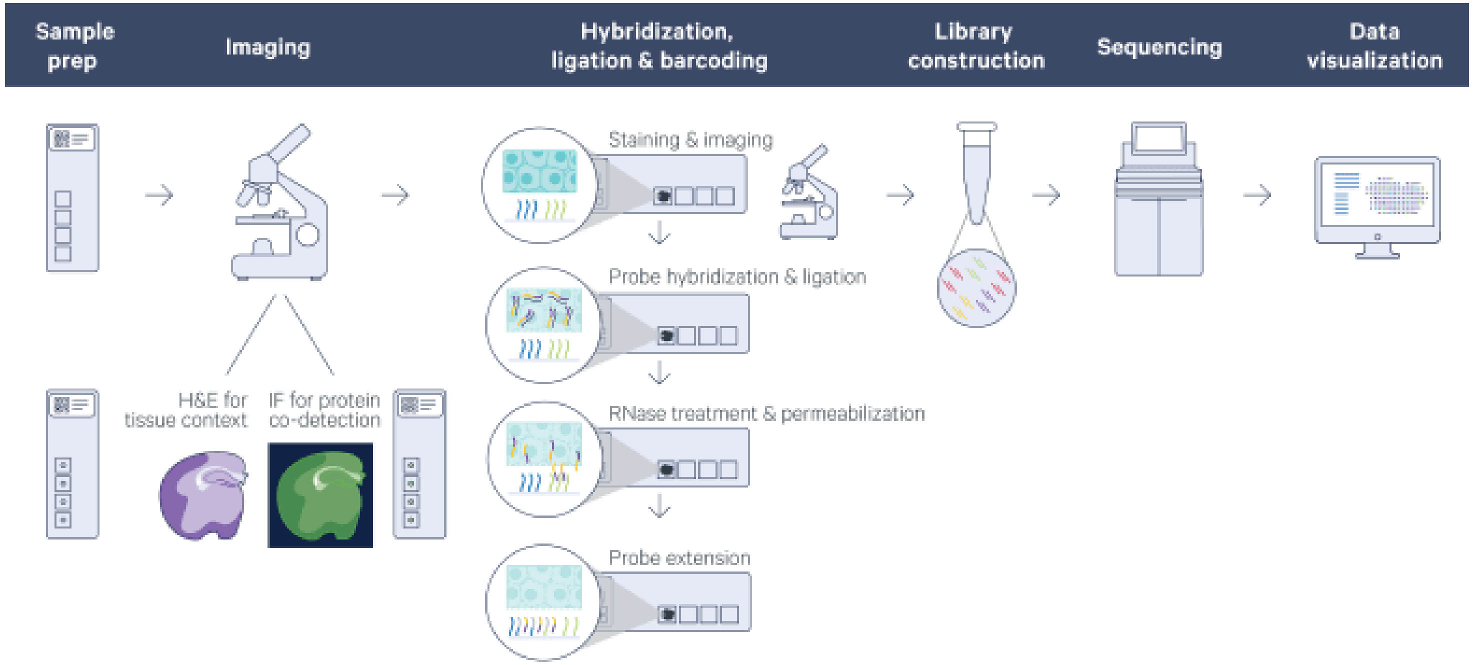


Longo et al, Nat Rev Gen, 2021

10X Visium

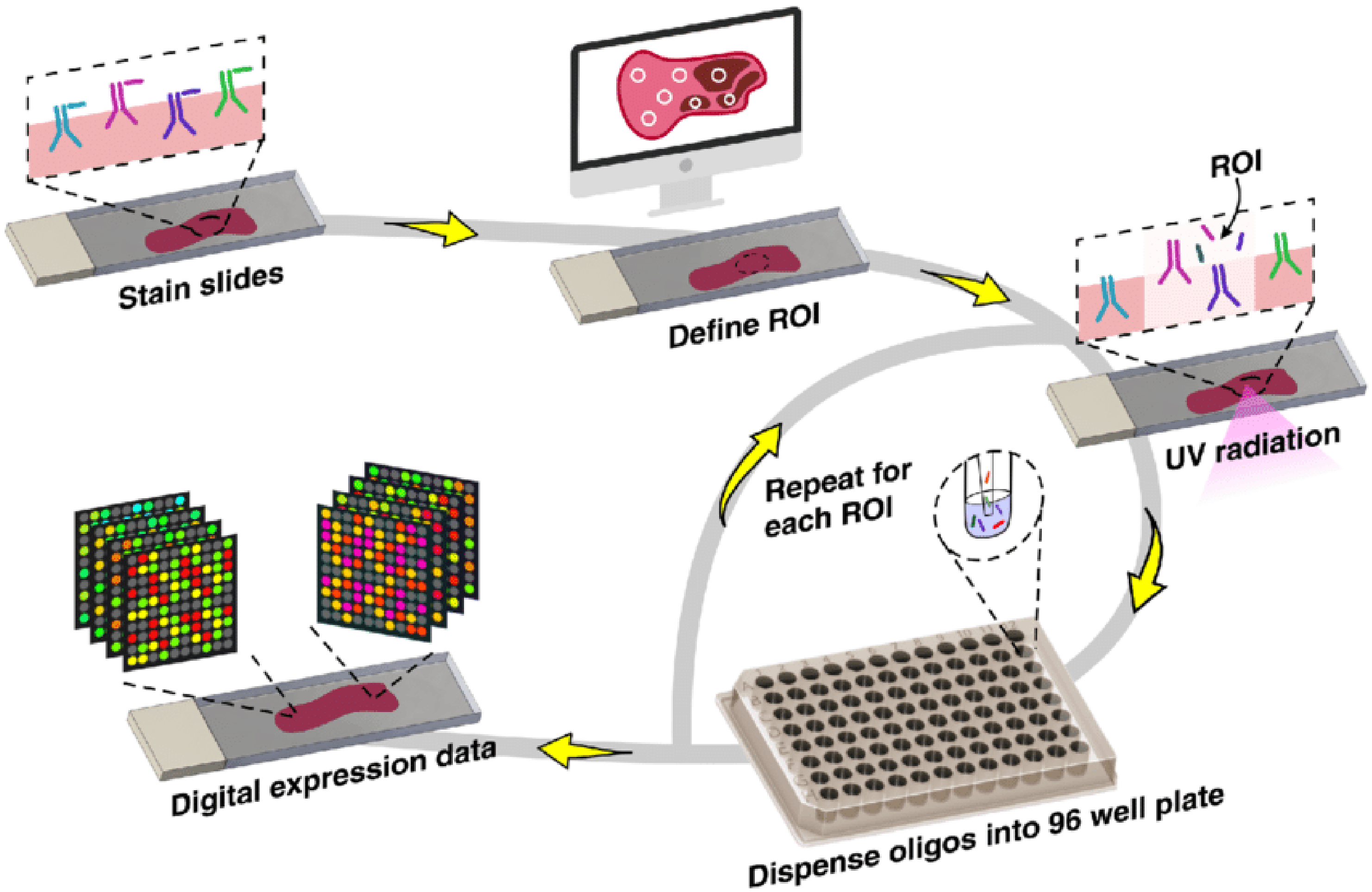
- First commercially available method
  - Barcoded spots printed on 6.5 mm x 6.5 mm slide
  - Spots are 55 µm diameter and spaced 100 µm apart
  - H&E and IF also available





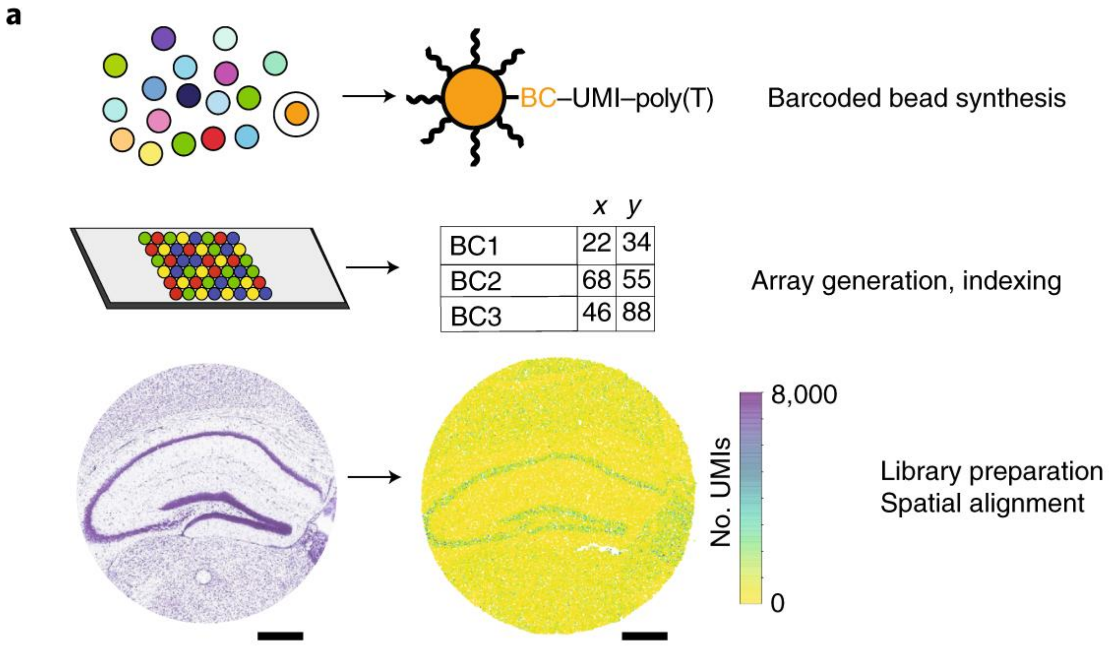
Nanostring GeoMx

- Use Abs (40) or oligo probes (genome wide) tagged with barcodes
- Cleave barcodes by UV
- Sequence using NGS
- ROIs are 10-600 µm diameter



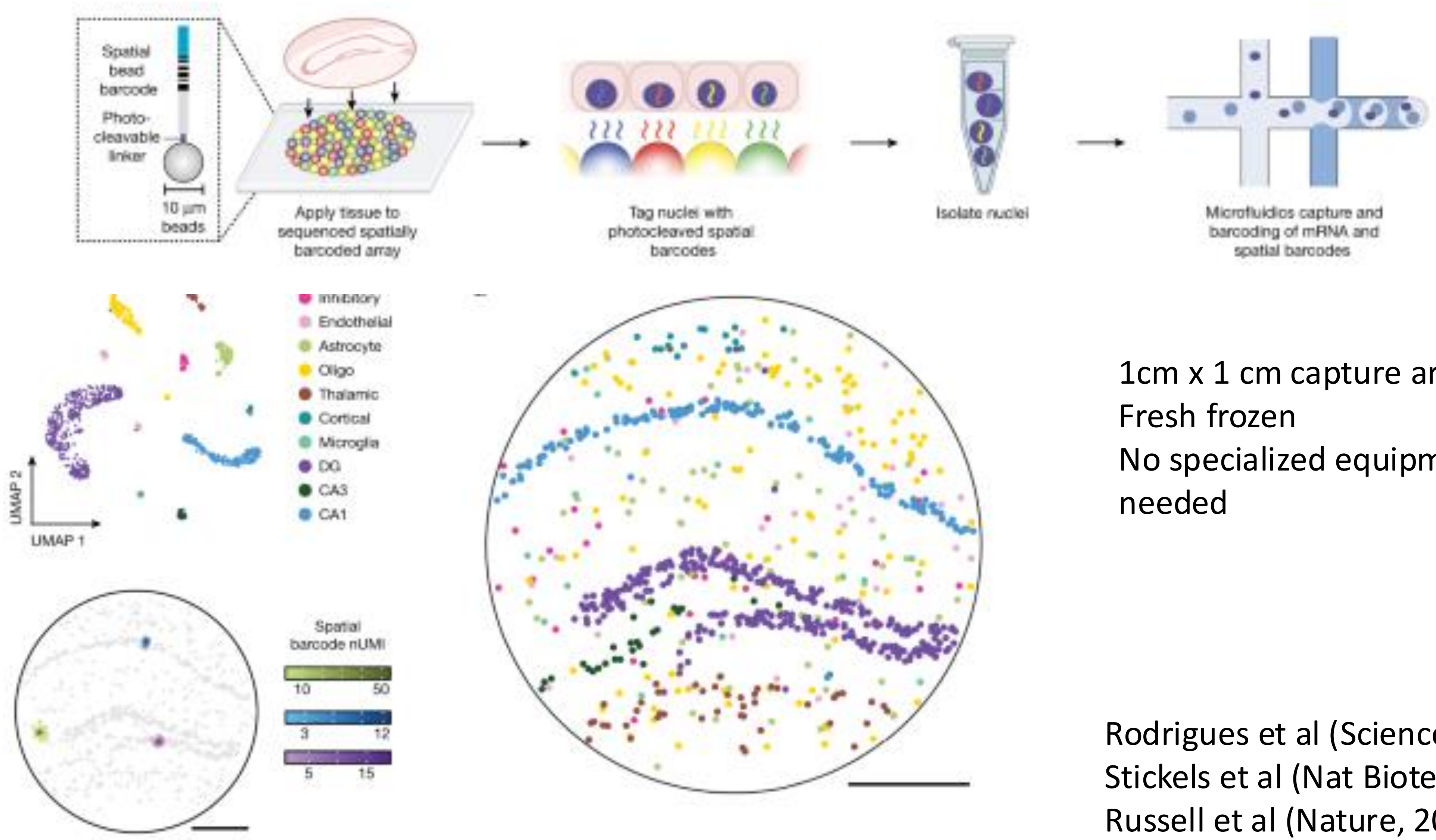
Slide-seq

- Beads with preprinted barcodes
  - 10 µm resolution
- Commercialized by Curio



Rodrigues et al (Science, 2019)
Stickels et al (Nat Biotech, 2021)

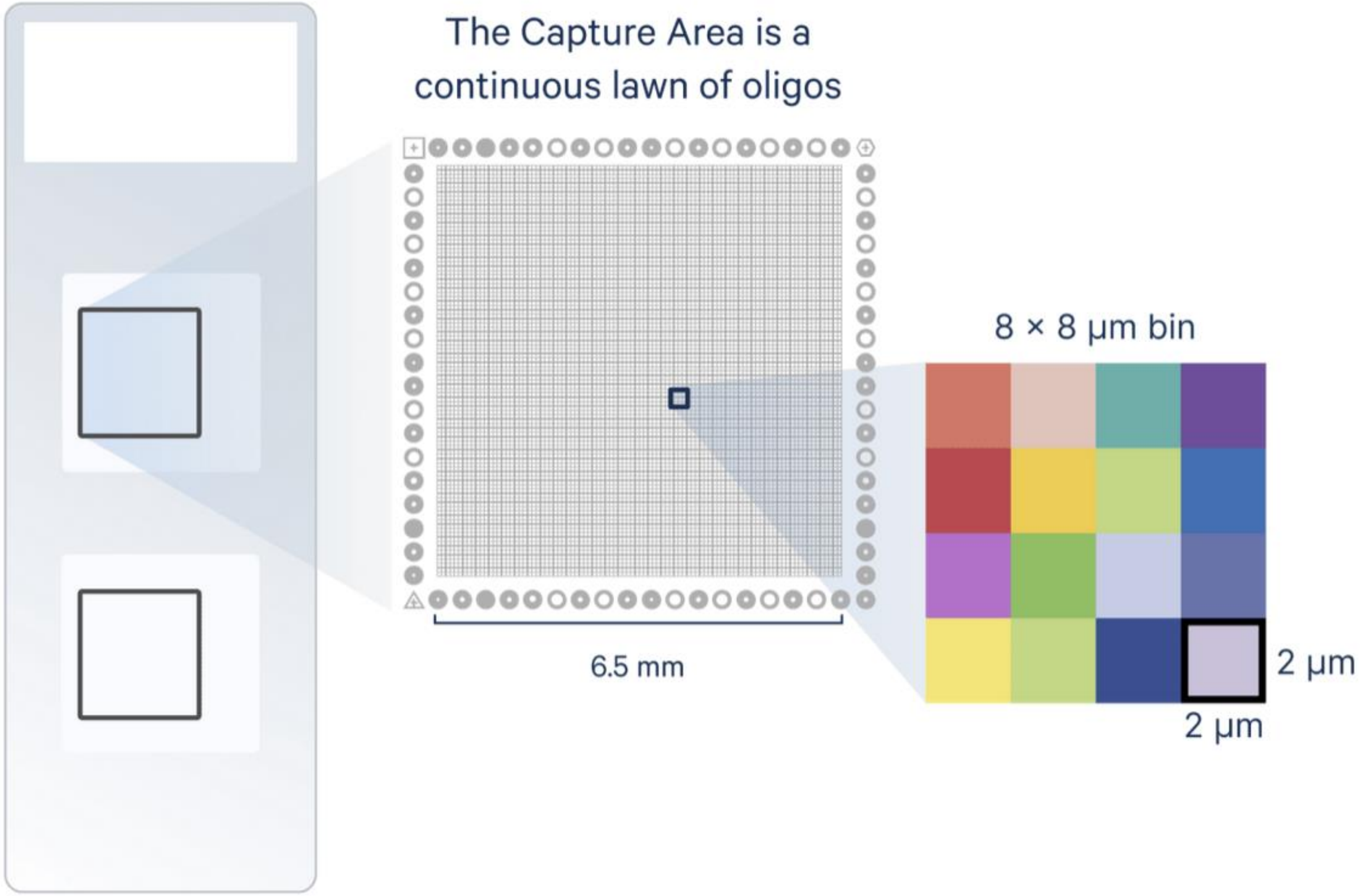
Slide-tags

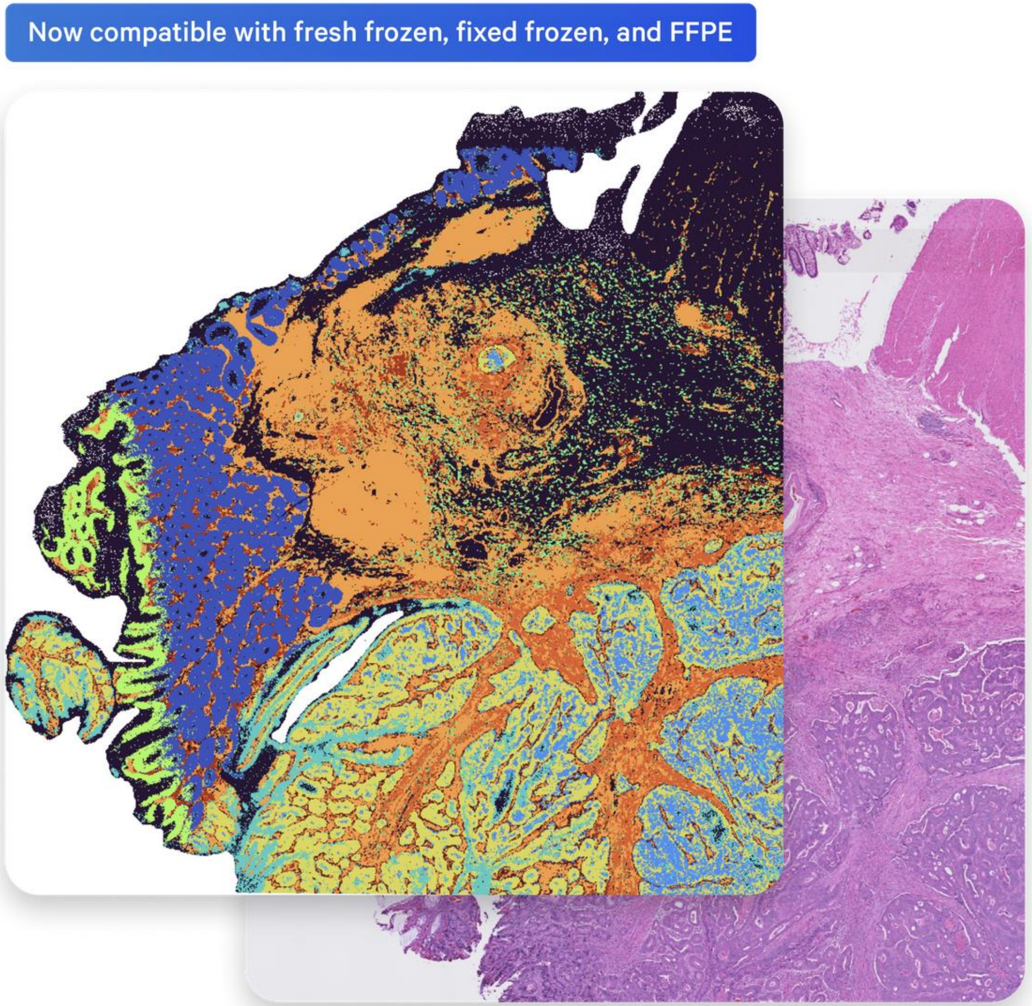


1cm x 1 cm capture area
Fresh frozen
No specialized equipment needed

Rodrigues et al (Science, 2019)
Stickels et al (Nat Biotech, 2021)
Russell et al (Nature, 2024)

Visium HD

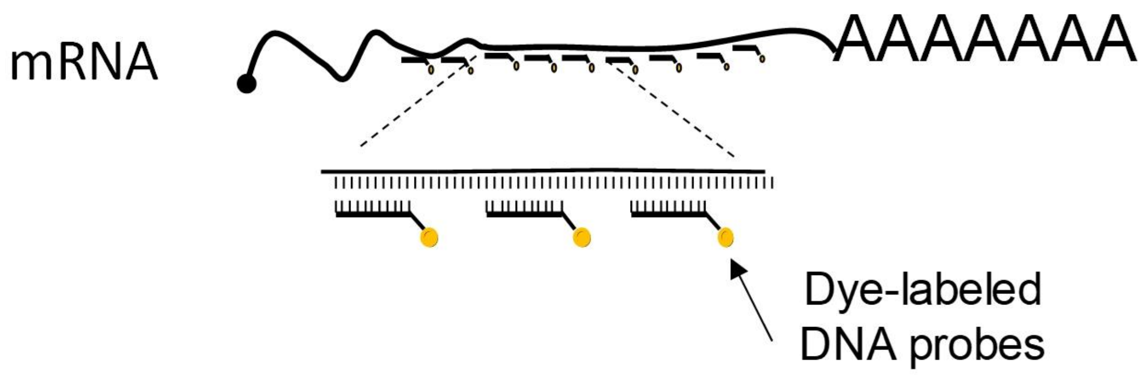


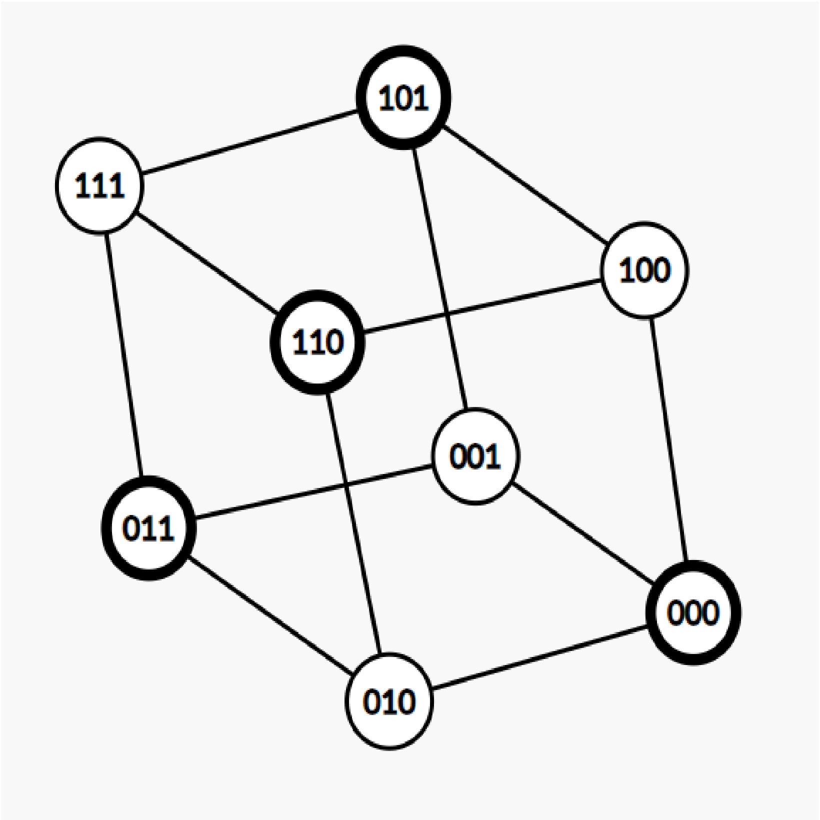


Oliveira et al (BioRxiv, 2024)

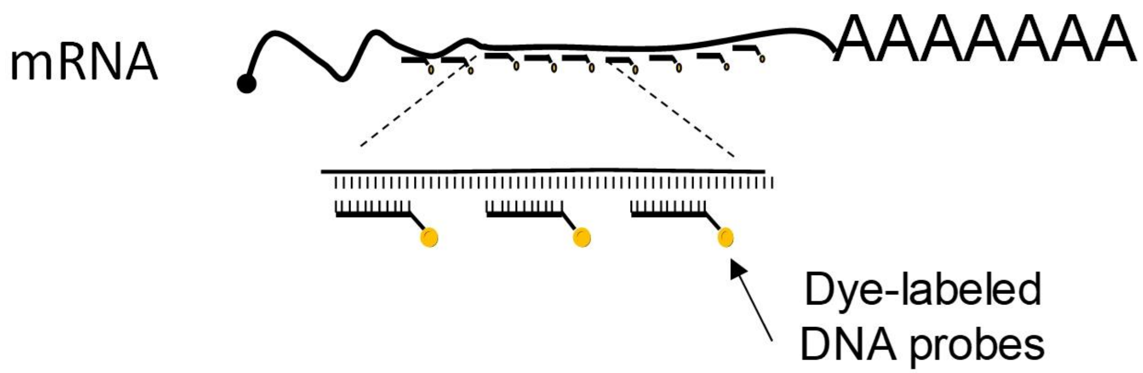
Error correcting codes

- Consider a binary code
- By adding extra bits (redundancy), we can ensure that errors are detected and/or corrected

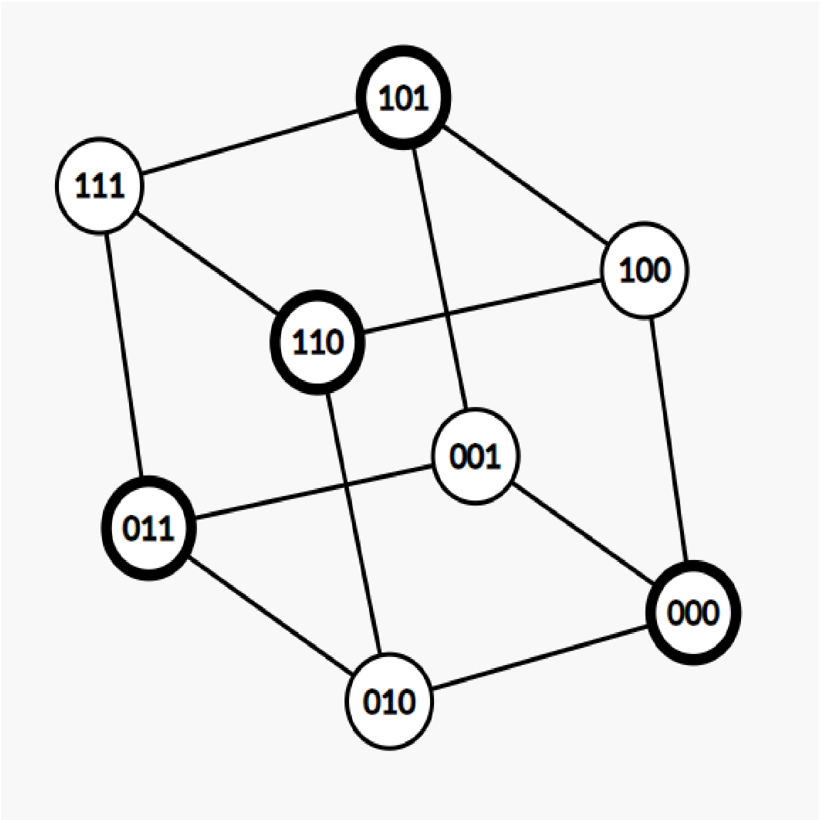




Error correcting codes

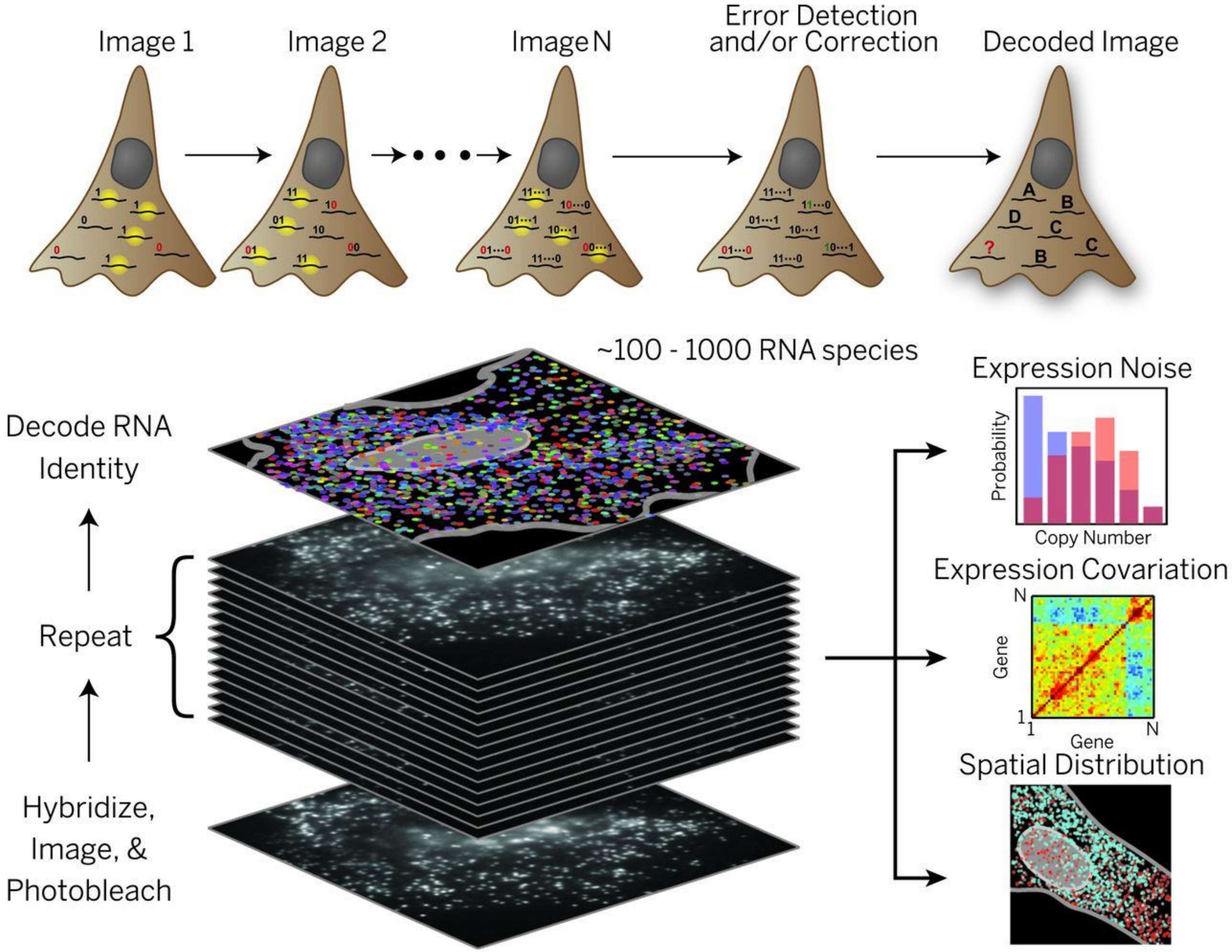


- Consider a binary code
- By adding extra bits (redundancy), we can ensure that errors are detected and/or corrected
- More colors would be really helpful:
  - 2^8 = 256
  - 4^8 = 2^16 = 65,536
  - Space is reduced since error probabilities are not symmetric



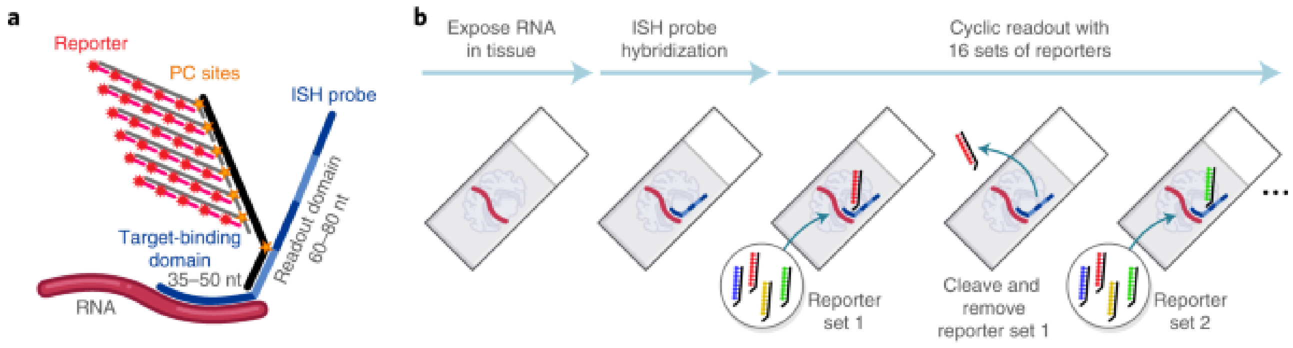
MERFISH

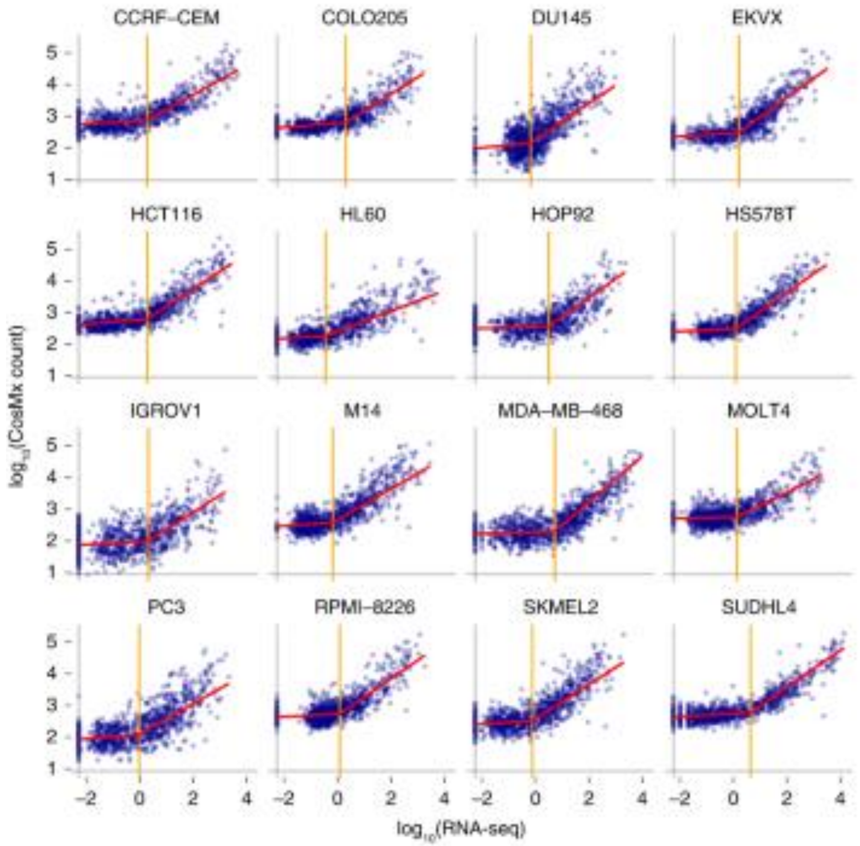
- Multiplexed error-robust fluorescence in situ hybridization
  - 1 cm^2 capture area
  - Can also do proteins
  - Entirely customized, can do up to 1k transcripts
  - High sensitivity and resolution
  - Run on any sample type (including dissociated cells)
  - Commercialized as MERSCOPE through Vizgen



Nanostring CosMx

- Follow-up for GeoMx based on Spatial molecular imaging (SMI)
  - Based on ISH
  - 5 probes/gene
  - No enzymes or amplifications
  - Can achieve single-cell resolution
  - 375 mm^2 imaging area
- Limited to 6k probes
  - Combinatorial code of 4 colors
  - Will be extended to 18k probes

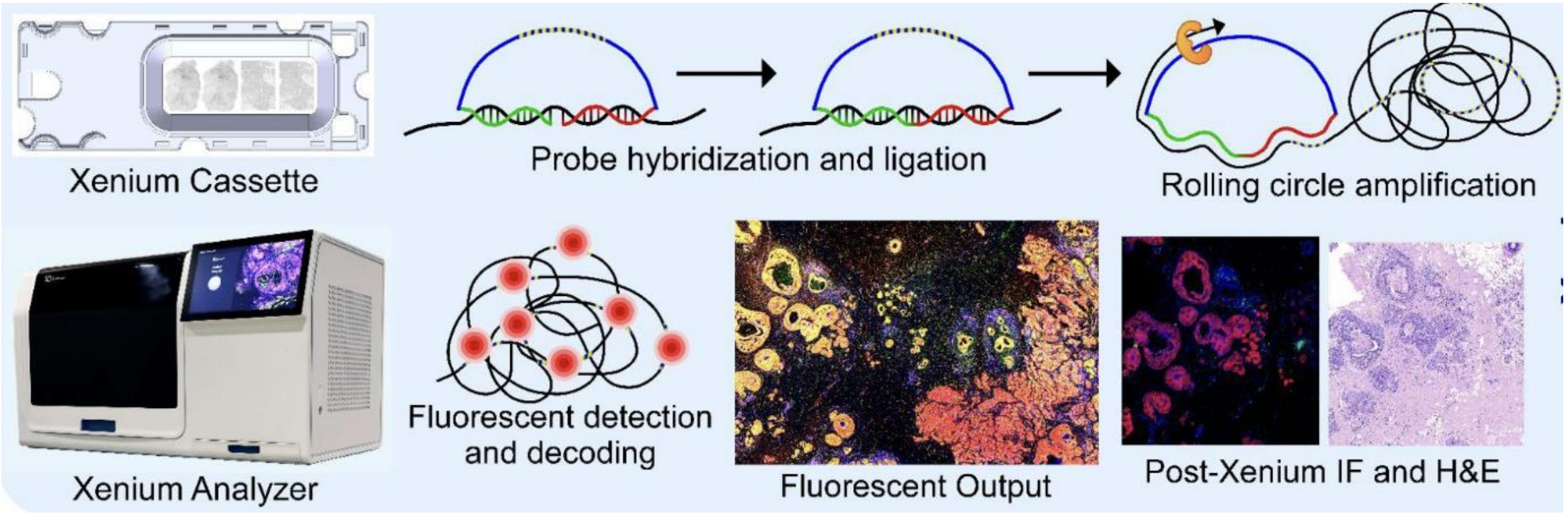




He et al, Nat Biotech, 2022

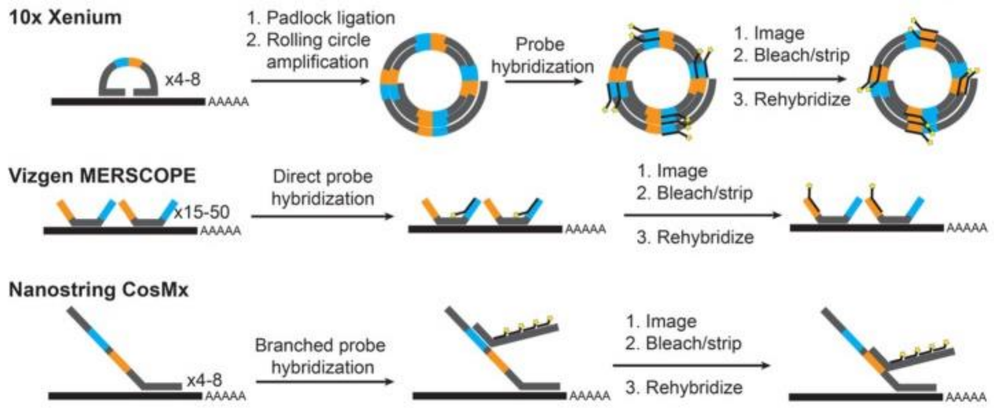
10X Xenium

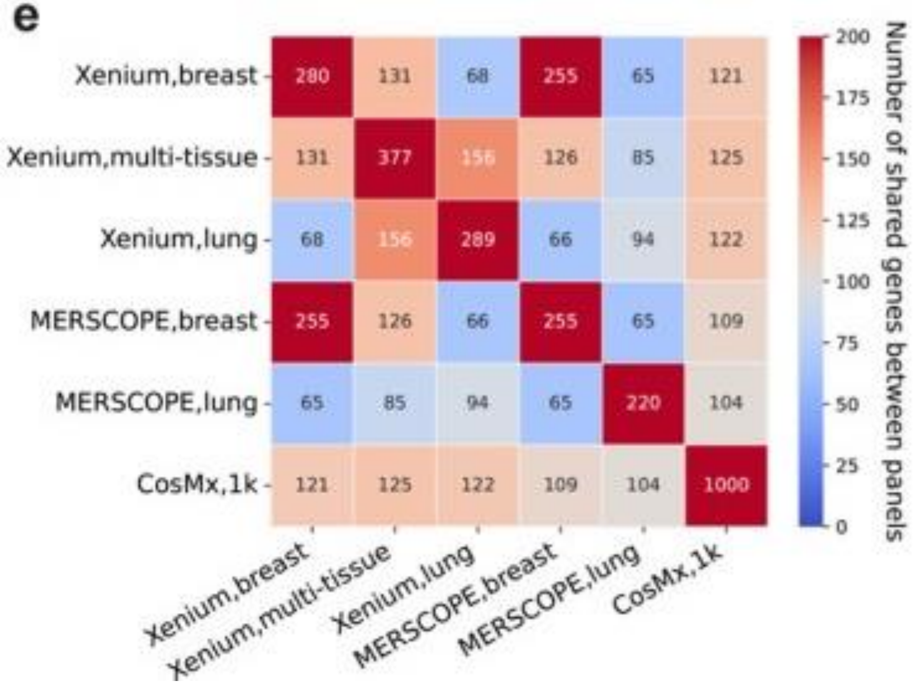
- Similar idea as Nanostring, use oligonucleotide probes and amplify
- Fresh frozen, FFPE
- Panel size ranging from 50 (fully customized) to 5,000 (pre-designed)



Benchmarking MERSCOPE, CosMx, Xenium

- Used breast and lung samples from same donors

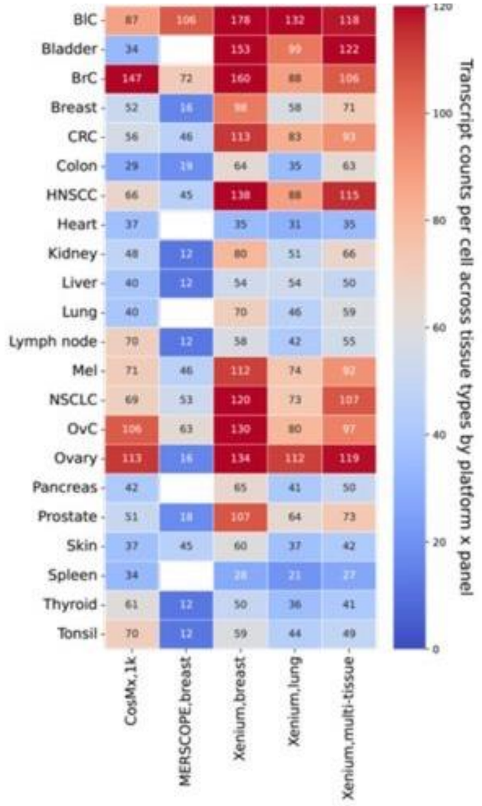


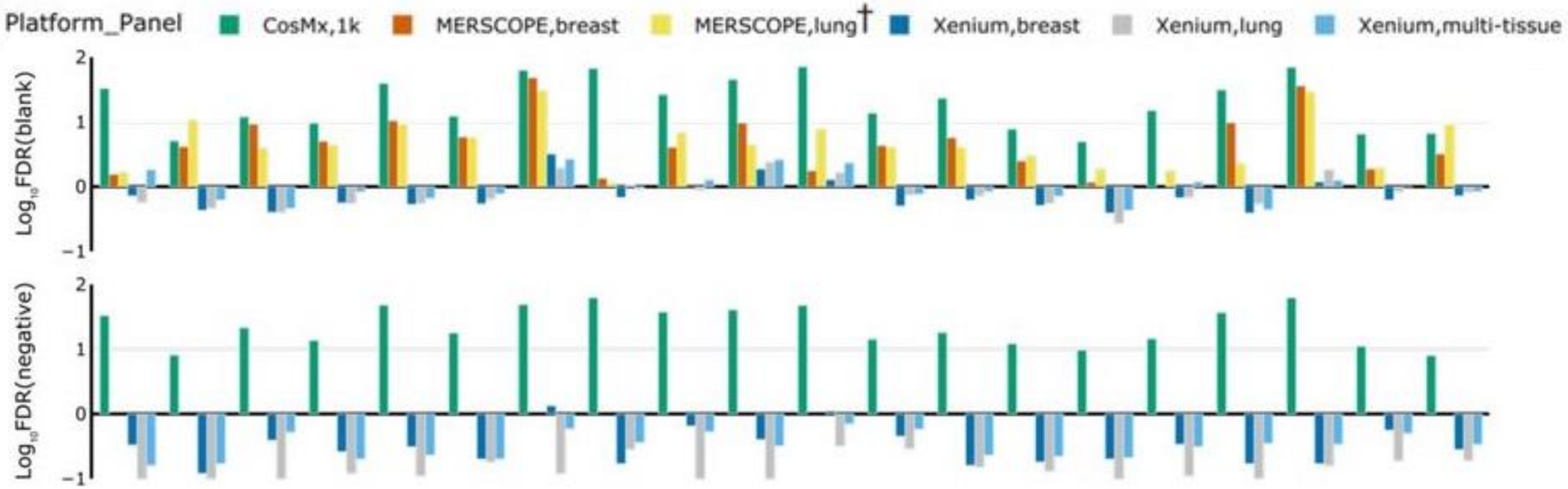


Wang et al, BioRxiv, 2023

Benchmarking MERSCOPE, CosMx, Xenium

- Overall: Xenium > MERSCOPE > CosMx





Wang et al, BioRxiv, 2023

Considerations when choosing platform

- Biological question
  - Spatial resolution vs transcriptome coverage
- Cell-typing vs gene programs
  - Designing probe panel
  - Cost!!!
  - Time – microscopy based methods can take >24h (sometimes days) to run
- Data analysis
  - Microscopy based workflows less established

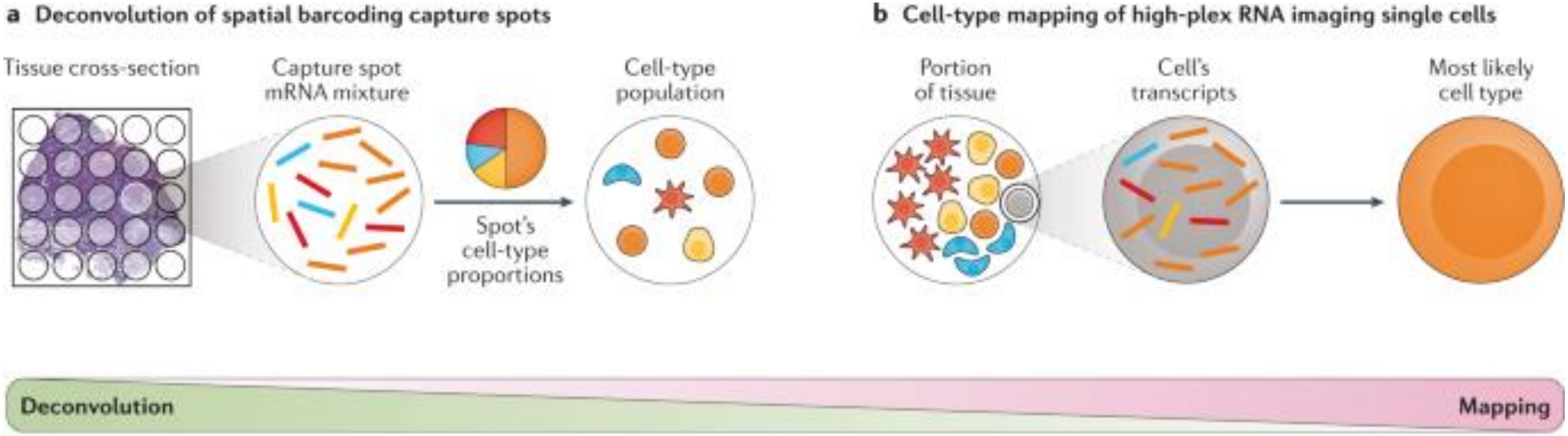
Computational problems

Computational workflow for spatial transcriptomics, sequencing based

- Sequencing based
  - Quality control
  - Demultiplexing
  - Align reads
  - Quantify genes/transcripts
    - Positions as metadata
  - Obtain gene x spot matrix
  - Use annotated scRNA-seq data to deconvolute spots
  - Downstream analysis
    - Spatial domains, neighborhood analysis, etc

- ISH based
  - Quality control
  - Fluorescent barcode decoding
  - Segmentation
  - Quantify genes/transcripts
  - Obtain gene x cell matrix
    - Positions as metadata
  - Use annotated scRNA-seq data to assign cell types
  - Downstream analysis
    - Spatial domains, neighborhood analysis, etc

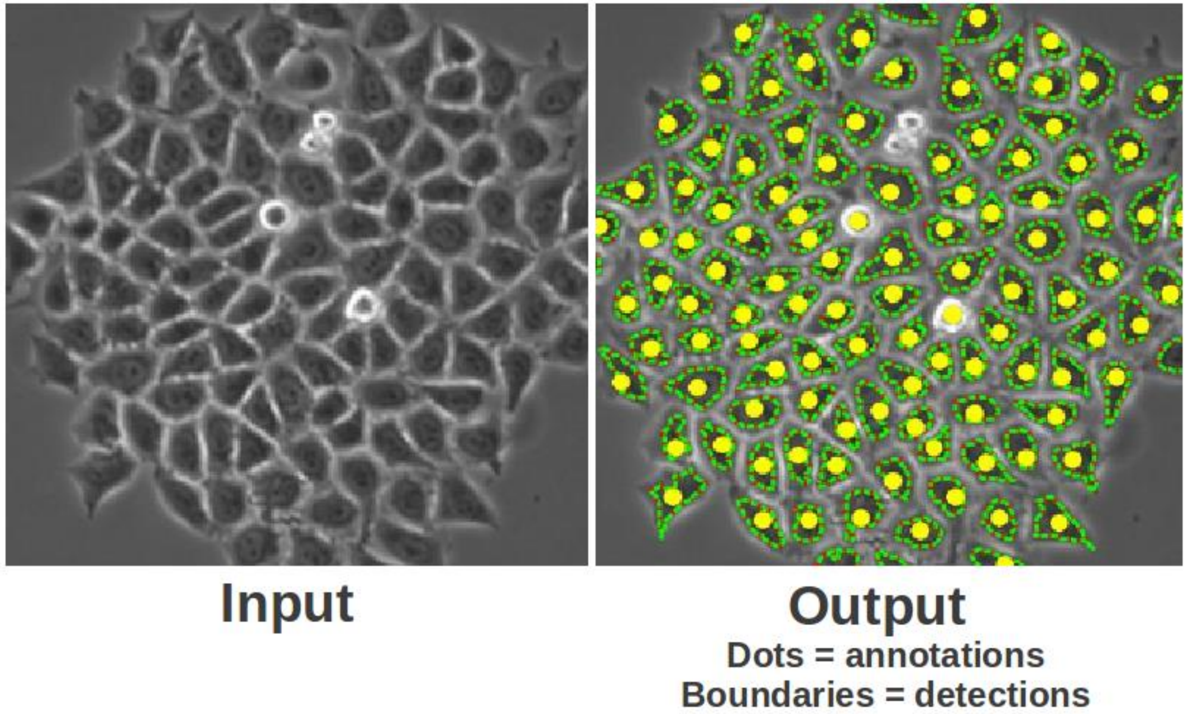
Deconvolution/mapping



Longo et al, Nat Rev Gen, 2021

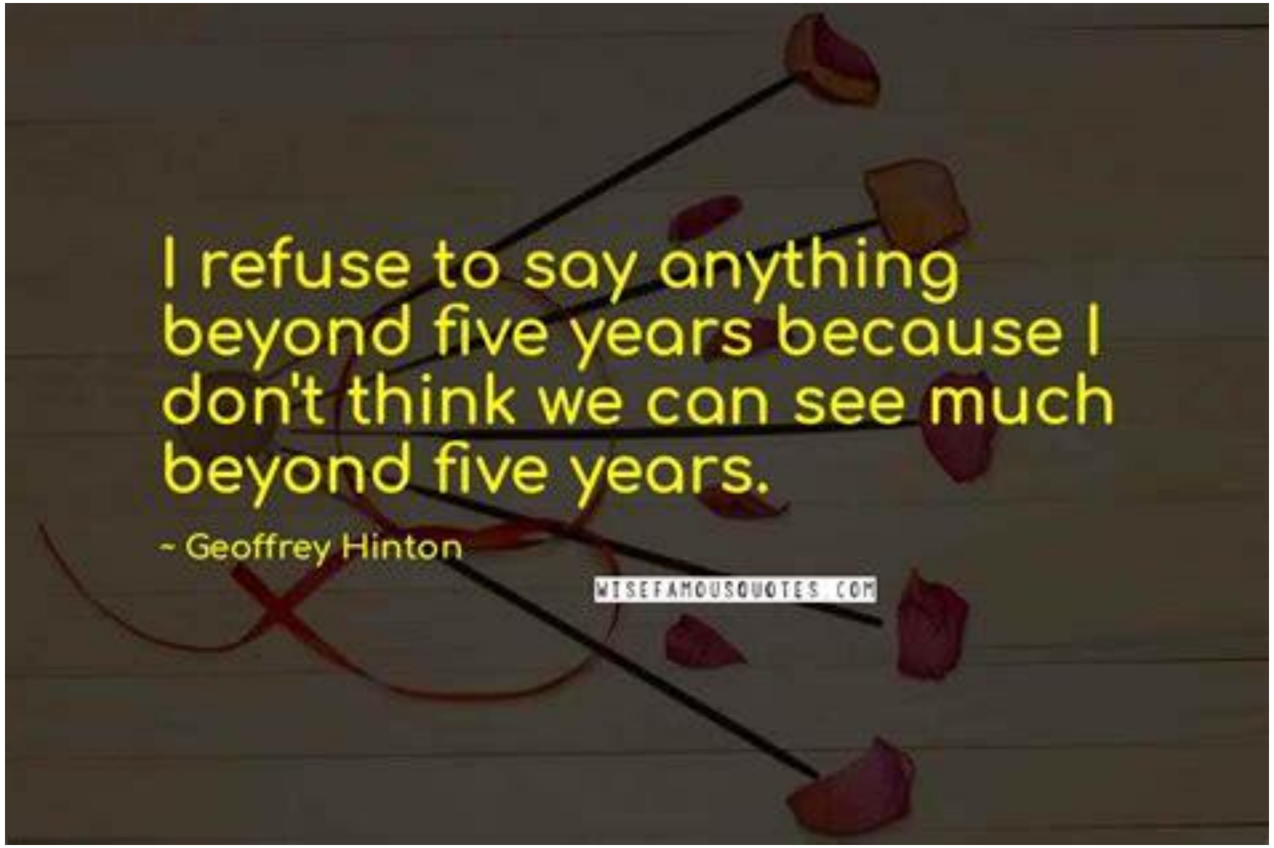
What is segmentation?

- To simplify the representation of an image into something that is more meaningful and easier to analyze. More precisely, image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics.
- Has been used for ordinary microscopy for decades
  - Vicar et al (BMC Bioinformatics, 2019) benchmark 38 methods
  - Since then many, many deep learning methods



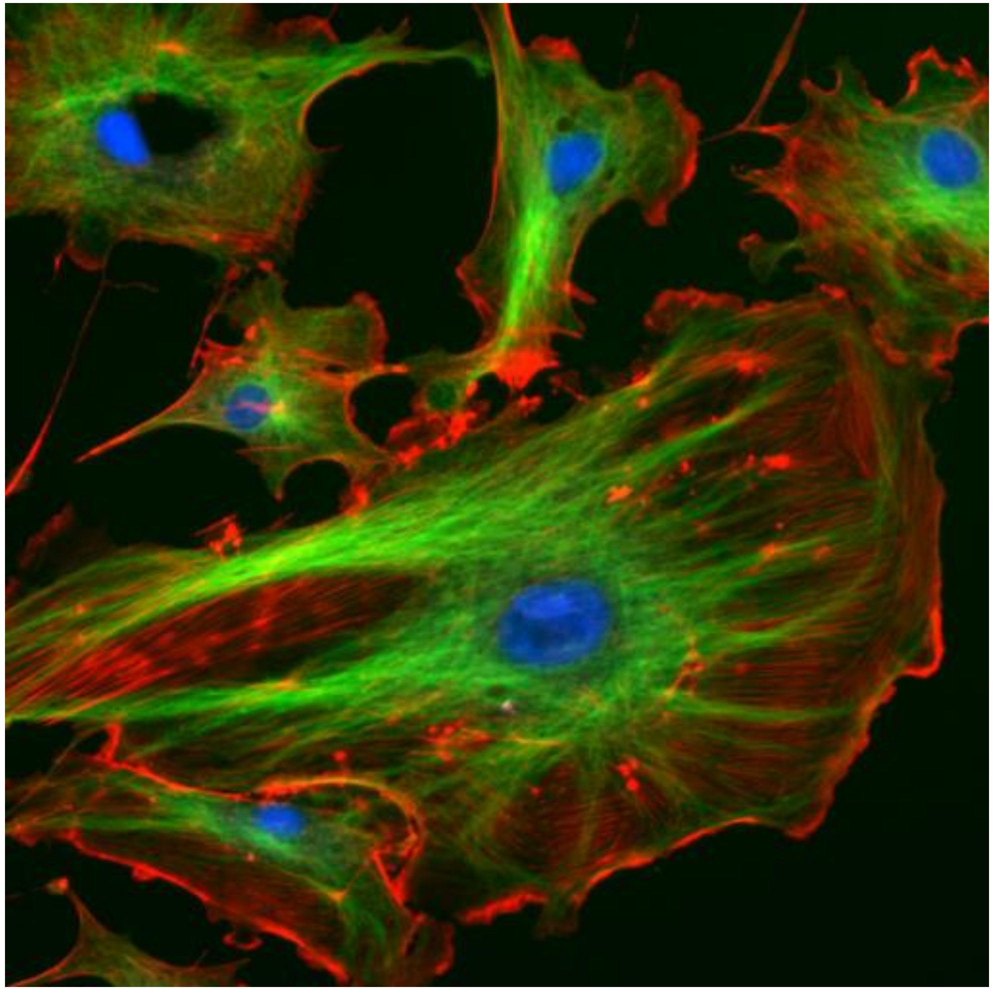
Geoffrey Hinton in 2016 on radiologists

- “People should stop training radiologists now. It’s just completely obvious within five years deep learning is going to do better than radiologists …. It might be 10 years, but we’ve got plenty of radiologists already.”



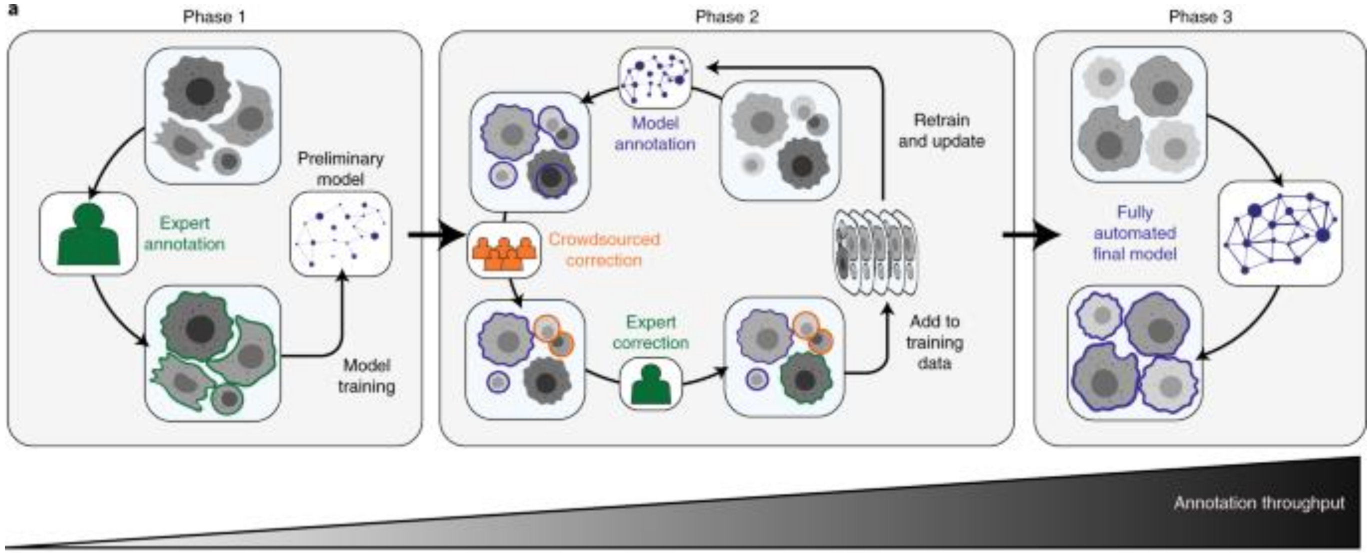
Segmentation strategies

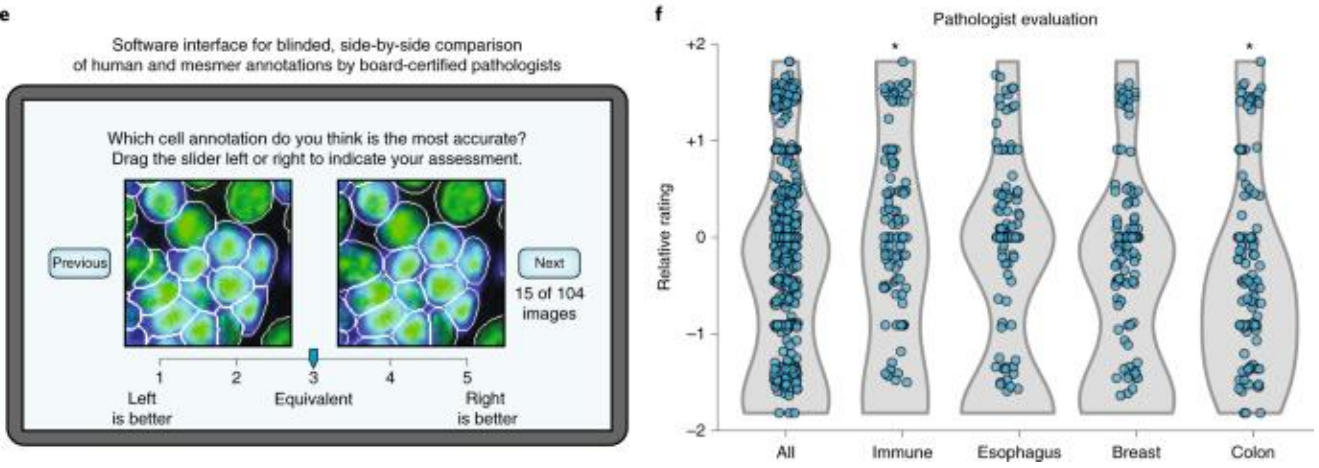
- Staining based (e.g. DAPI)
  - Mesmer
  - Cellpose
- Transcript based
  - Baysor
  - Clustermap

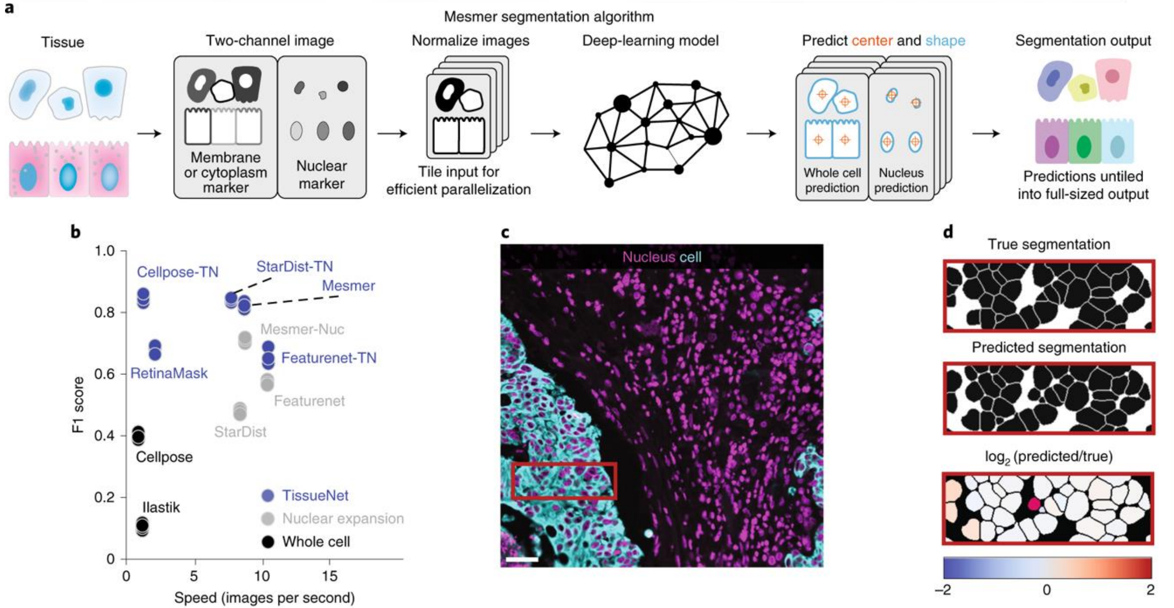


Mesmer retains human in the loop

- TissueNet contains >1M cell/nuclear pairs
- Mesmer is deep learning model

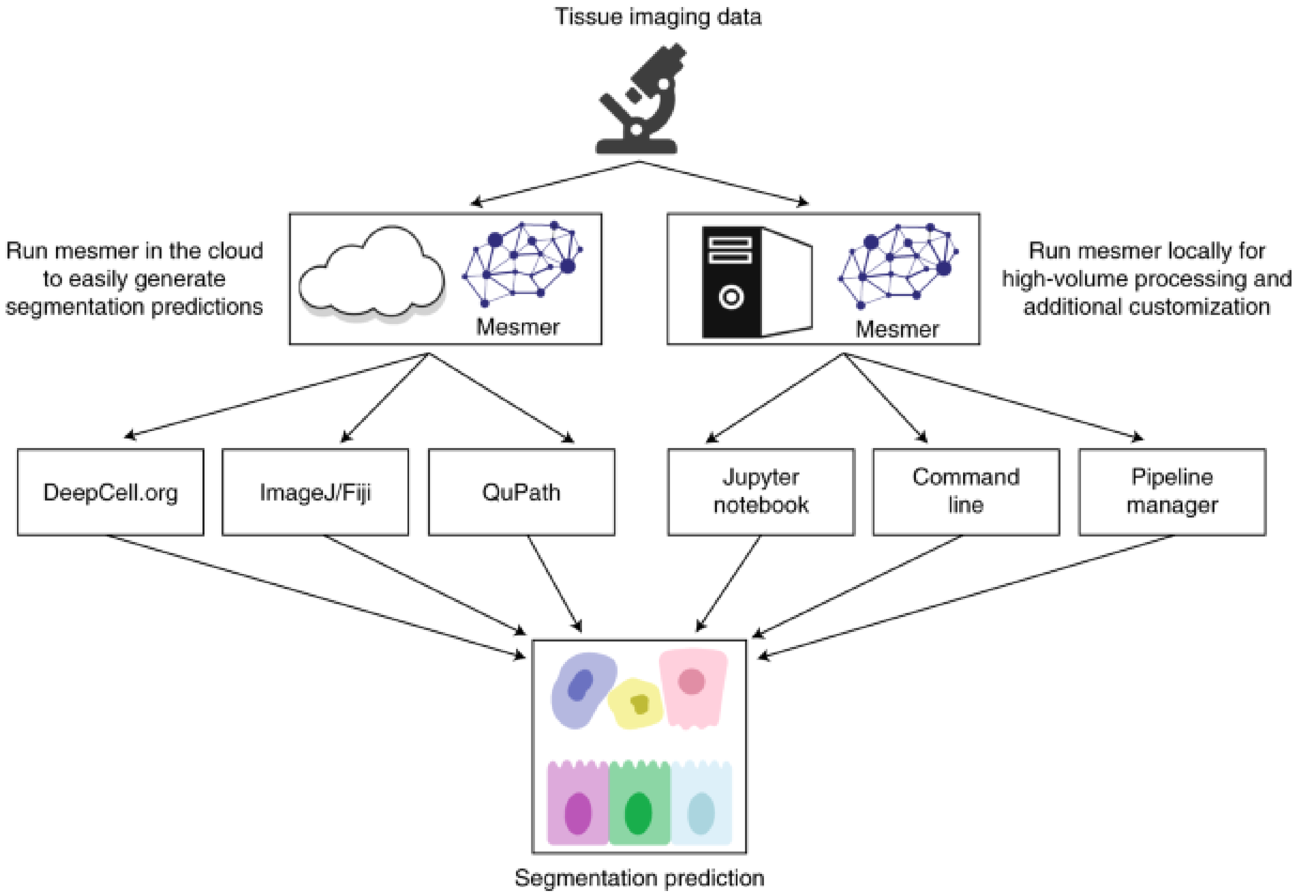


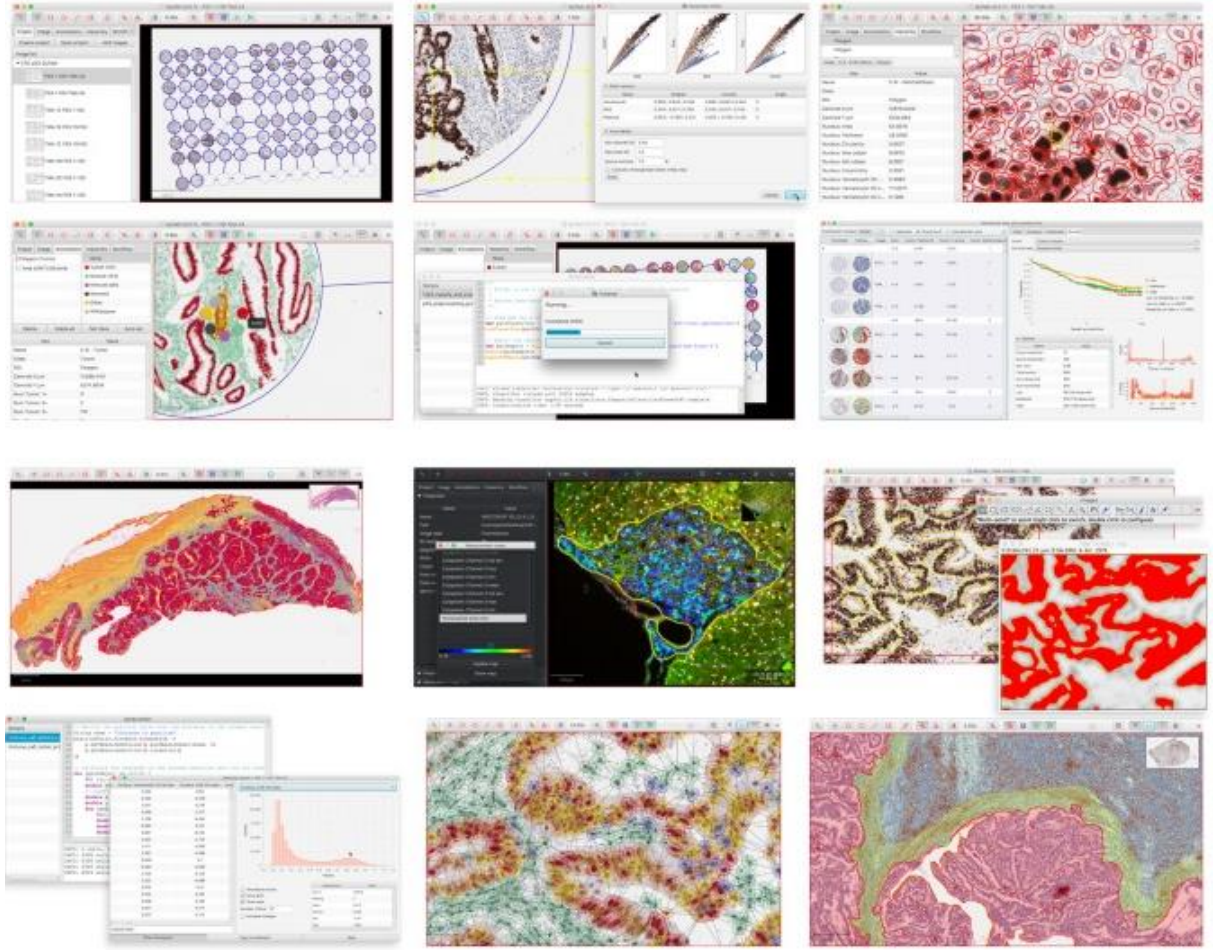




Greenwald et al, Nat Biotech, 2022

QuPath and DeepCell.org for fine tuning



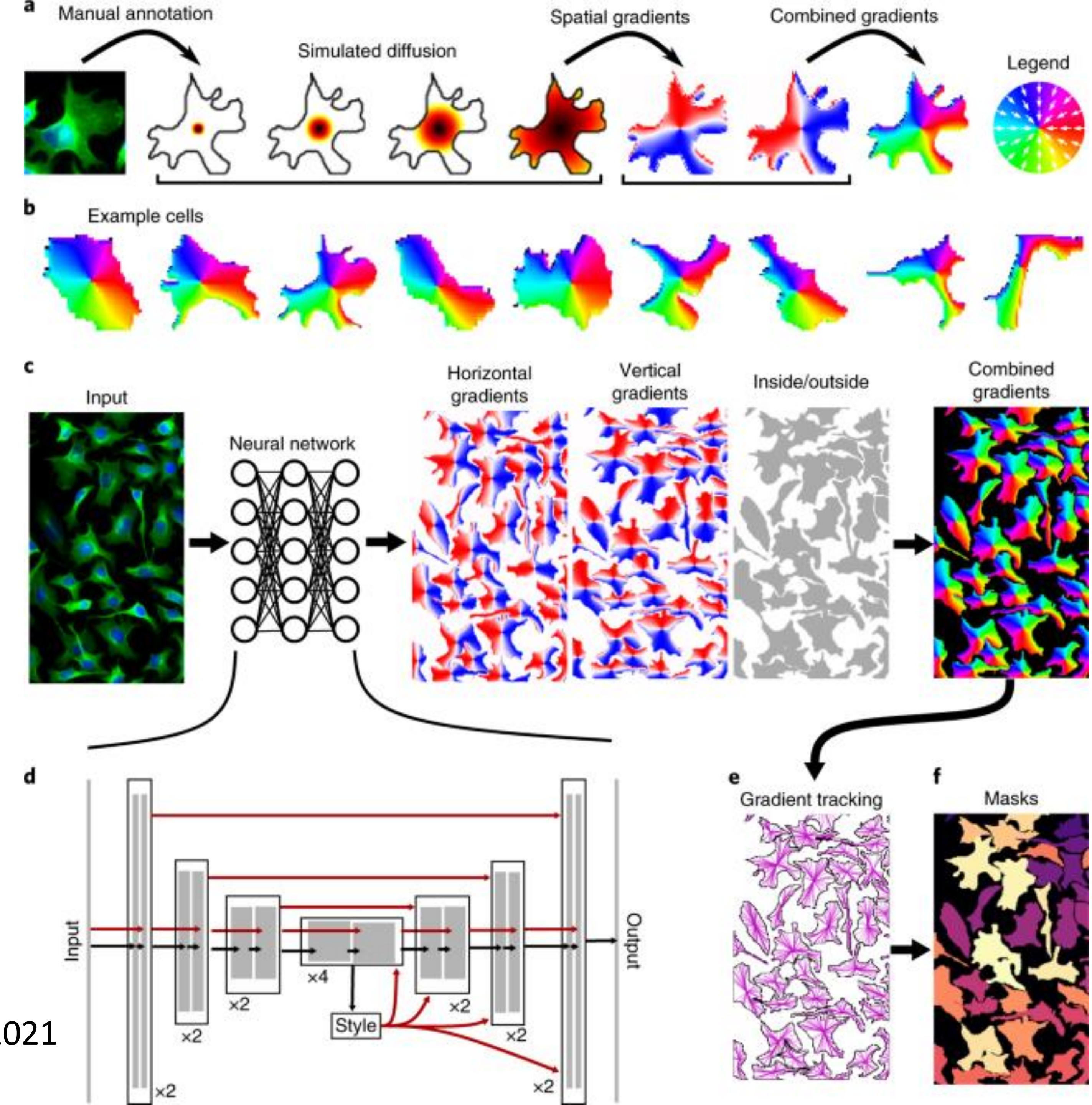


Bankhead et al (Scientific Reports, 2017)

Cellpose for image segmentation

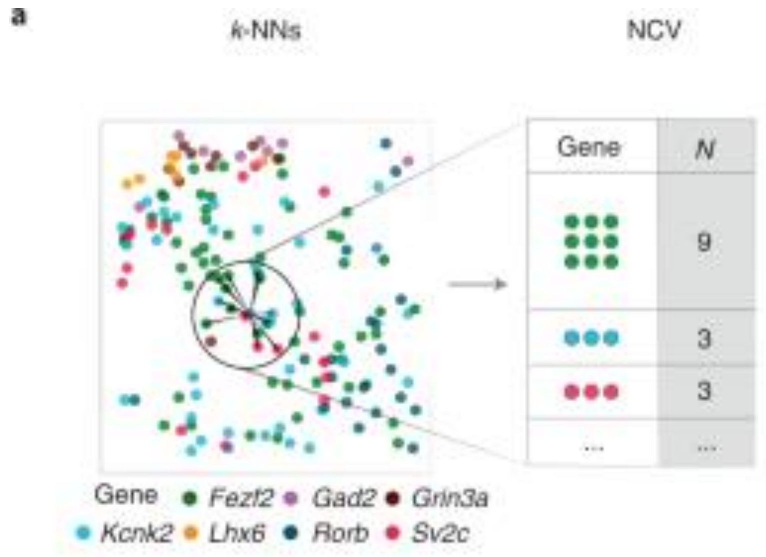
- Deep learning
  - Trained on >70k objects
  - Predict spatial gradients
  - Use gradients to assign pixels to cells

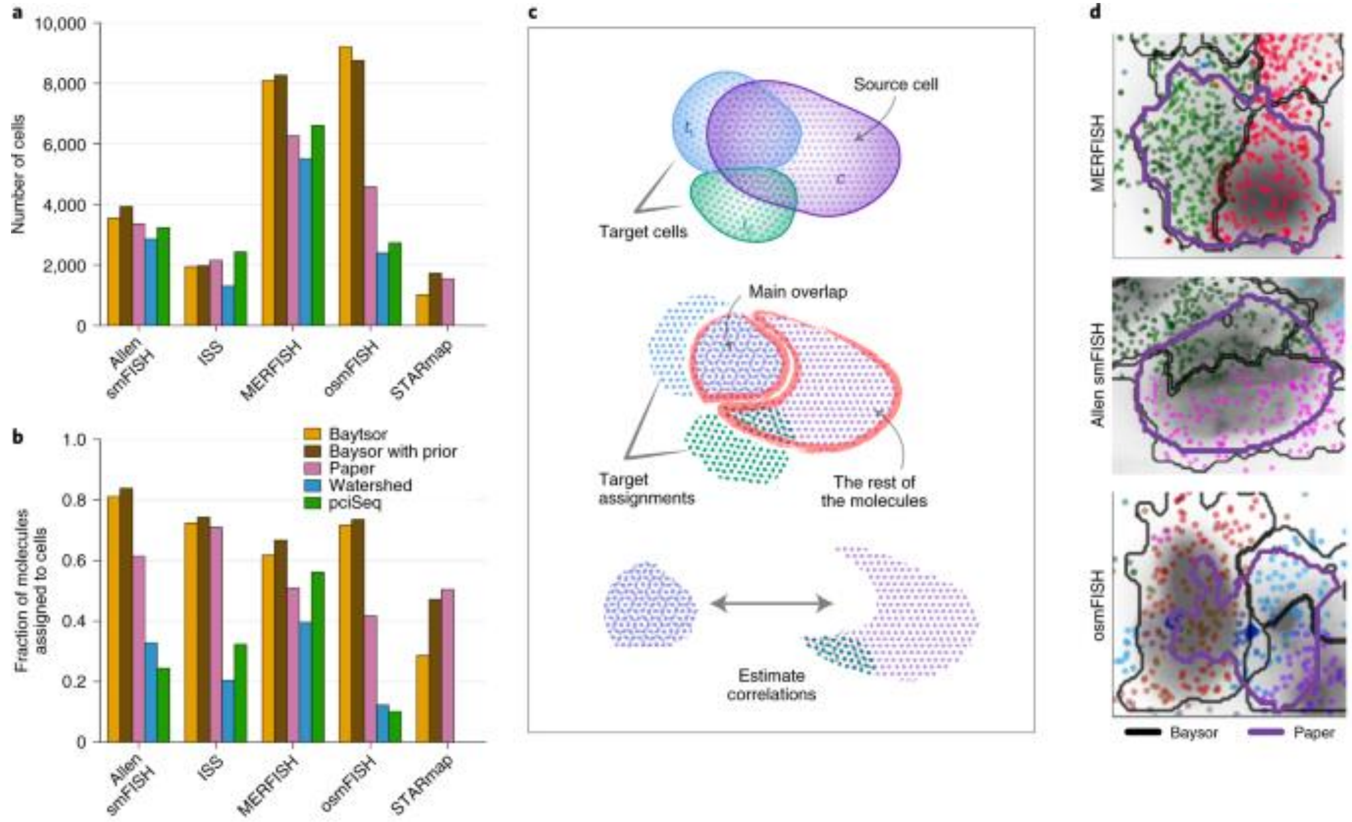
Stringer et al, Nat Methods, 2021



Baysor uses transcript density

- Build kNN graph from transcripts
  - No segmentation needed

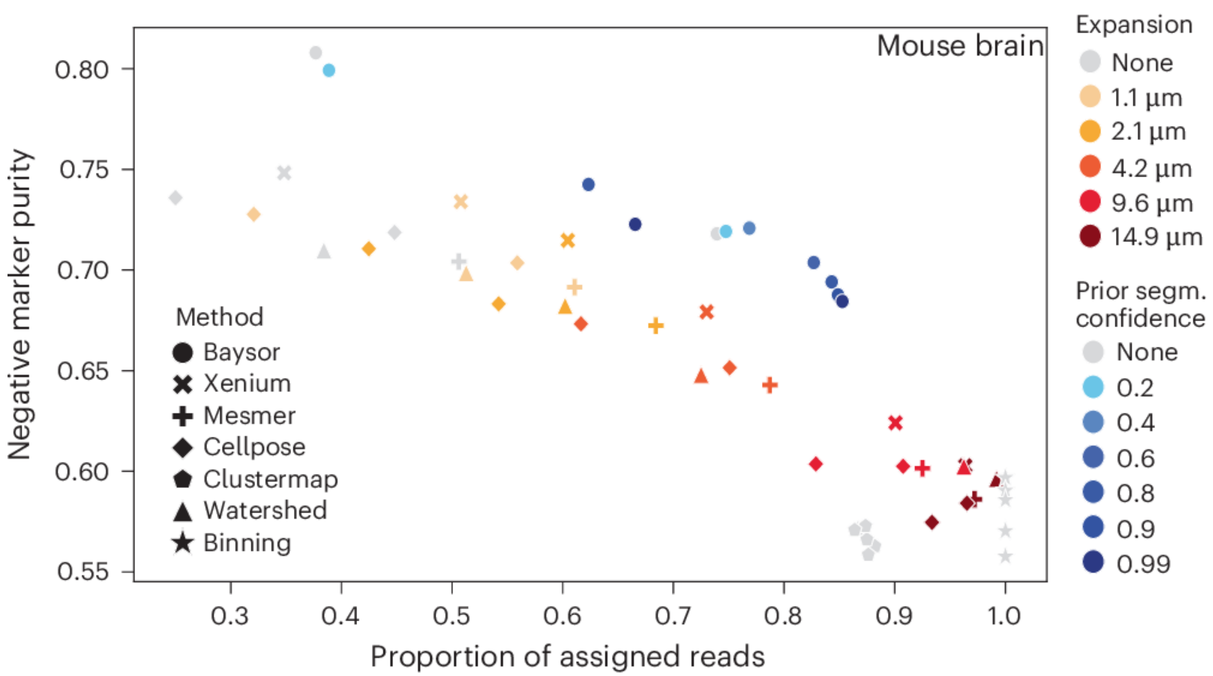


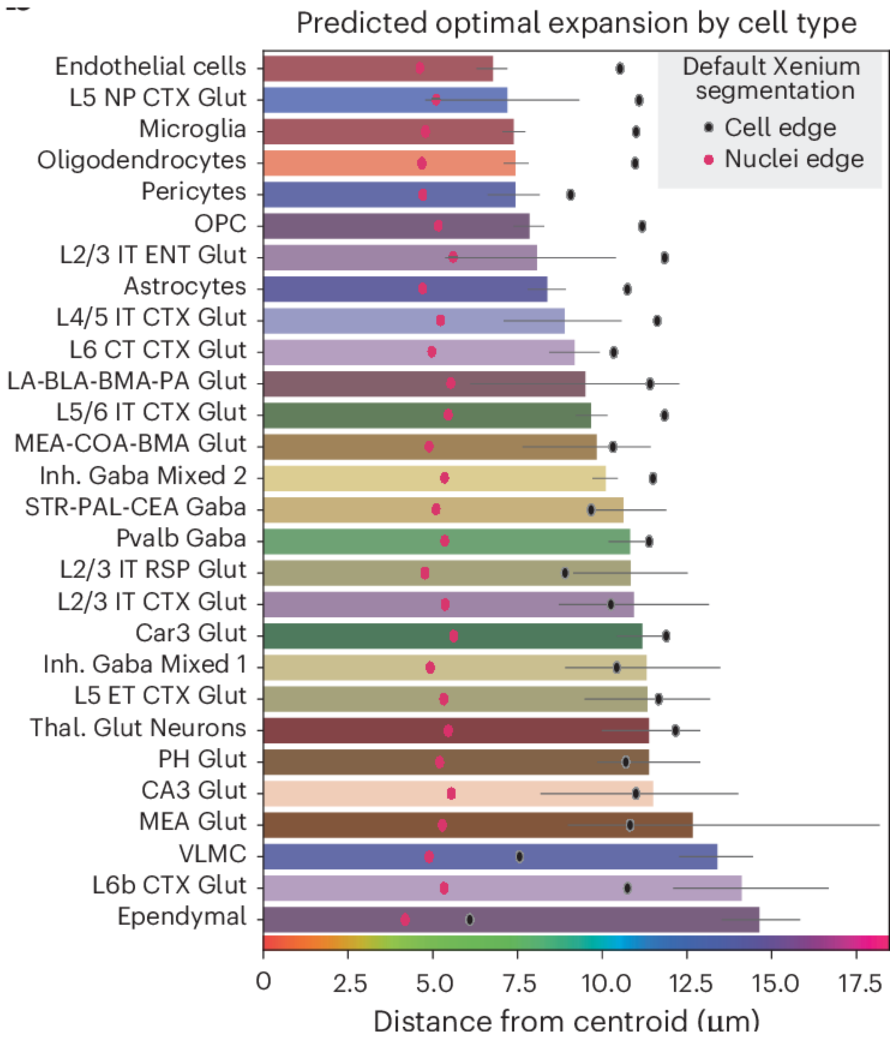


Petukhov et al (Nat Biotech, 2022)

Comparison of segmentation for Xenium

- Methods requiring prior on cell radius struggle across cell types

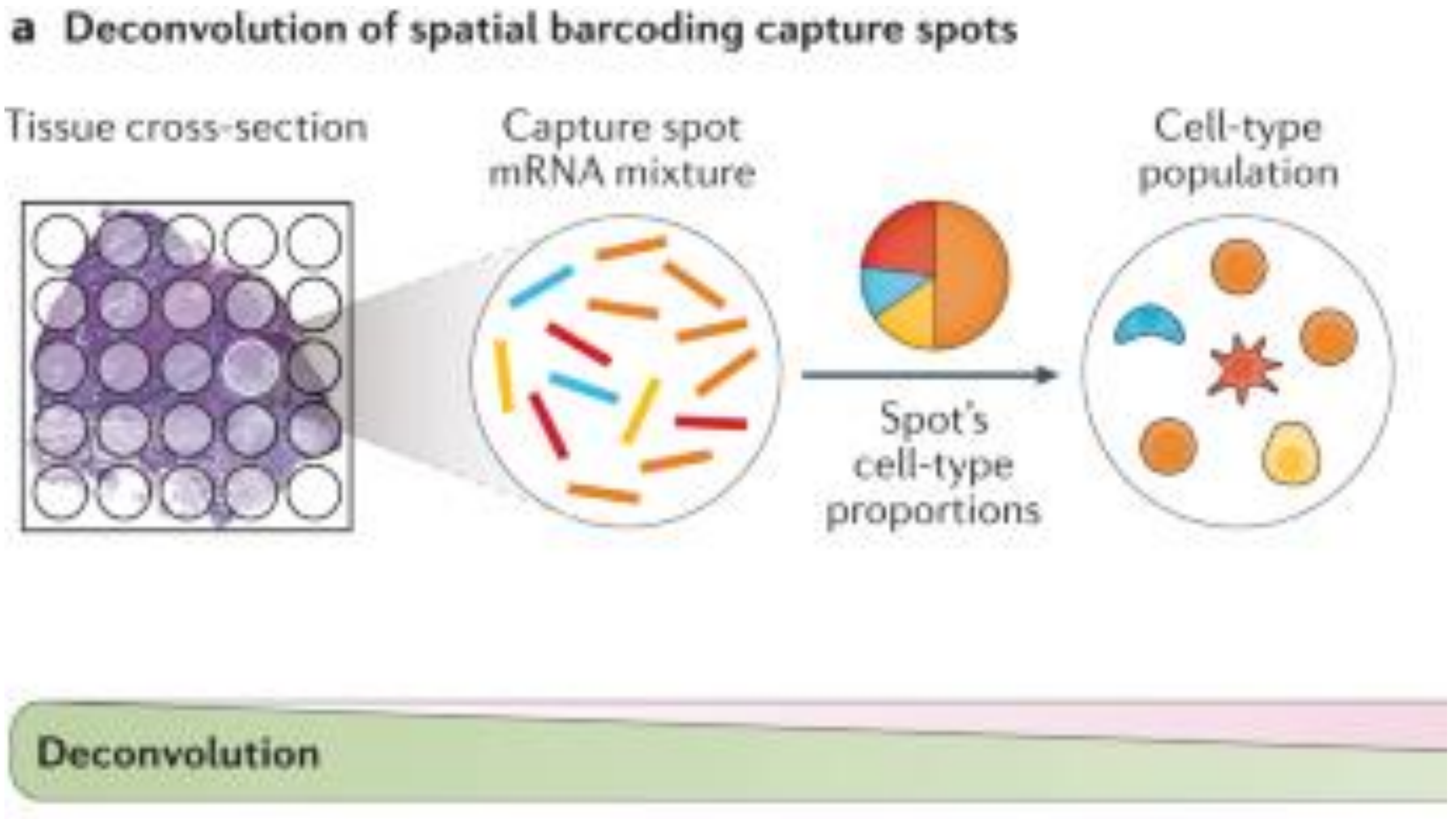




Salas et al (Nat Methods, 2025)

Deconvolution

- Probabilistic
  - Cell2location
  - Stereoscope
  - STdeconvolve
  - Adroit
- NMF
  - SPOTlight
  - spatialDWLS
- Deep learning
  - DestVI
  - Tangram
  - DSTG



Cell2location

- Bayesian model, decomposes count matrix
  - Hierarchical approach allows to ‘borrow’ information across space

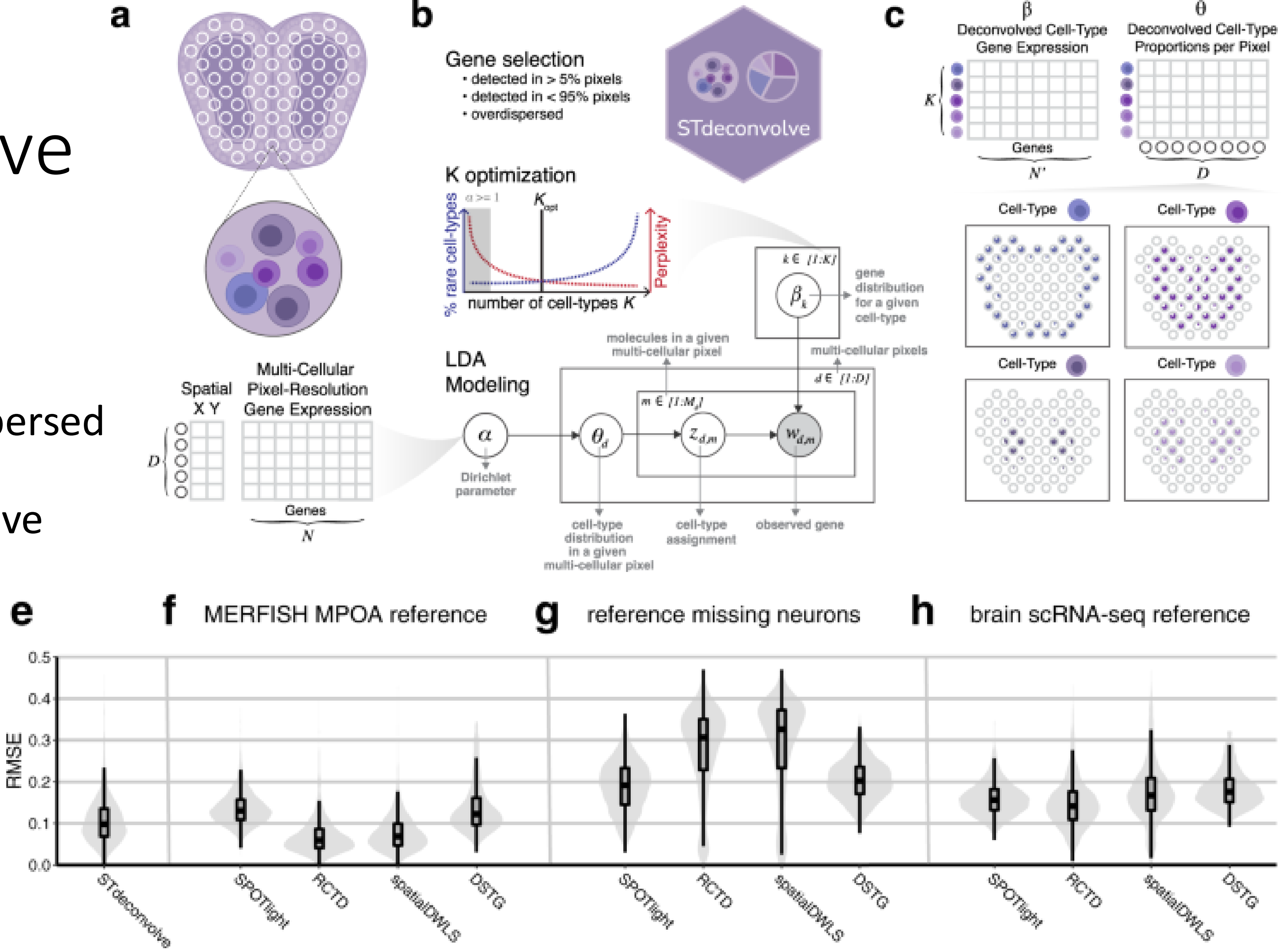
$$d_{s,g} \sim NB\left(\mu_{s,g}, \alpha_{e,g}\right)$$

$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f}\, g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}}$$

Kleshchevnikov et al, Nat Biotech, 2022

STdeconvolve

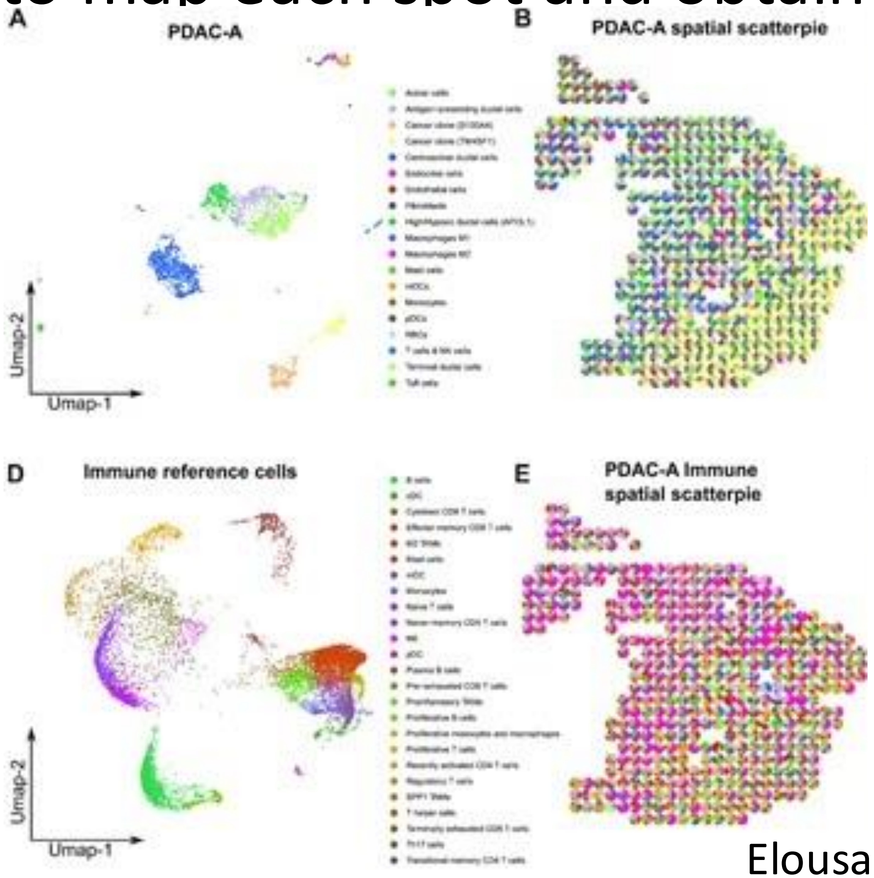
- Latent Dirichlet Allocation (LDA)
    - Select overdispersed genes, likely to be informative

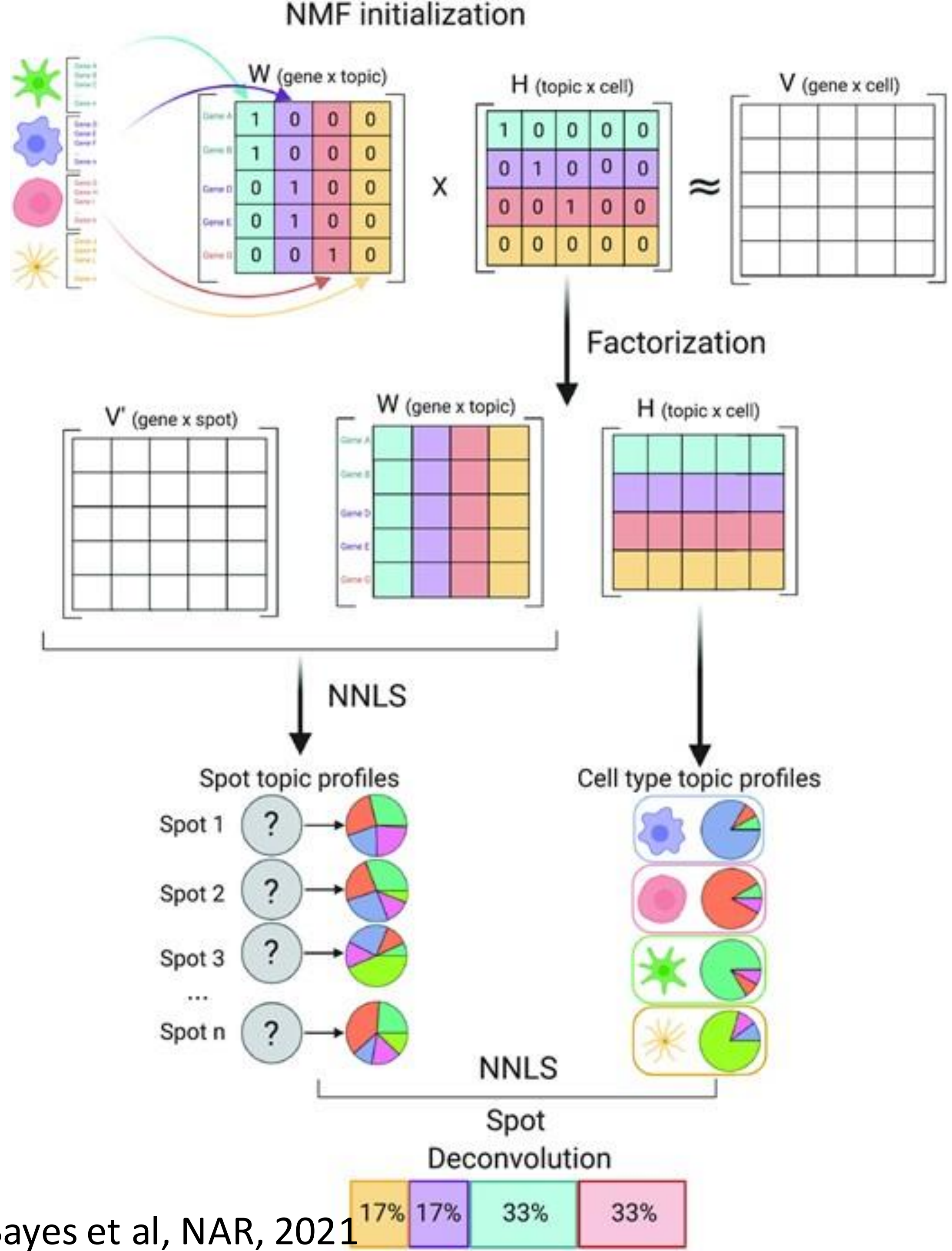


Miller et al, Nat Comms, 2021

SPOTlight

- Seeded NMF for sparse solutions
- Use NNLS to map each spot and obtain weights

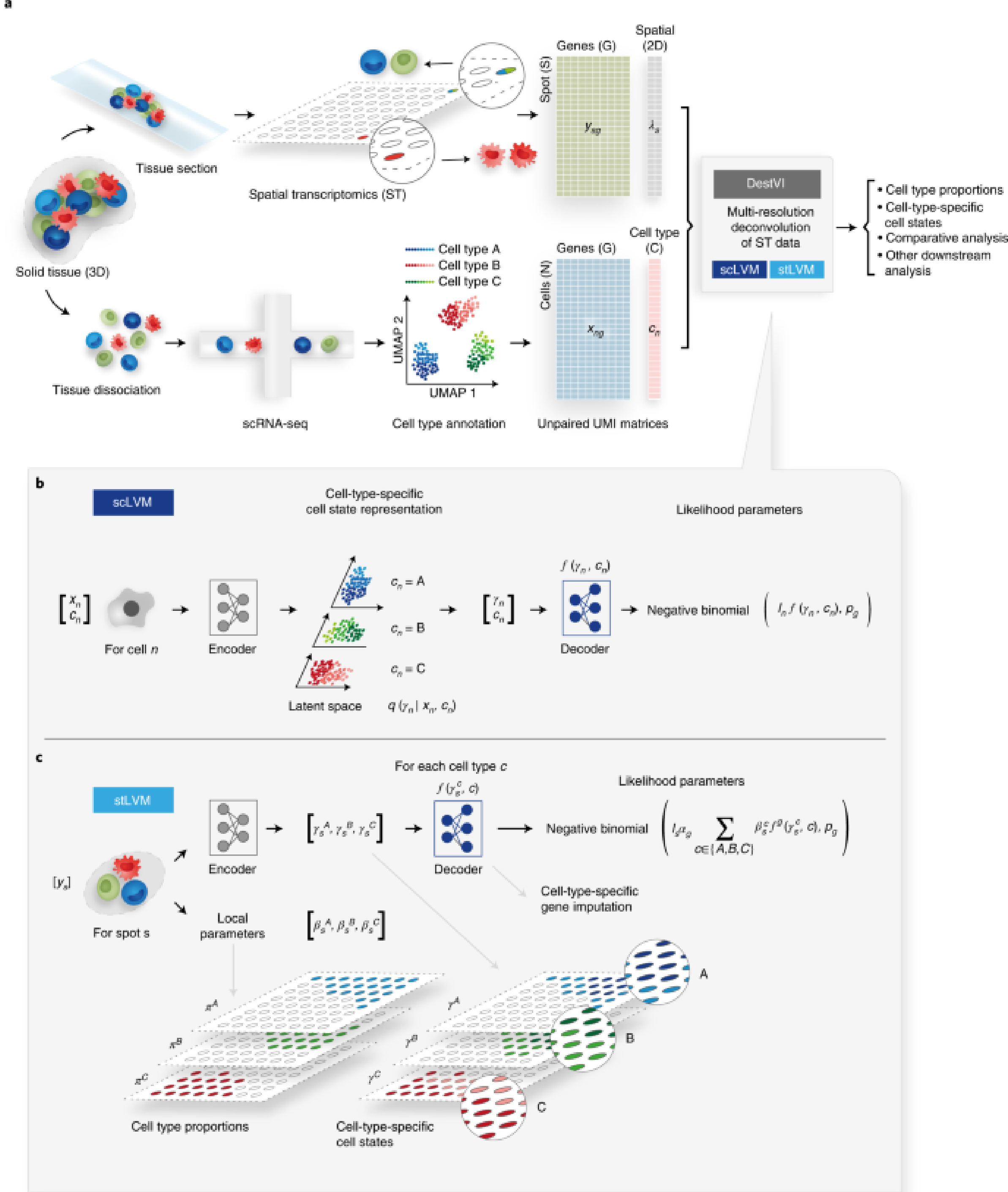




Elousa-Bayes et al, NAR, 2021

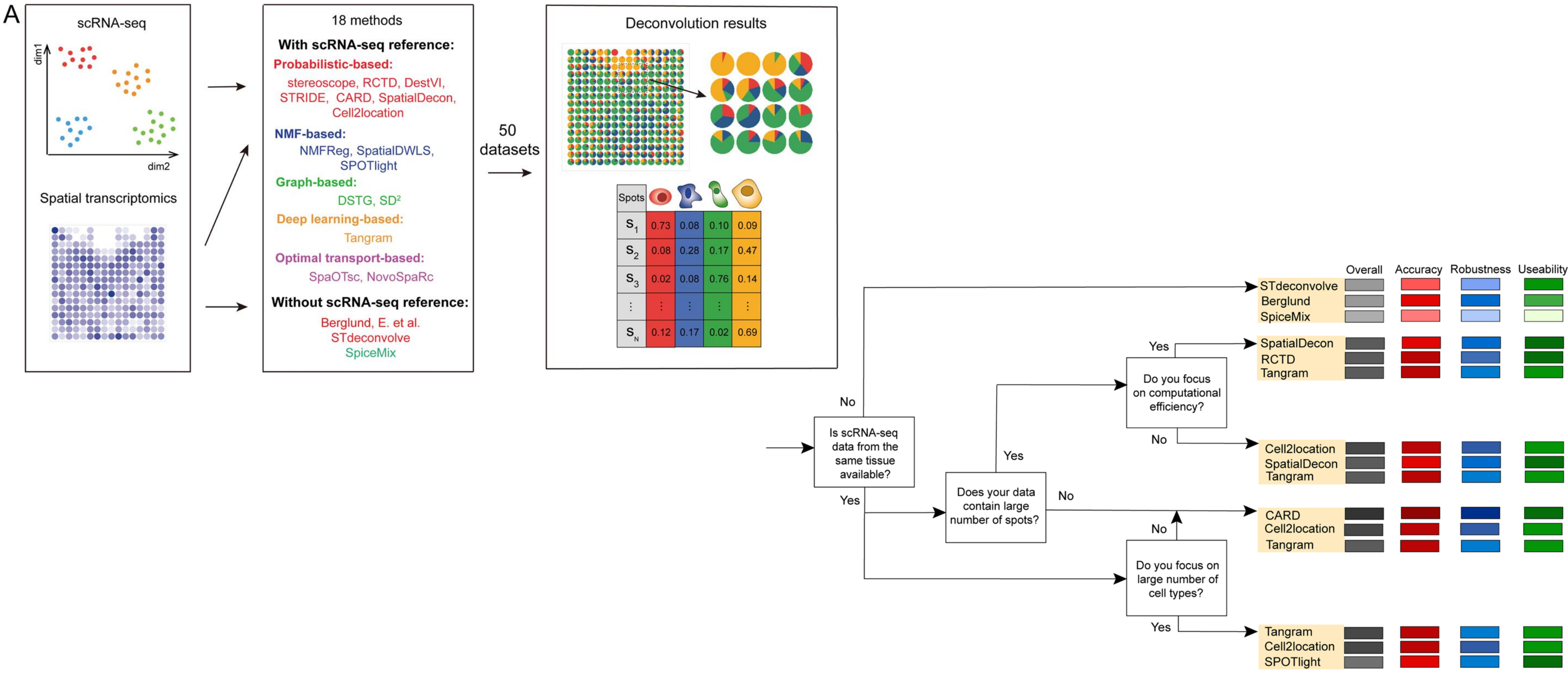
DestVI

- Part of the scVI framework
- Two latent variable models
    - Genes in a cell follow NB distribution
    - Genes in a spot also NB



Lopez et al, Nat Biotech, 2022

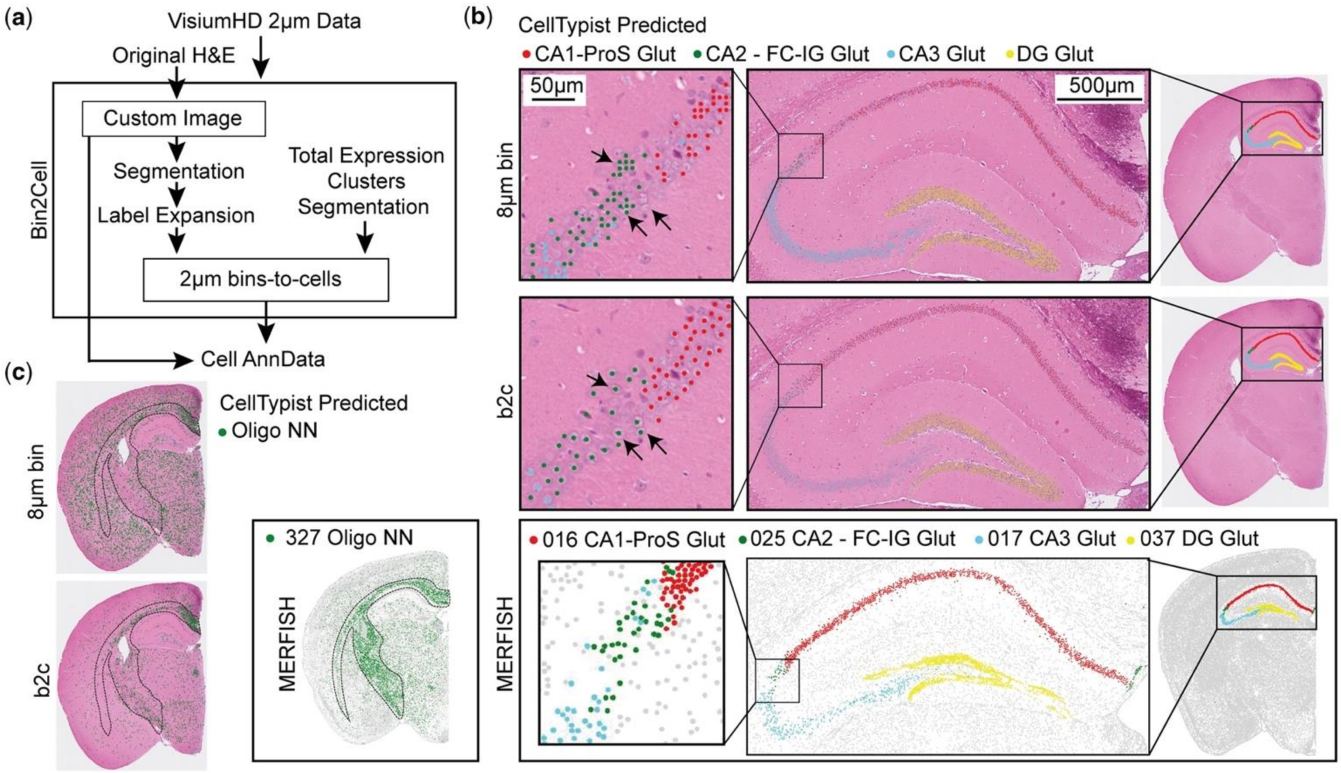
Benchmark of deconvolution methods

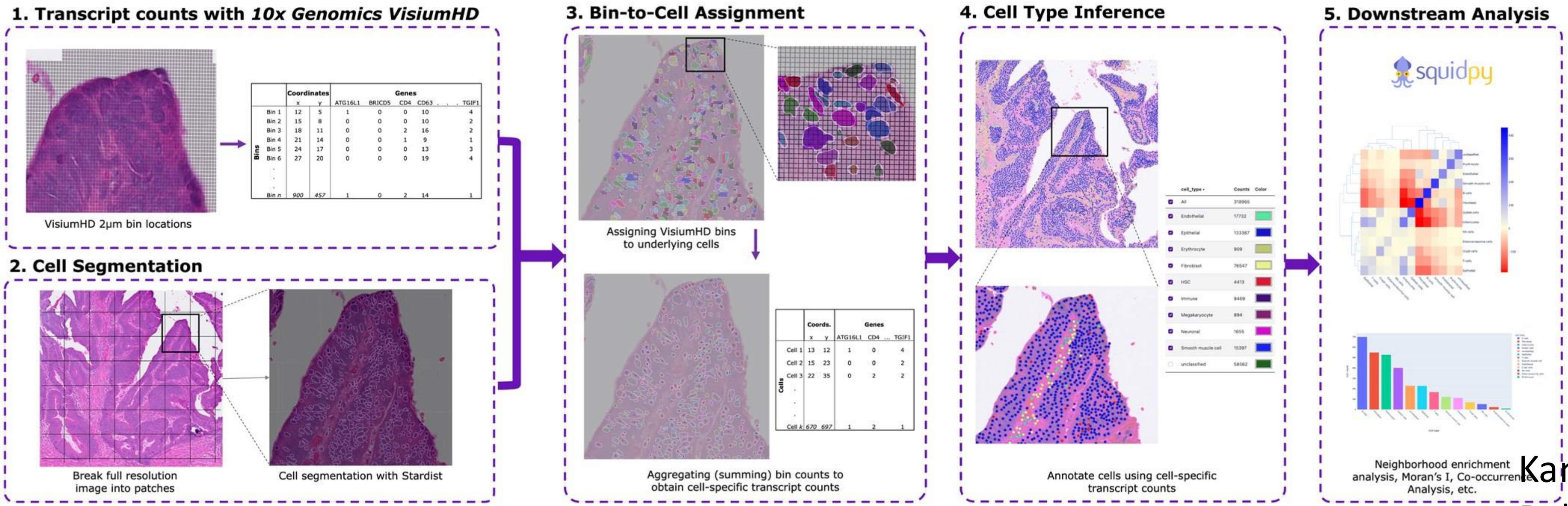


Li et al, Nat Comms, 2023

Visium HD processing

- More in 2026!



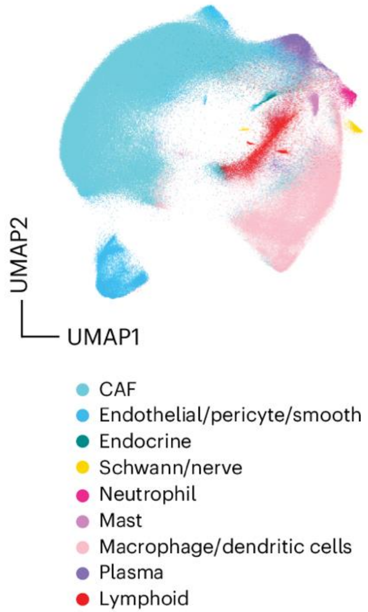


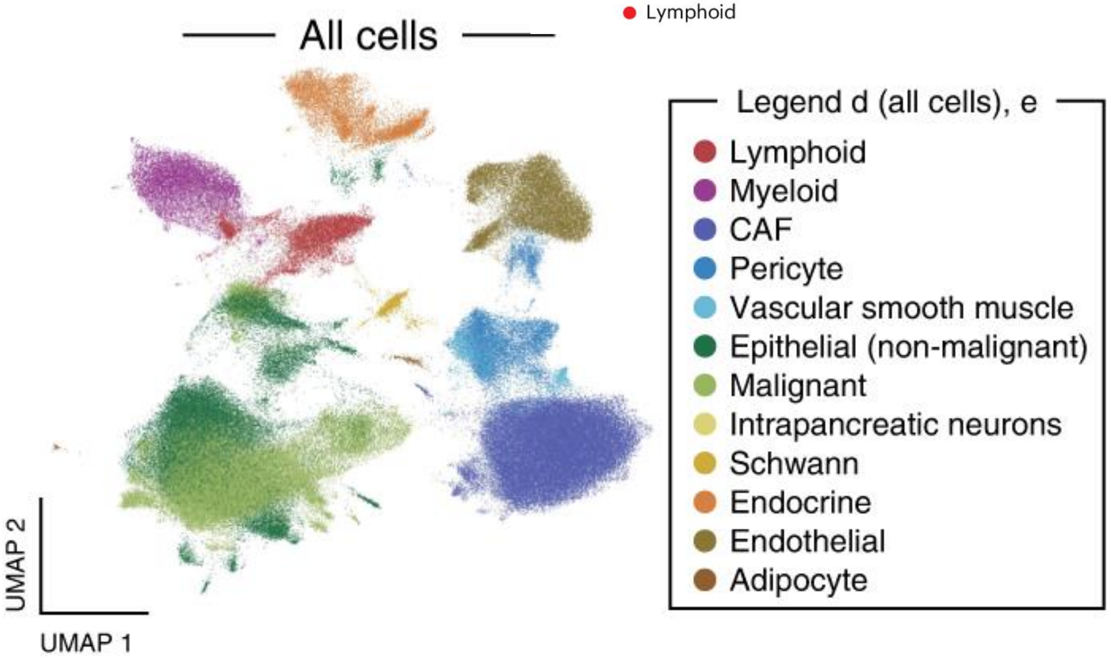
Kamel et al (Bioinformatics, 2025)

Polanski et al (Bioinformatics, 2024)

Annotation of probe based methods

- Limited panel size makes it more challenging to cluster
- Small number of genes and transcripts detected
  - Thresholds used ~50-100

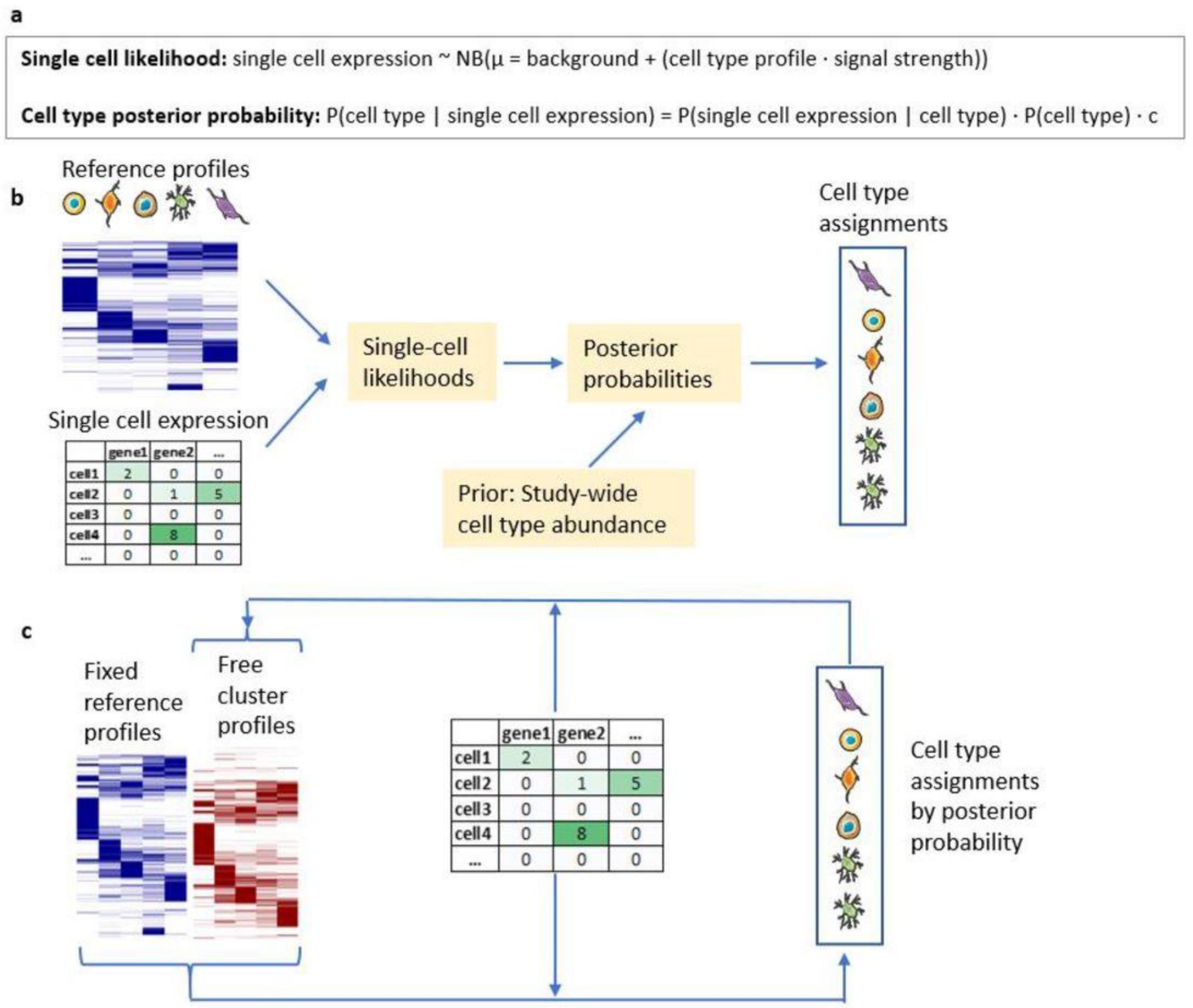




Hwang et al, Nat Genetics, 2022,
Shiau et al, Nat Genetics, 2024

InSituType

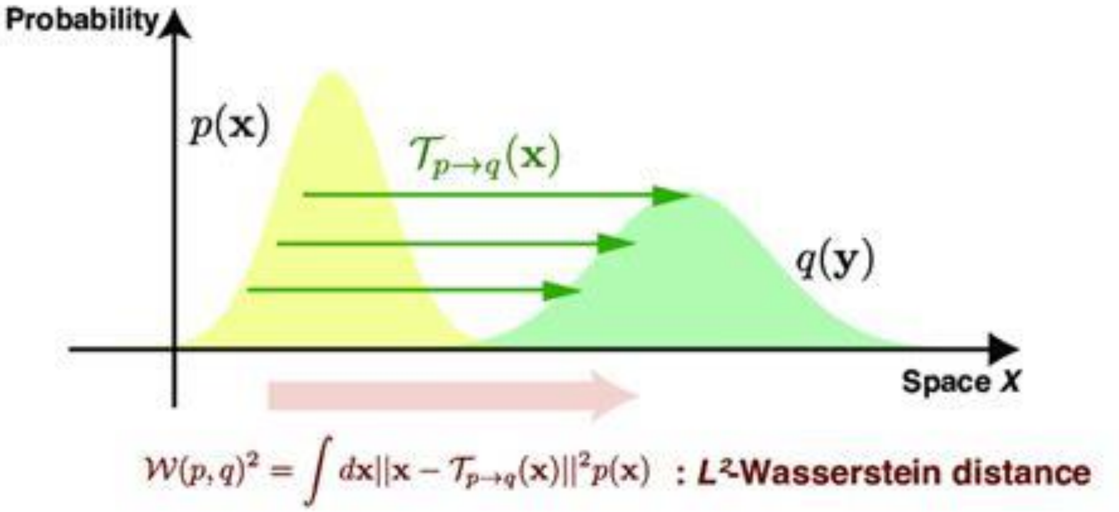
- Developed for CosMx
- Maximum likelihood approach

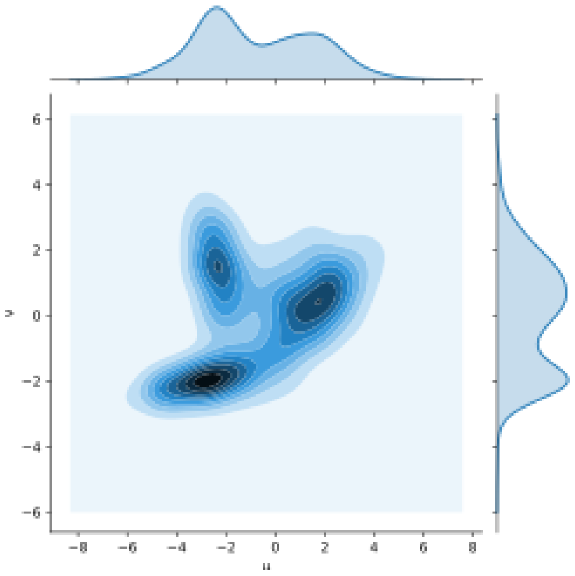


Danaher et al (BioRxiv, 2023)

Optimal transport (OT) theory

- Find the best plan for moving a resource from A to B
  - Cost function associated with moving each unit
  - Additional constraints, e.g. regularization



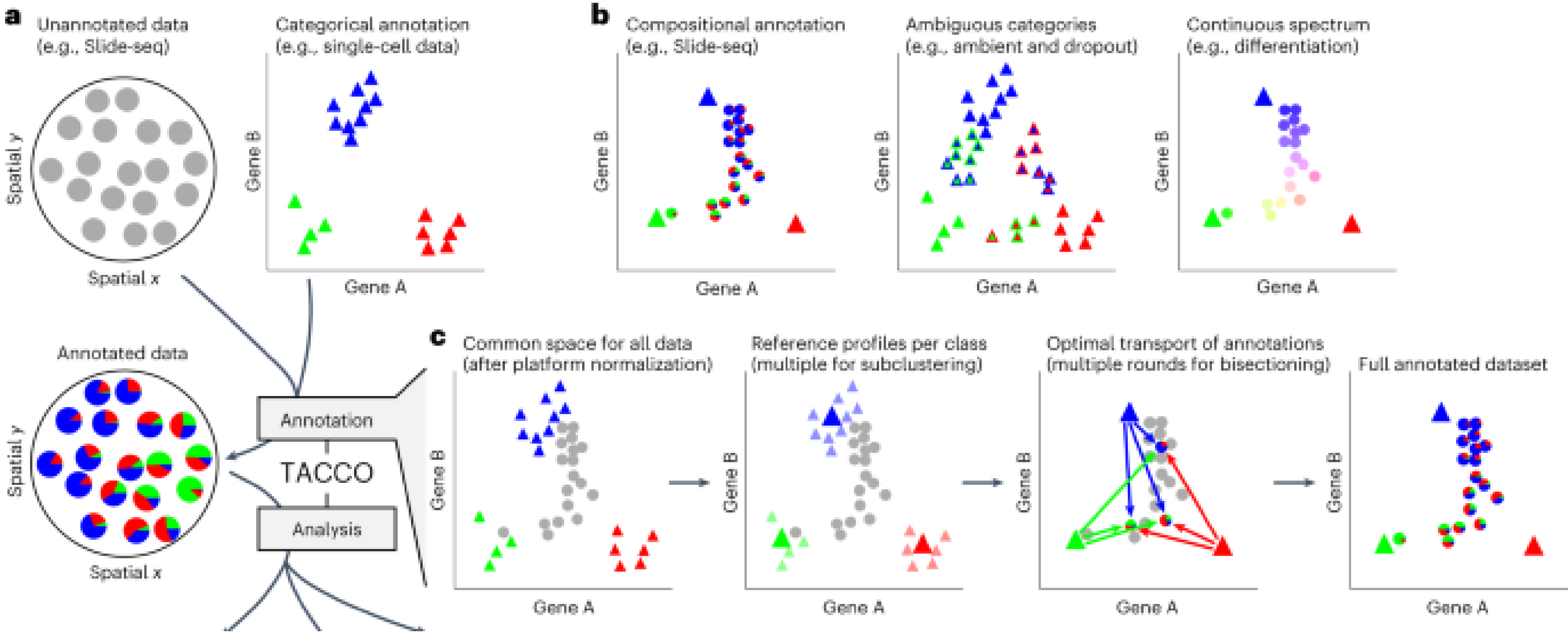


$$W_p(\mu,\nu) = \left(\inf_{\gamma\in\Gamma(\mu,\nu)} \mathbf{E}_{(x,y)\sim\gamma} d(x,y)^p \right)^{1/p}$$

Joint distribution

TACCO

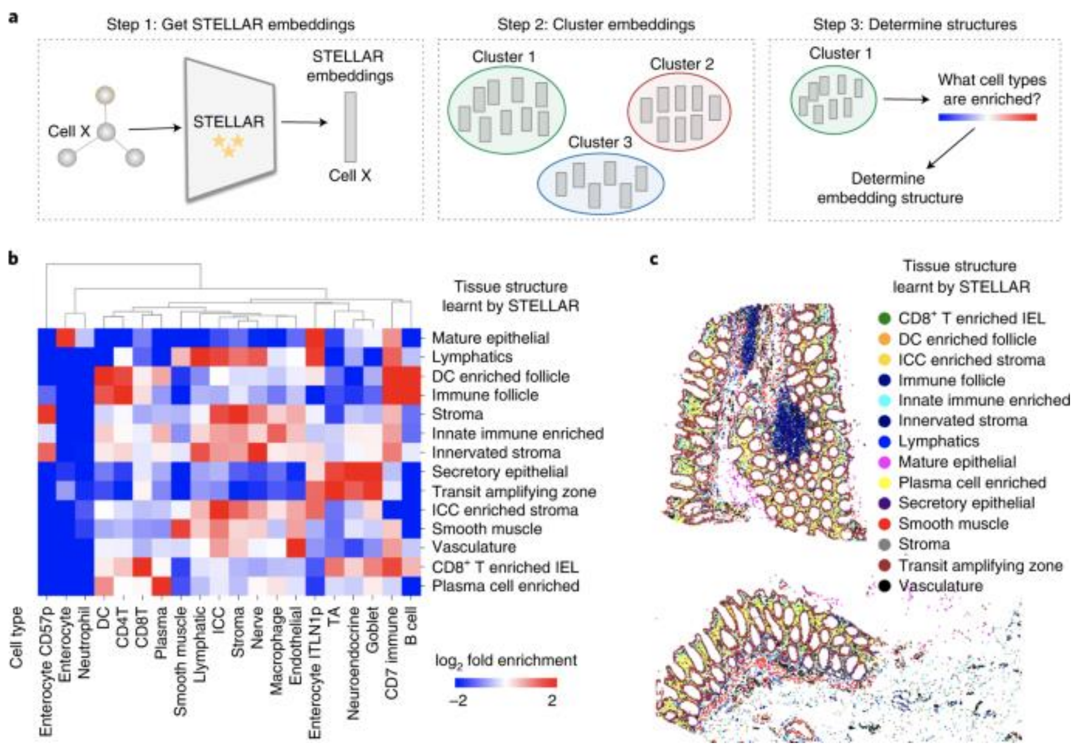
- OT to relate unannotated spatial data to scRNAseq
  - Both decomposition and annotation

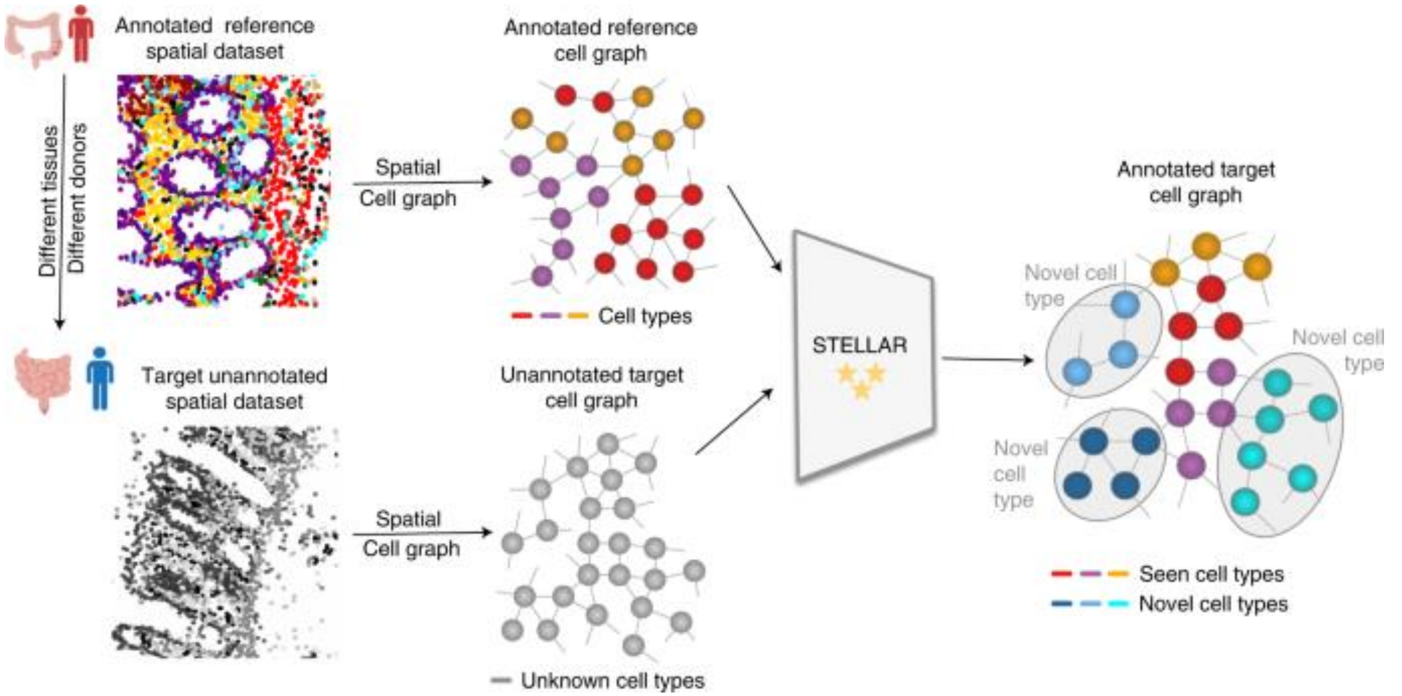


Mages et al (Nat Biotech, 2023)

STELLAR

- Geometric deep learning for annotation and cell type discovery
  - GCN
  - Can find new cell types



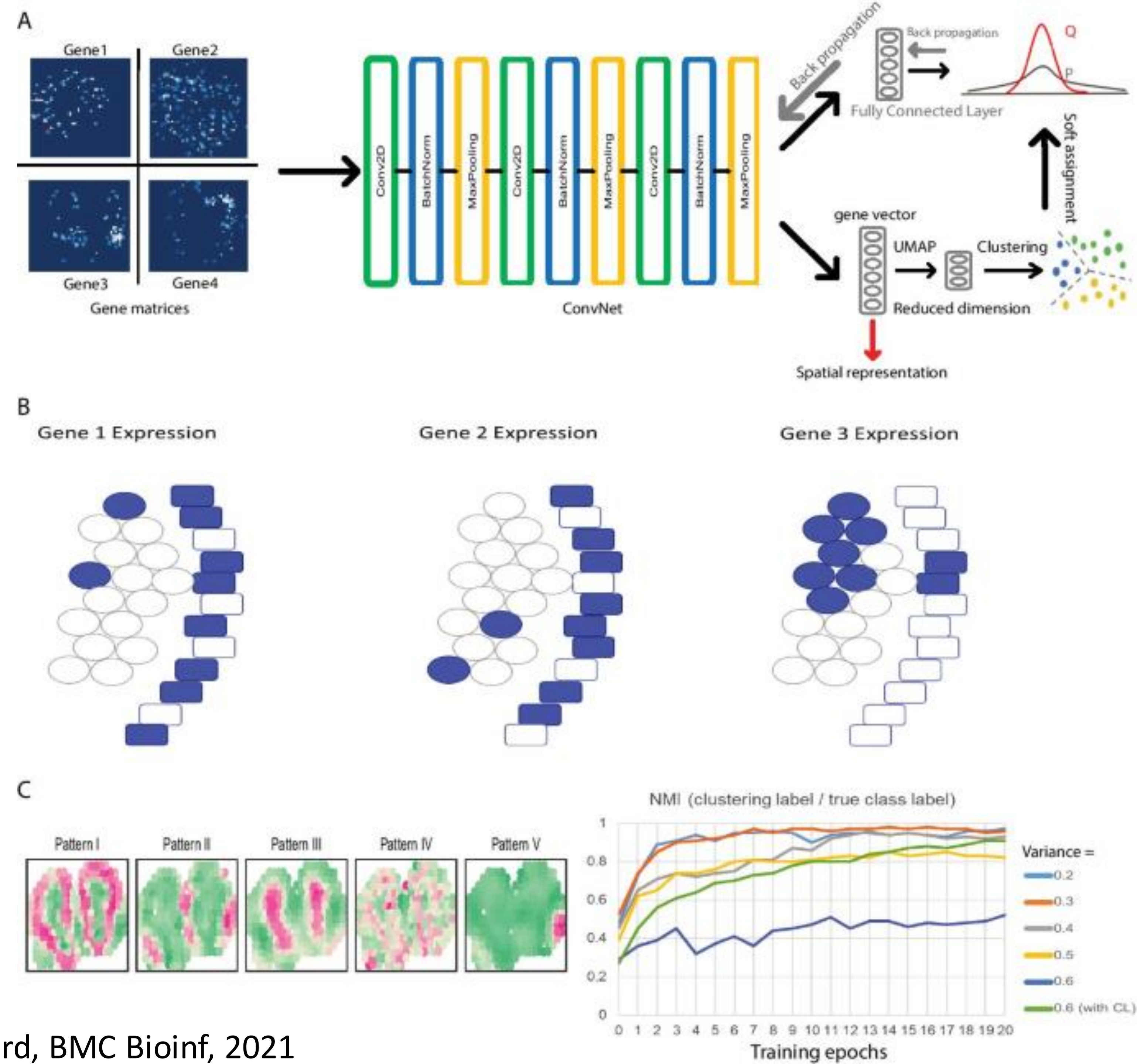


Spatial clustering

DEGs, variable genes, imputation

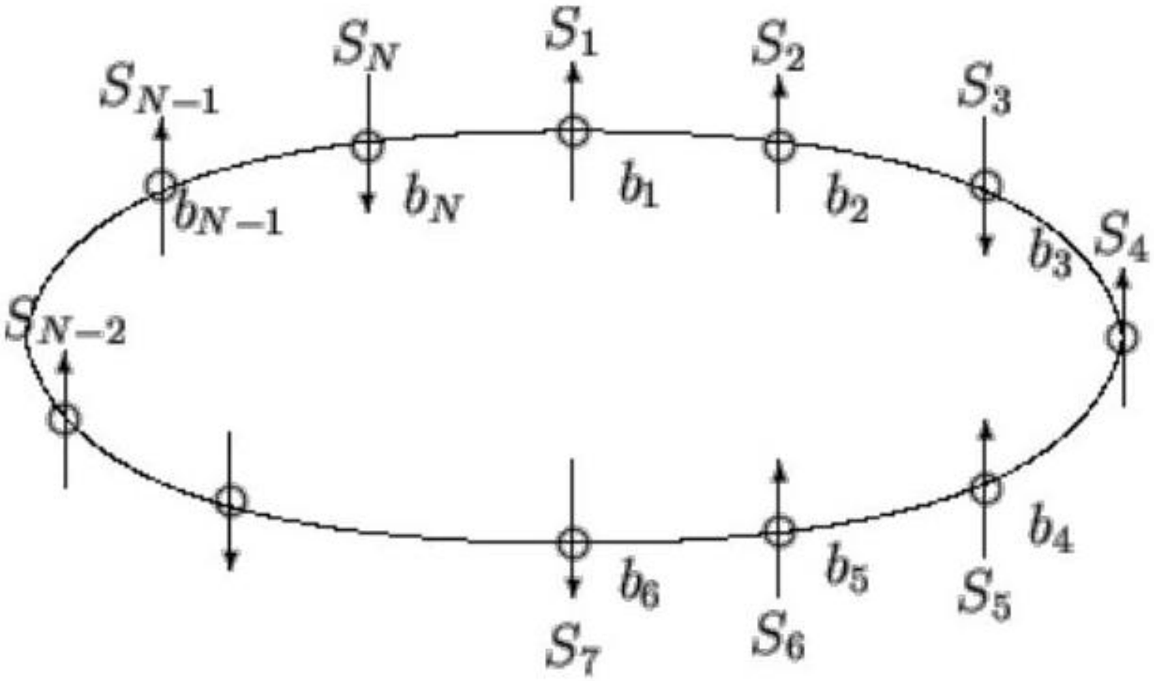
CoSTa

- Convolutional neural network
- Finds genes with similar patterns

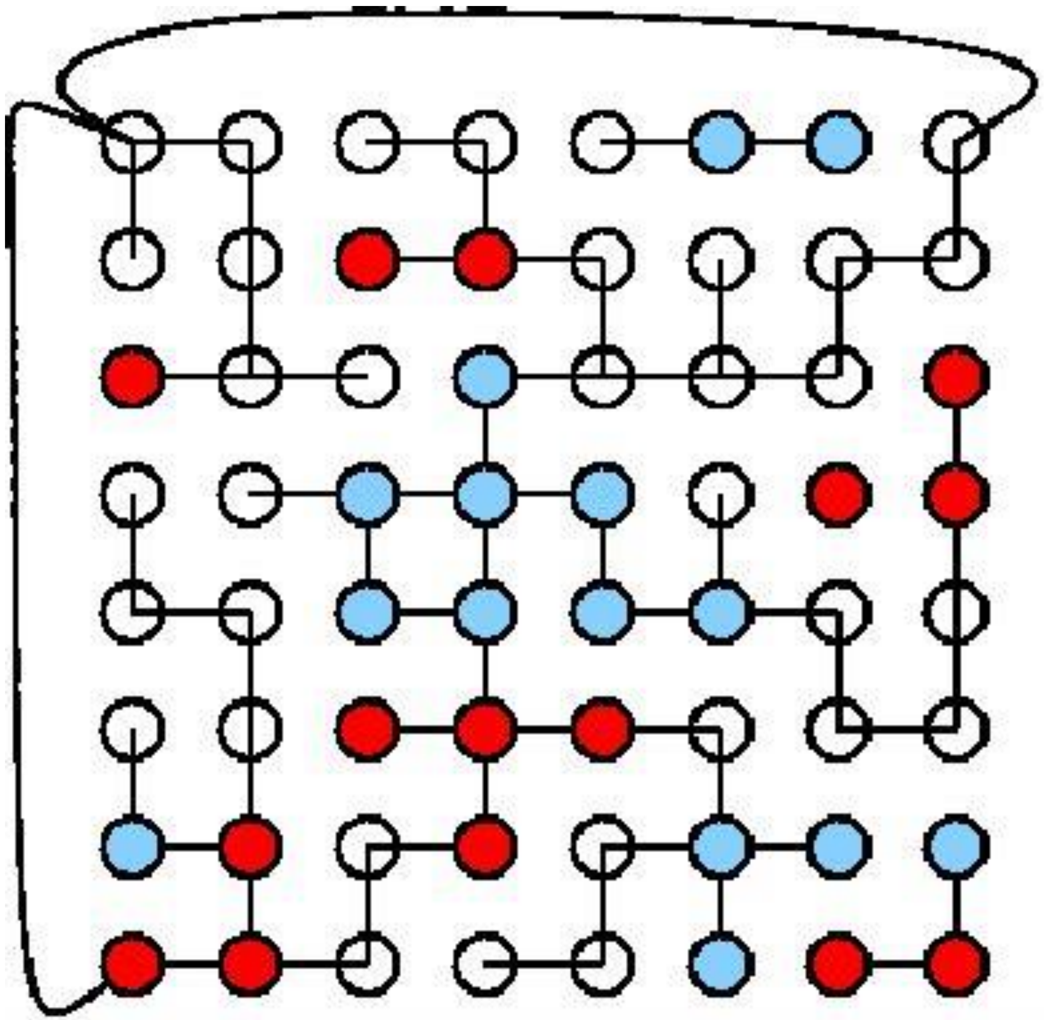


Xu and McCord, BMC Bioinf, 2021

Ising and Potts models

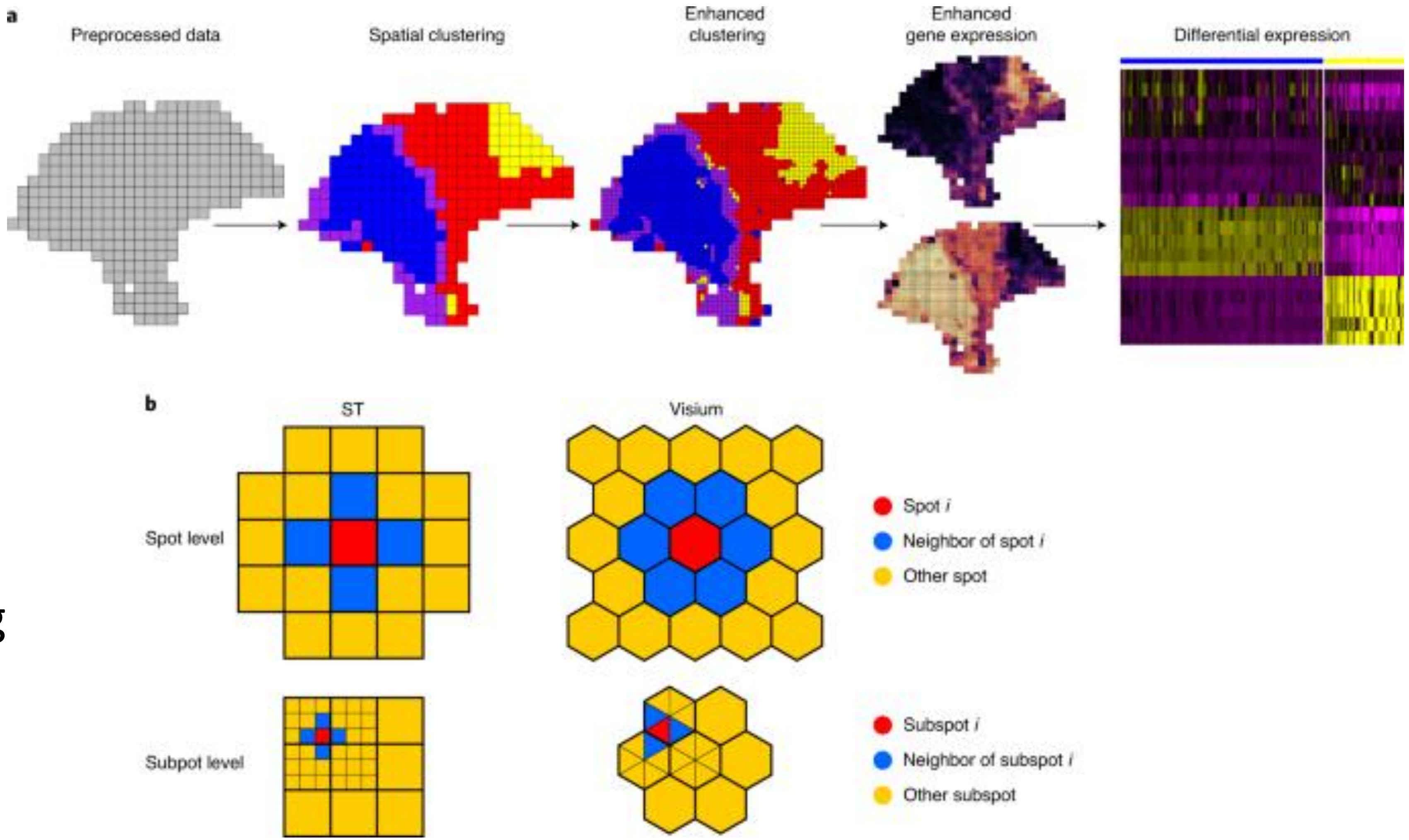


$$H(\sigma) = -\sum_{\langle ij \rangle} J_{ij}\sigma_i\sigma_j - \mu \sum_j h_j \sigma_j$$



BayesSpace

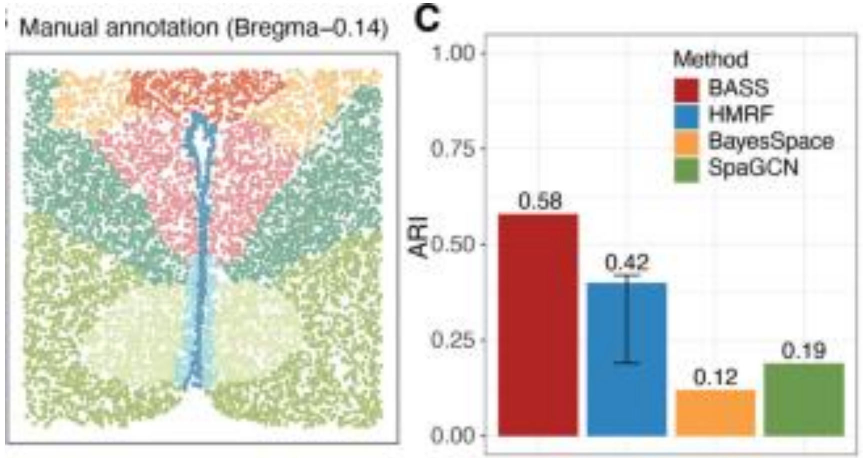
- Use neighborhood information to deconvolve and cluster
  - Use Potts model w Gaussian dists
  - Fit parameters using MCMC

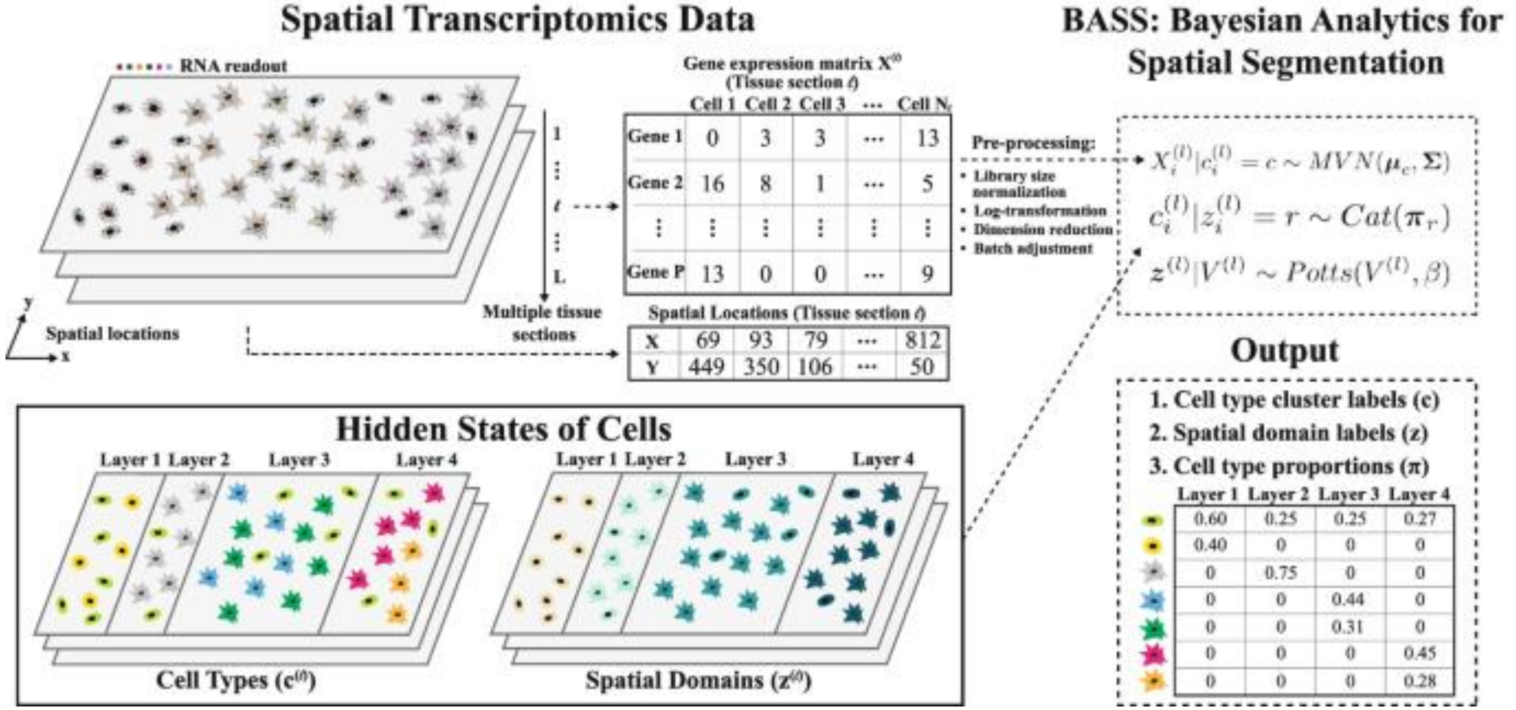


Zhao et al, Nat Biotech, 2021

Bayesian Analytics for Spatial Segmentation

- Hidden Markov random field model (Potts)
  - Gibbs sampling to fit

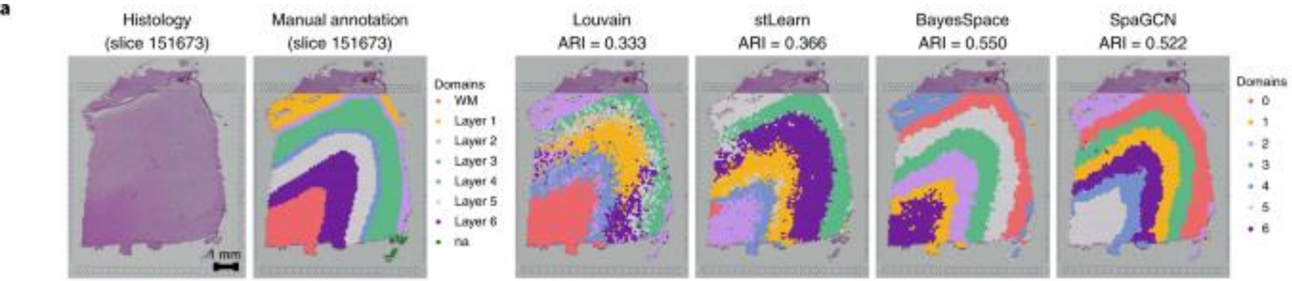


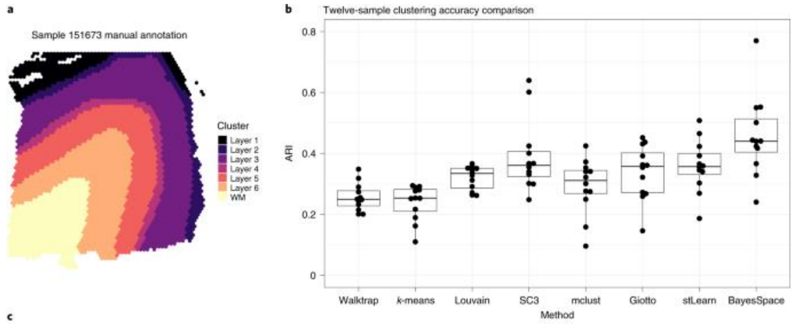


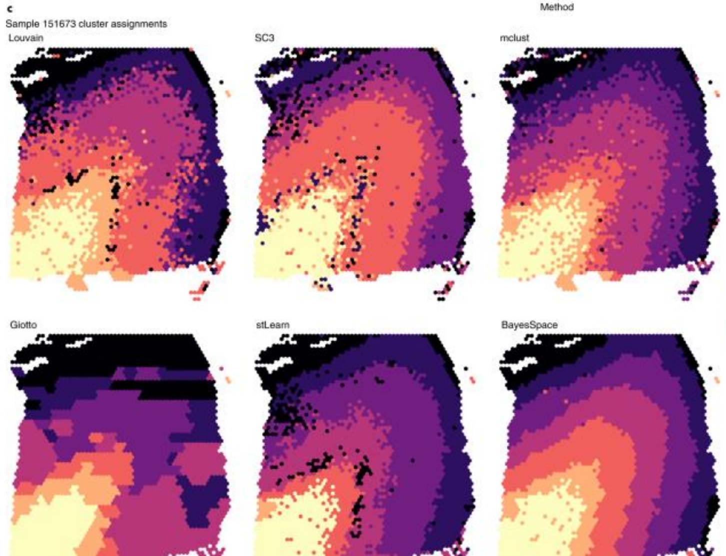
$$\Pr\left(z^{(l)} | V^{(l)}, \beta\right) = \frac{1}{C^{(l)}(\beta)} \exp\left\{\beta \sum\nolimits_{i\sim i'} I\left(z_i^{(l)} = z_{i'}^{(l)}\right)\right\},$$

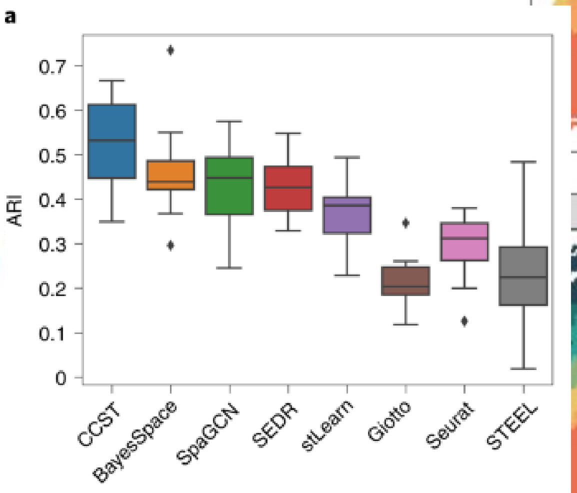
Li and Zhuo, Genome Bio, 2022

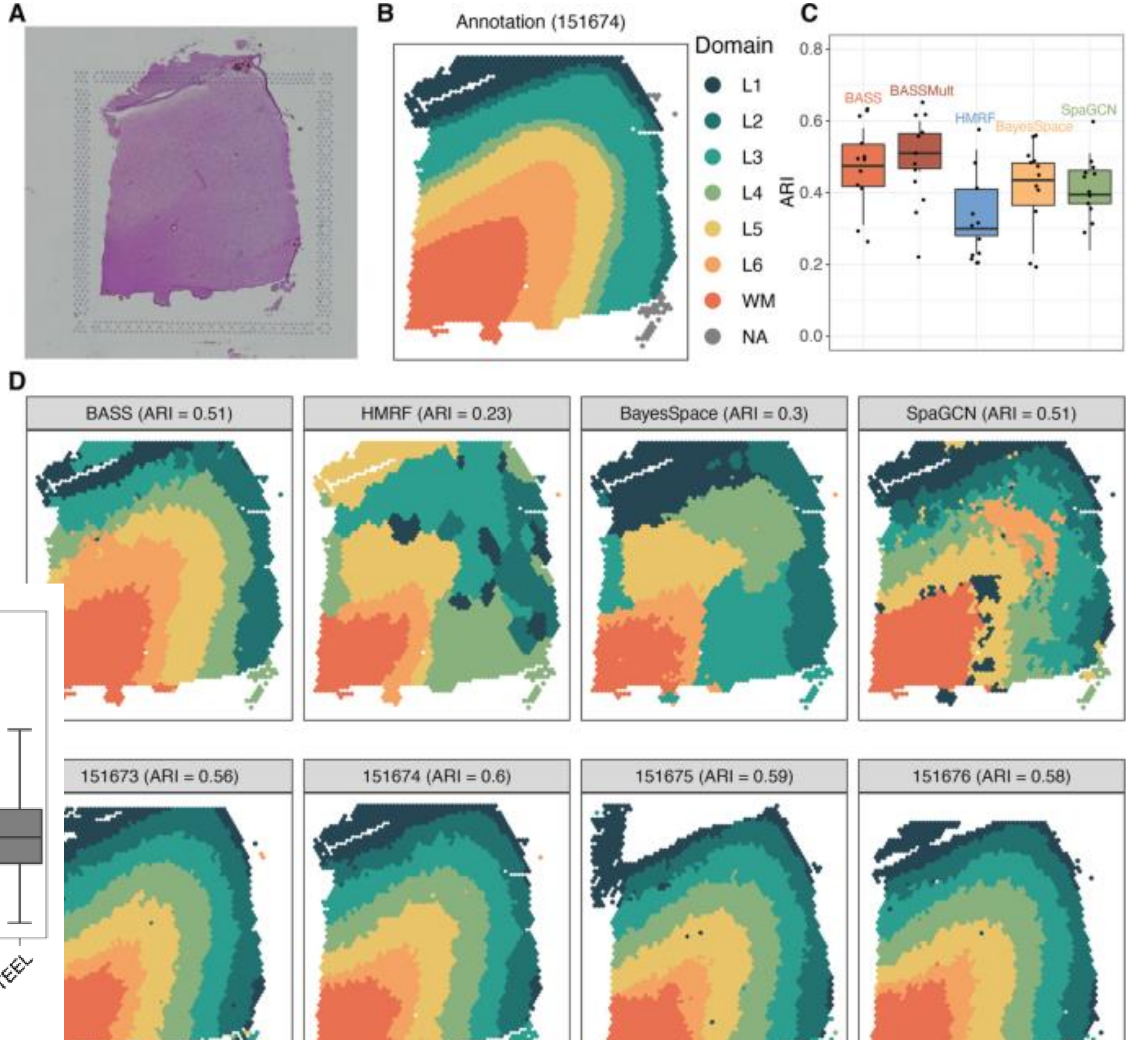
BayesSpace vs SpaGCN vs BASS vs CCST



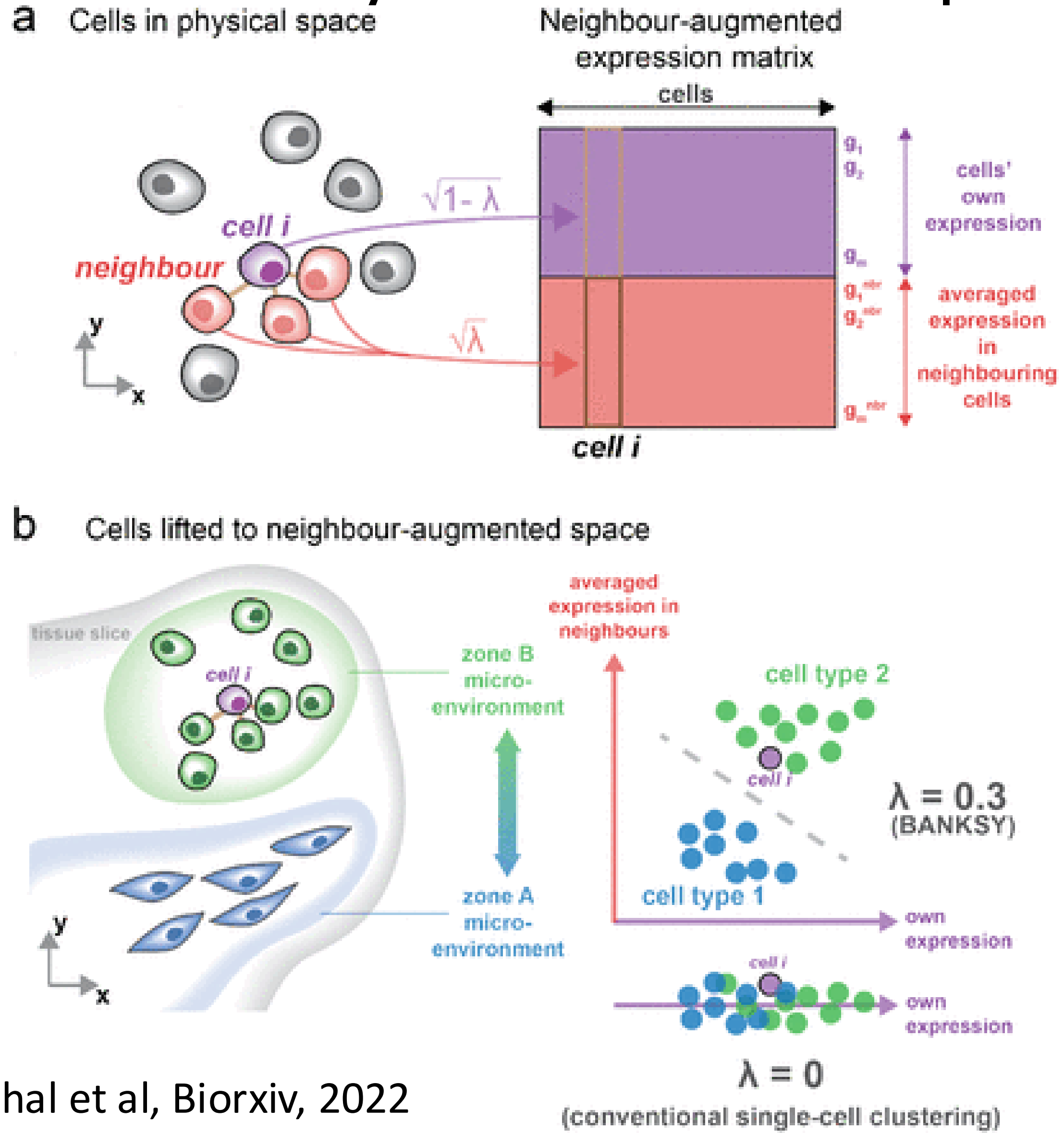


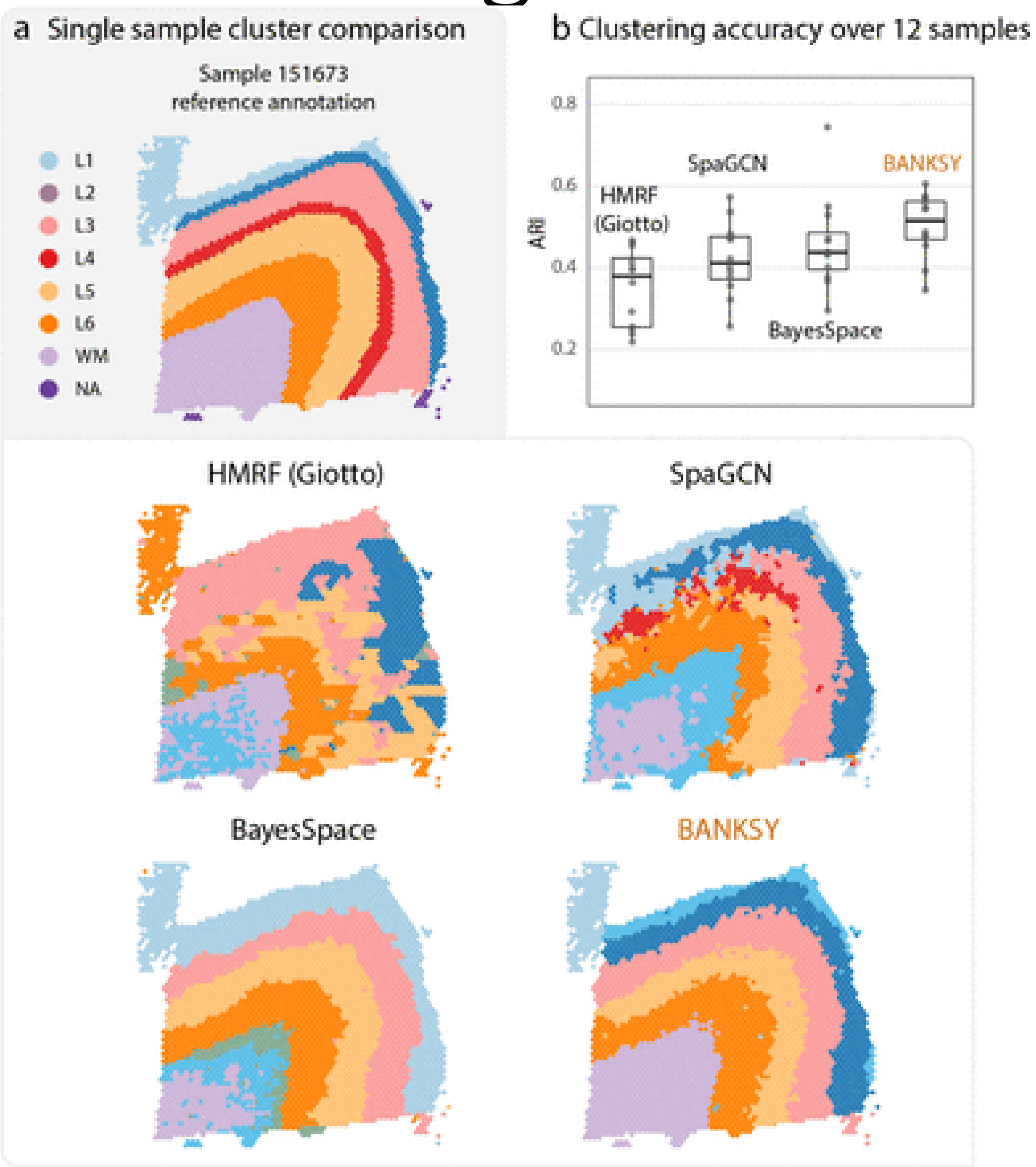






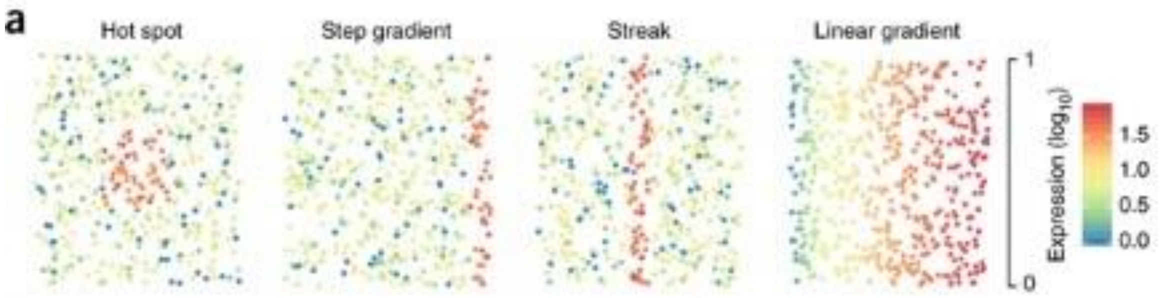
Banksy uses the expression of neighbors



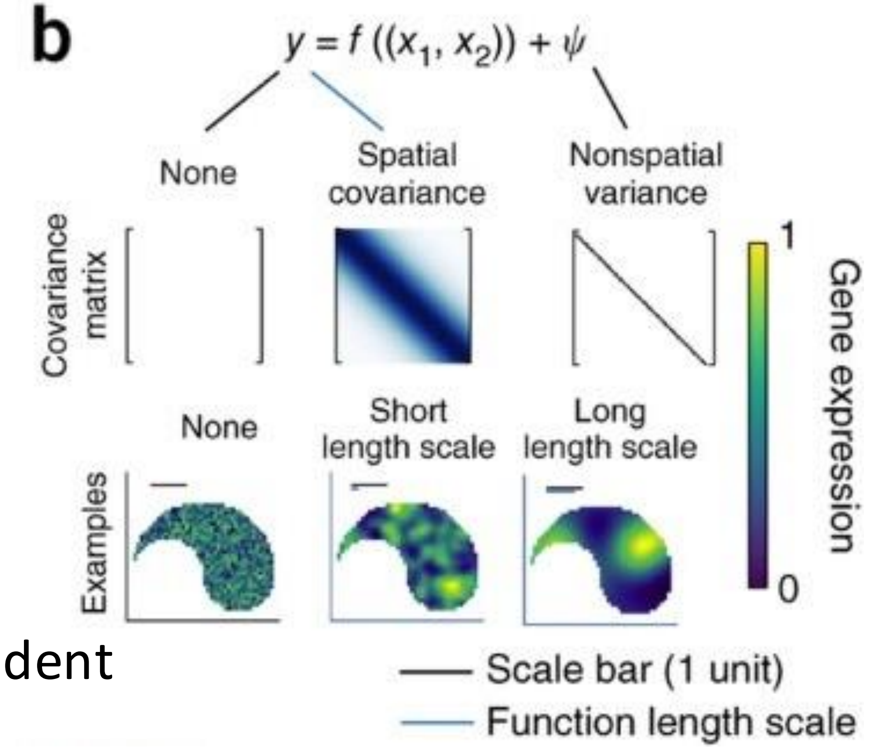


Singhal et al, Biorxiv, 2022

Differential expression

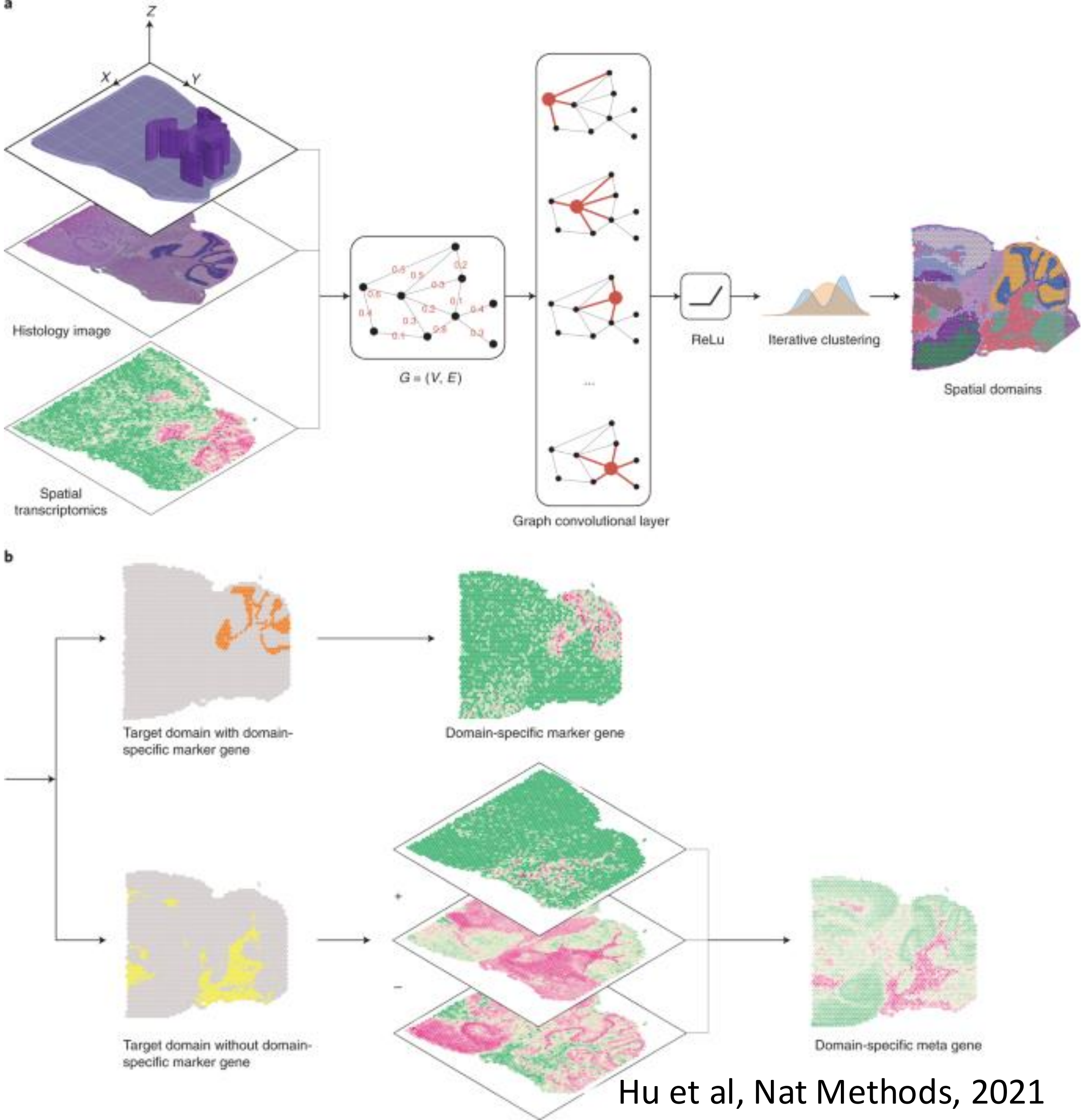


- Trendsceek
  - Use marked point process
  - Compare genes based on spatial mean, variance and correlation
  - Find hotspots using KDE
- SpatialDE
  - Use linear model to decompose variance into spatial vs non-spatial components
- SPARK
  - Use non-parametric test to find if expression is independent of spatial location
  - Most favorable scaling with number of cells



SpaGCN integrates histology information

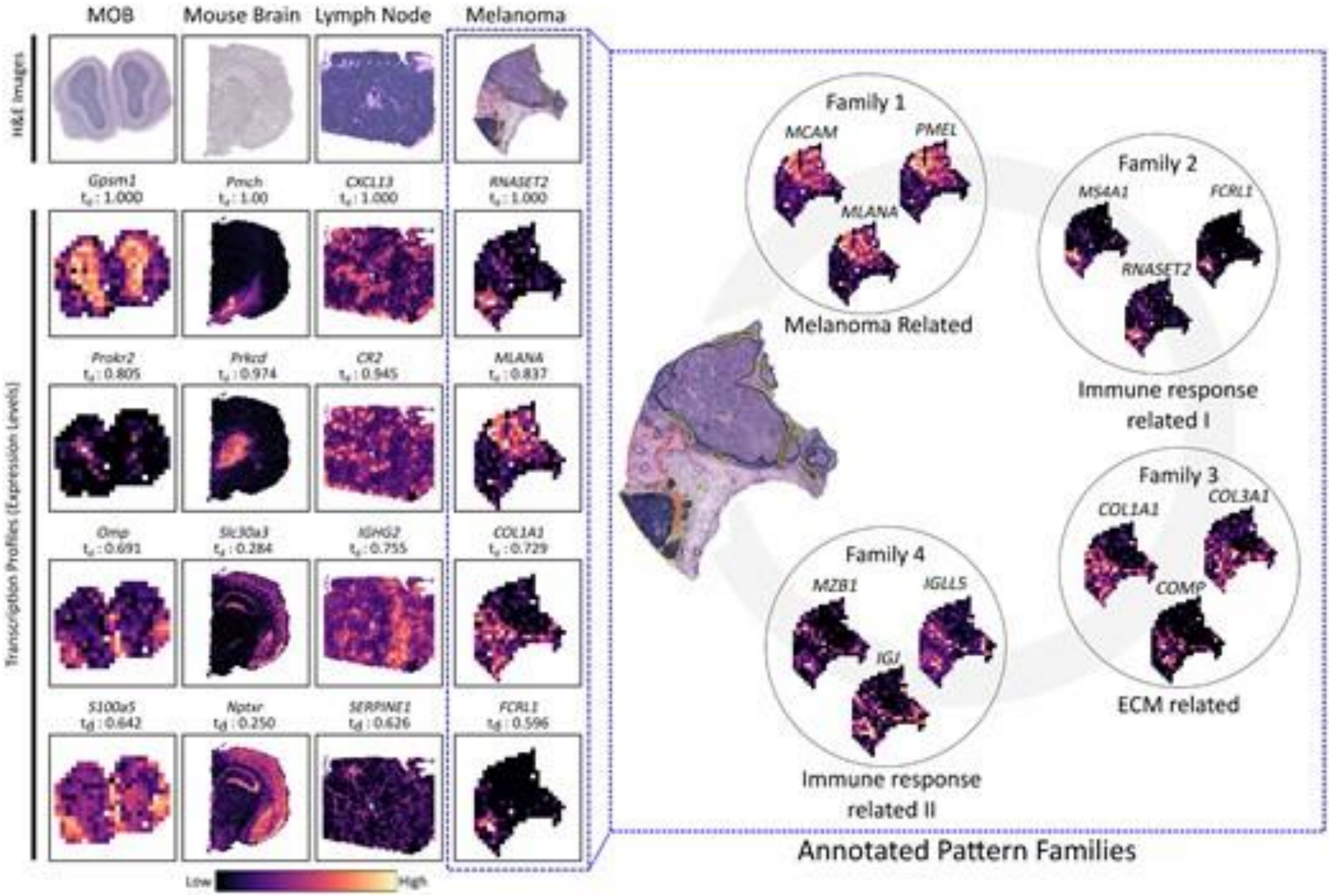
- Identify coherent domains using graph convolutional network
  - Integrate H&E and ST
  - Jointly find spatial domains and spatially variable genes
  - Use Gaussian kernel and Euclidean distance to build graph



Hu et al, Nat Methods, 2021

sepal

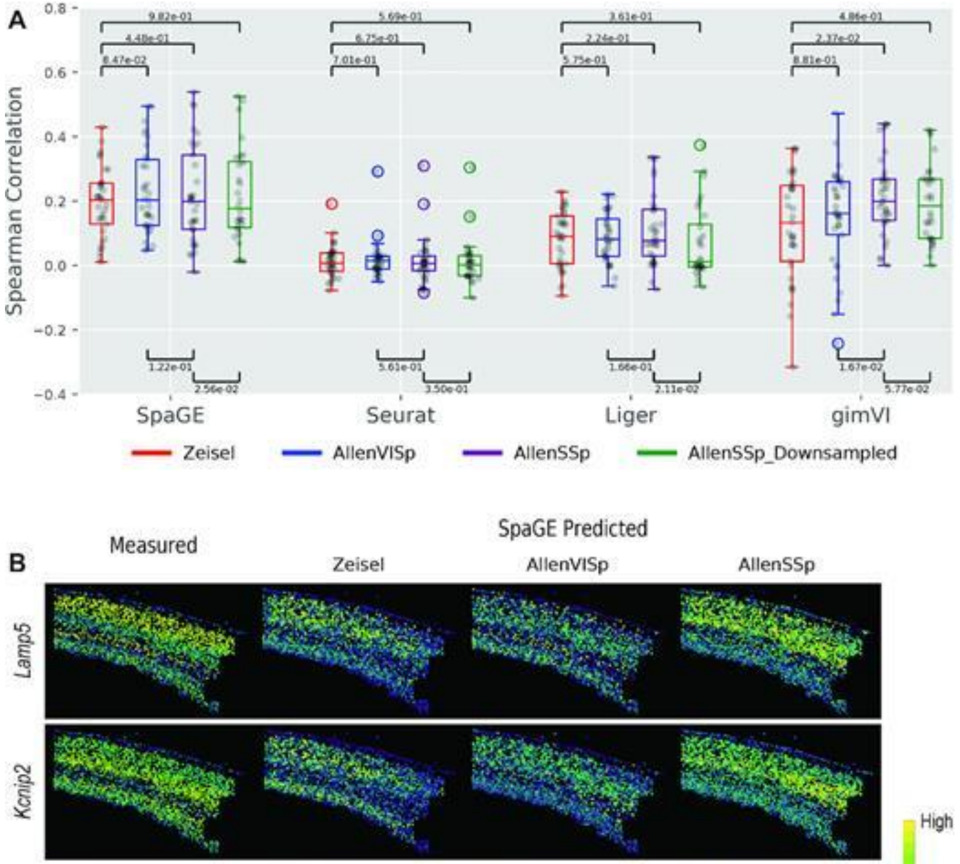
- Diffusion based approach to identify distinct patterns
  - Long time to reach stationarity -> more structure
  - Cluster top genes to identify groups with similar patterns

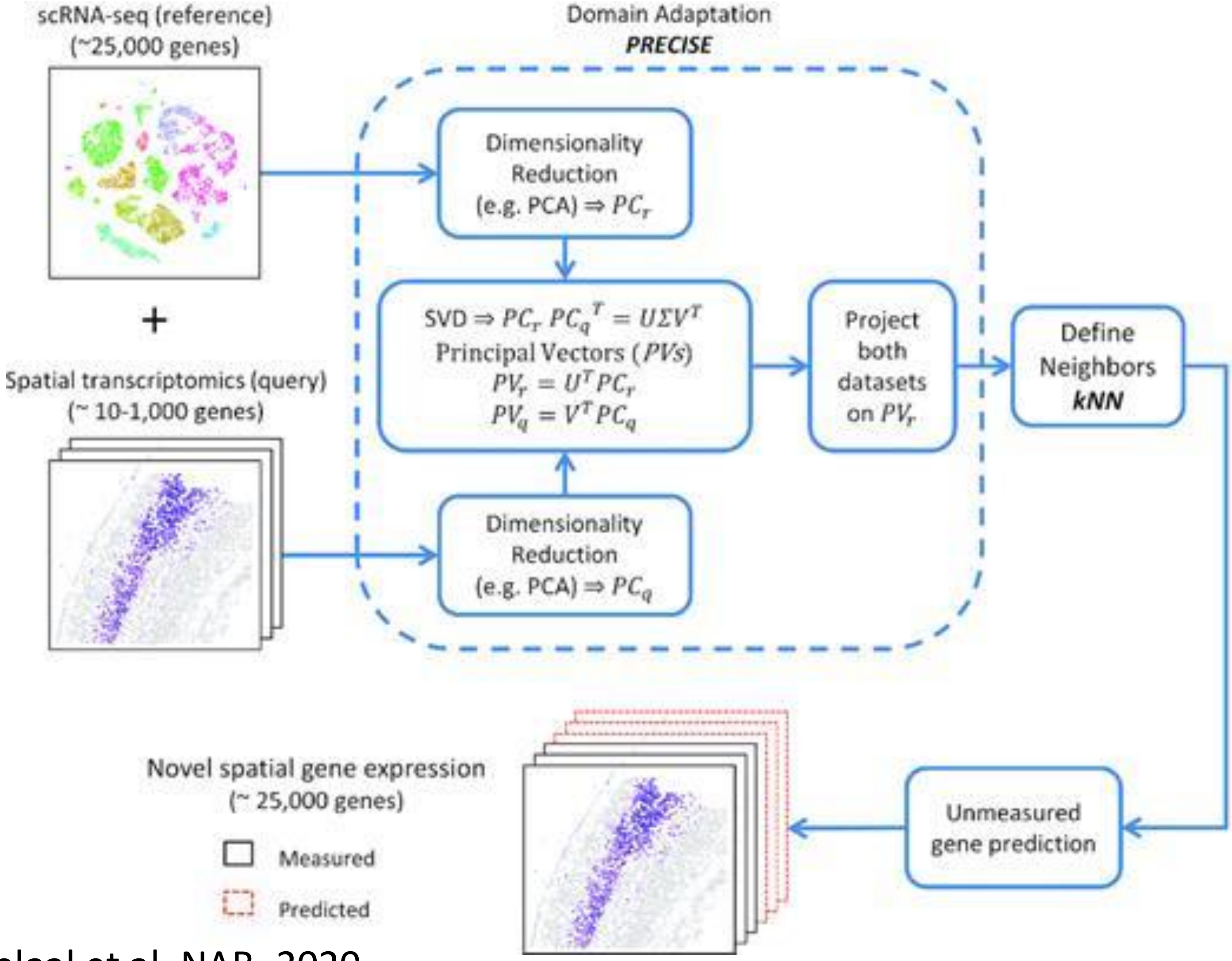


Andersson and Lundeberg, Bioinformatics, 2021

SpaGE uses scRNAseq data for imputation

- Align datasets in latent space
- Use knn graph for imputation



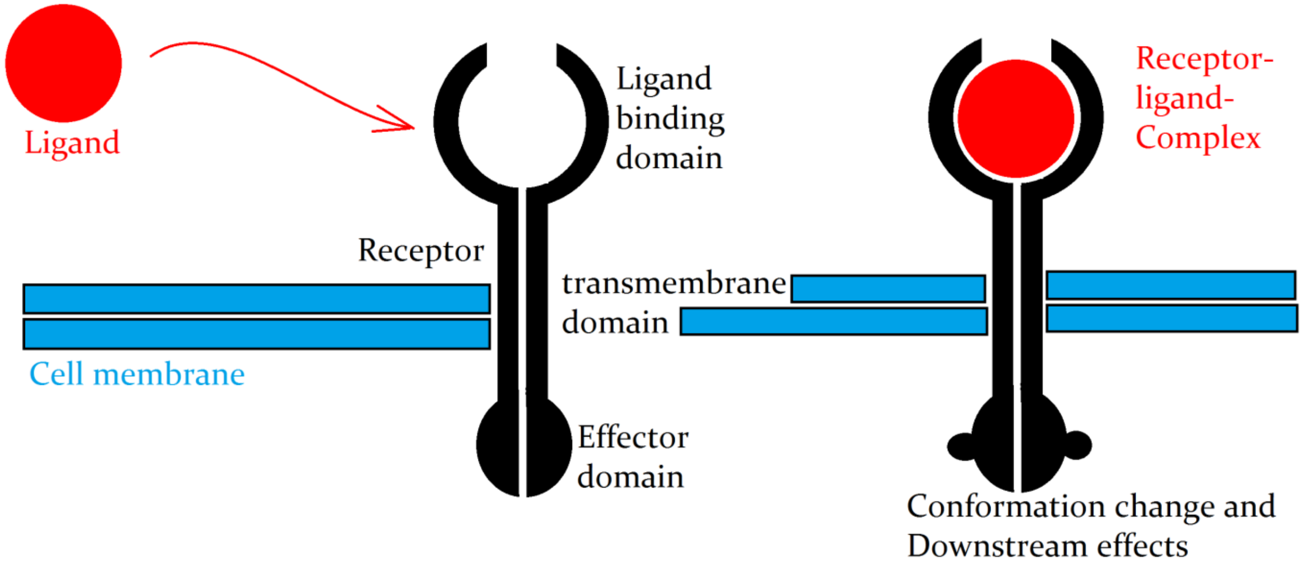


Abdelaal et al, NAR, 2020

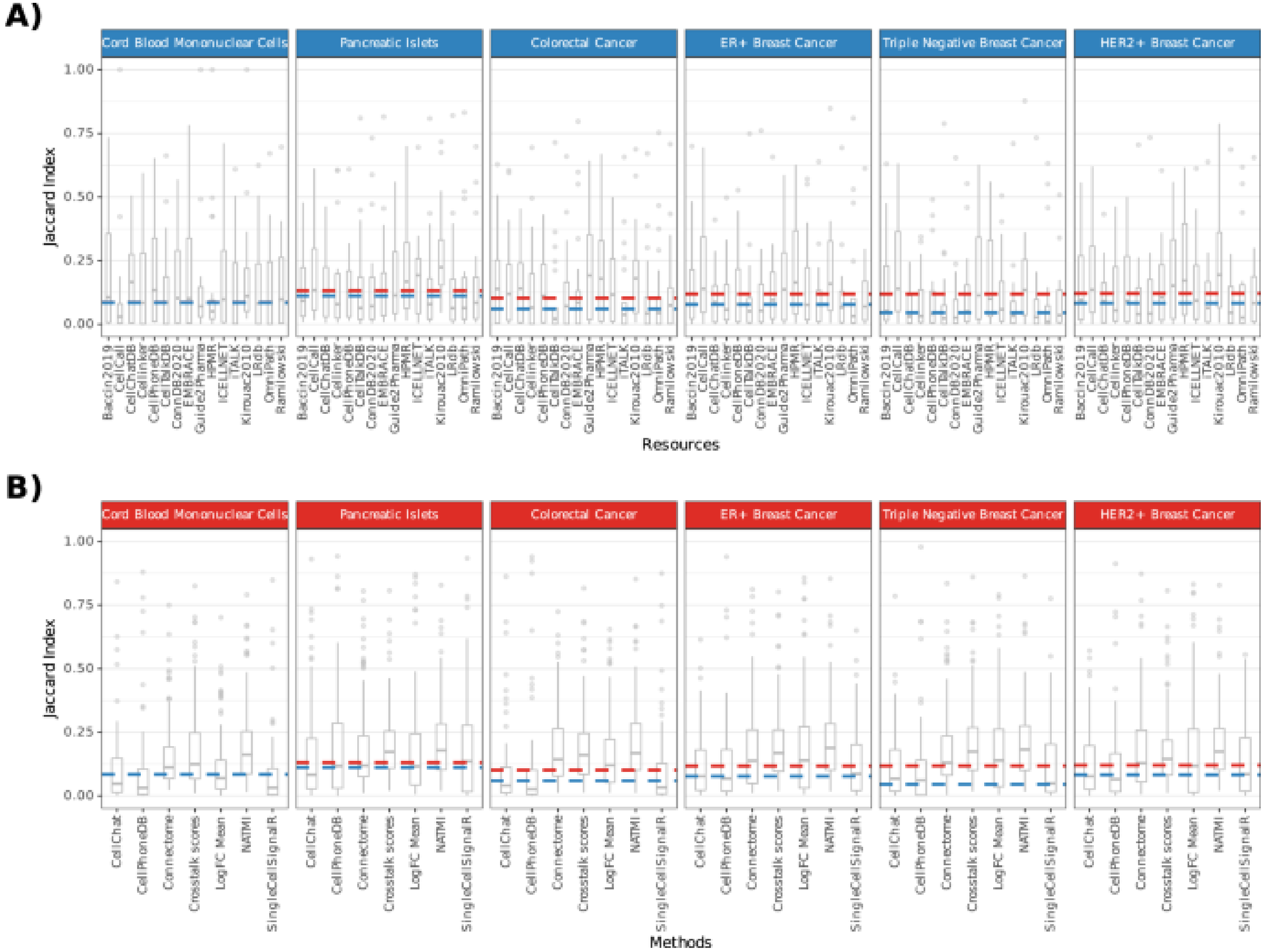
Cell-cell interactions

Ligand-receptor communication

- Detect pairs of cell types expressing high ligands and matching receptors
- First step towards inferring spatial organization

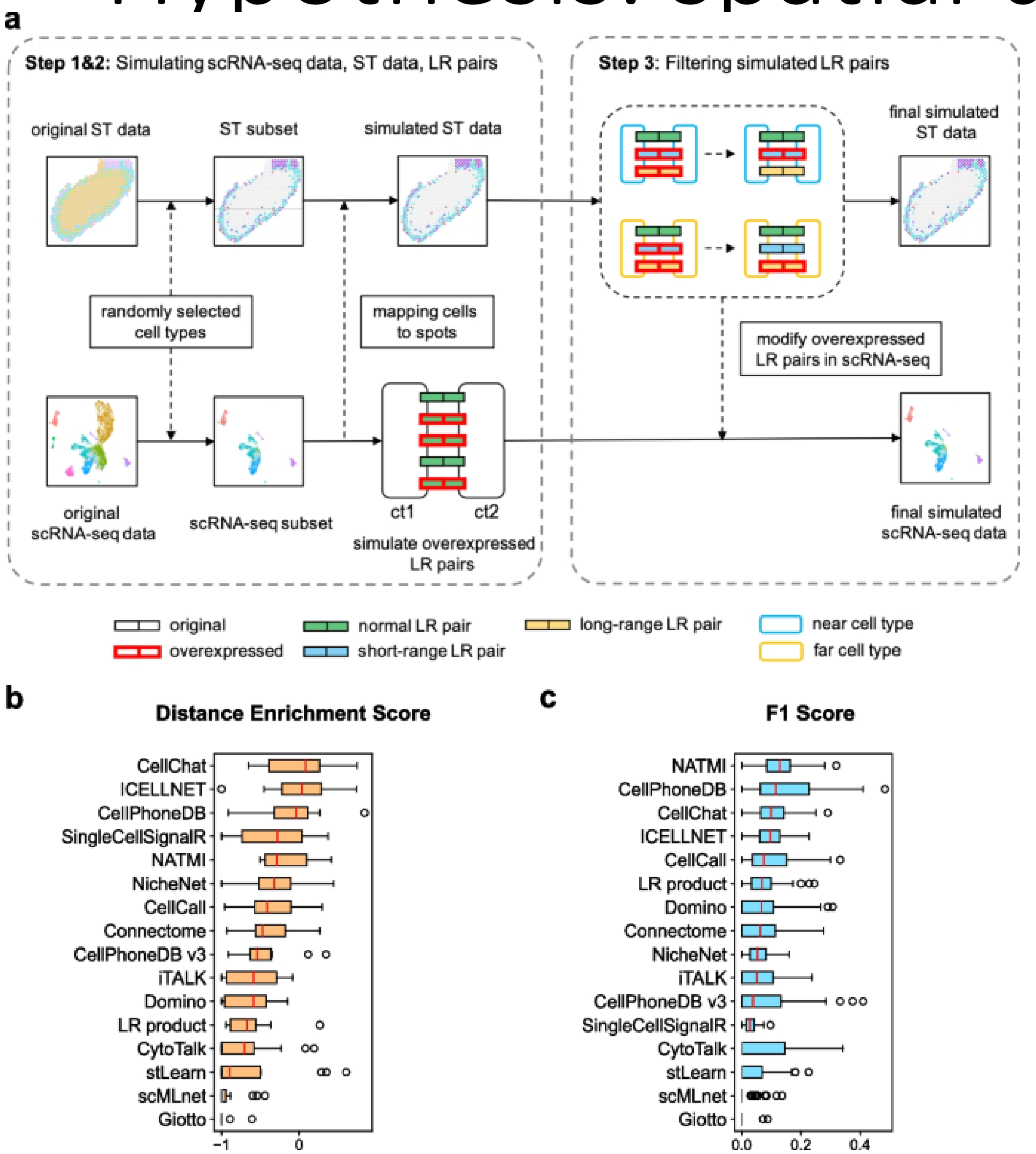


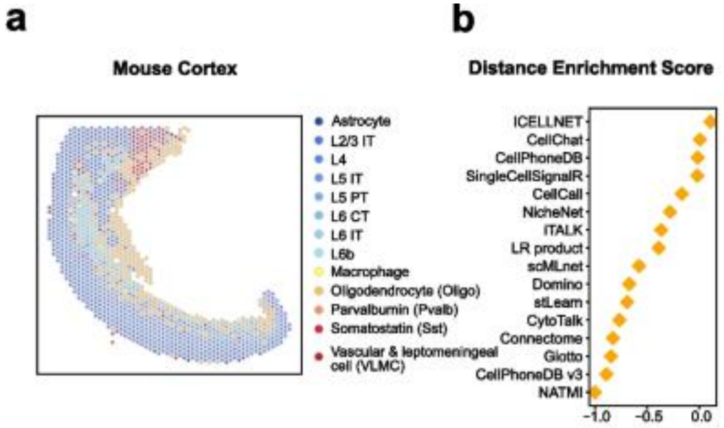
Result depends on method and database

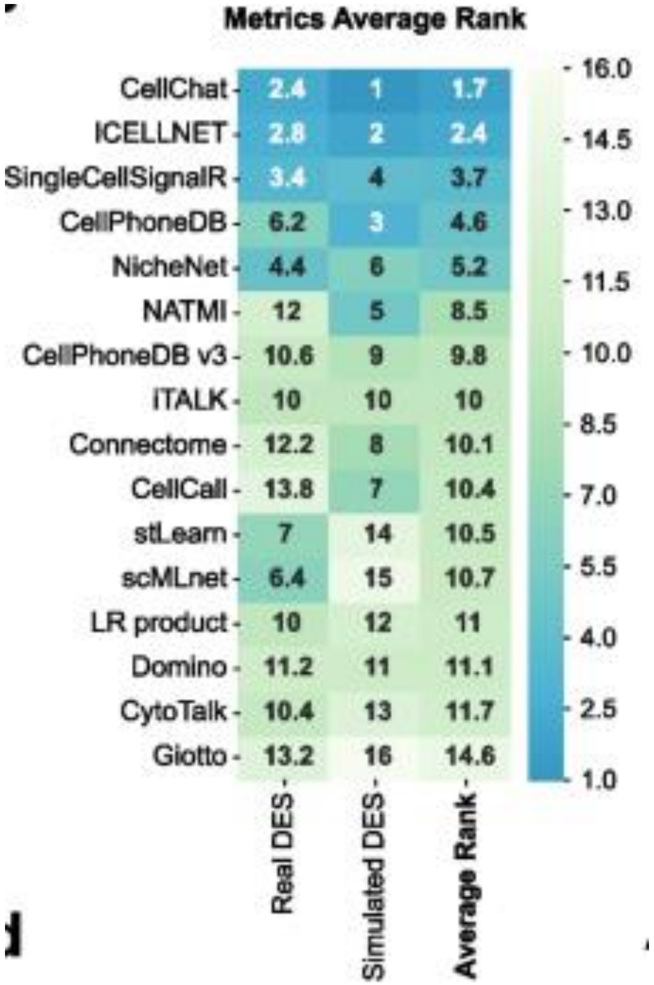


Dimitrov et al, Nature Communications, 2022

Hypothesis: spatial data improves predictions





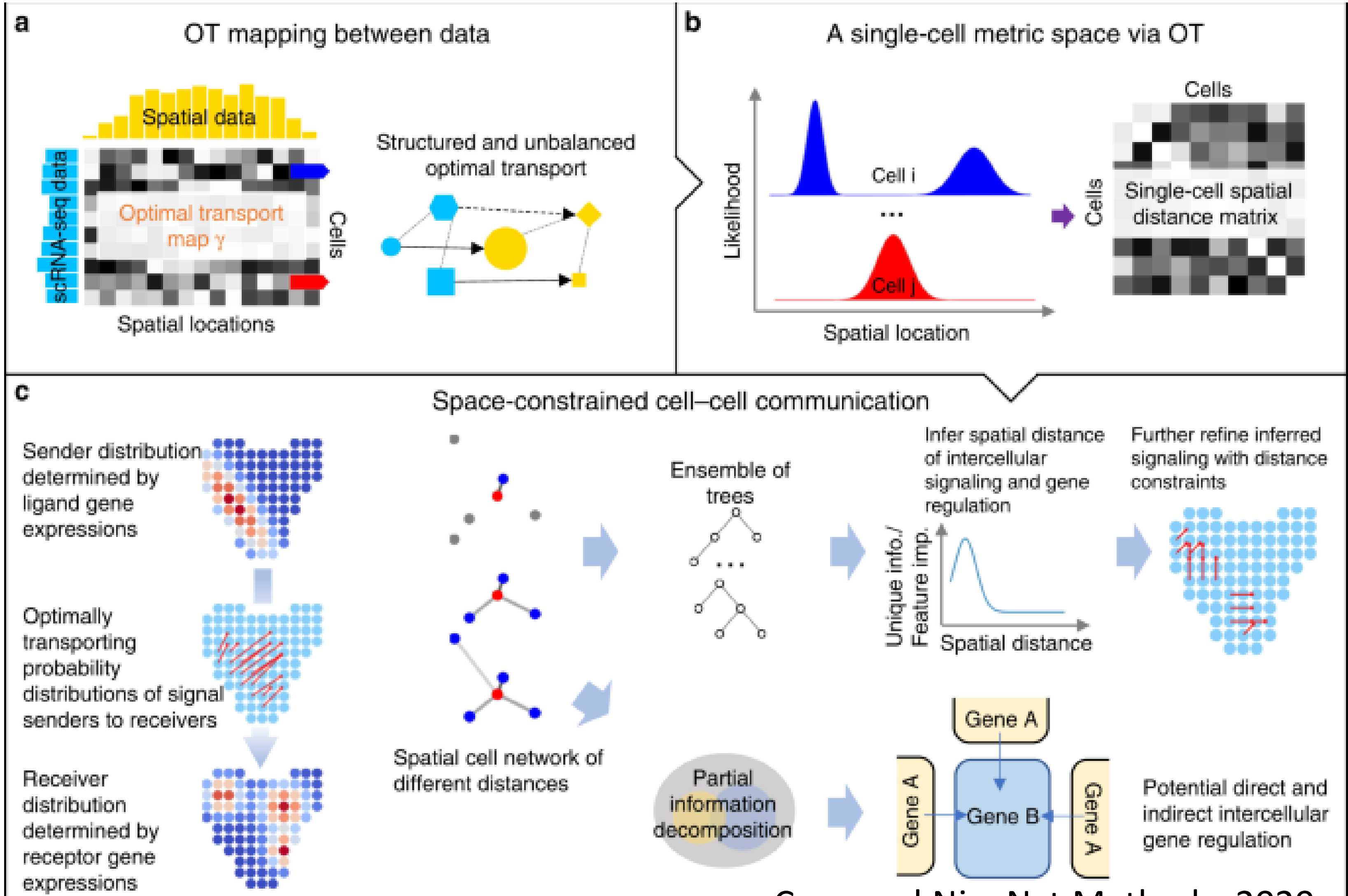


Liu et al (Genome Biology, 2022)

SpaOTsc uses optimal transport

- Distance for
  - Gene expression and location
  - Gene expression btwn cells
  - Spatial distances
- Also considers expression of downstream targets

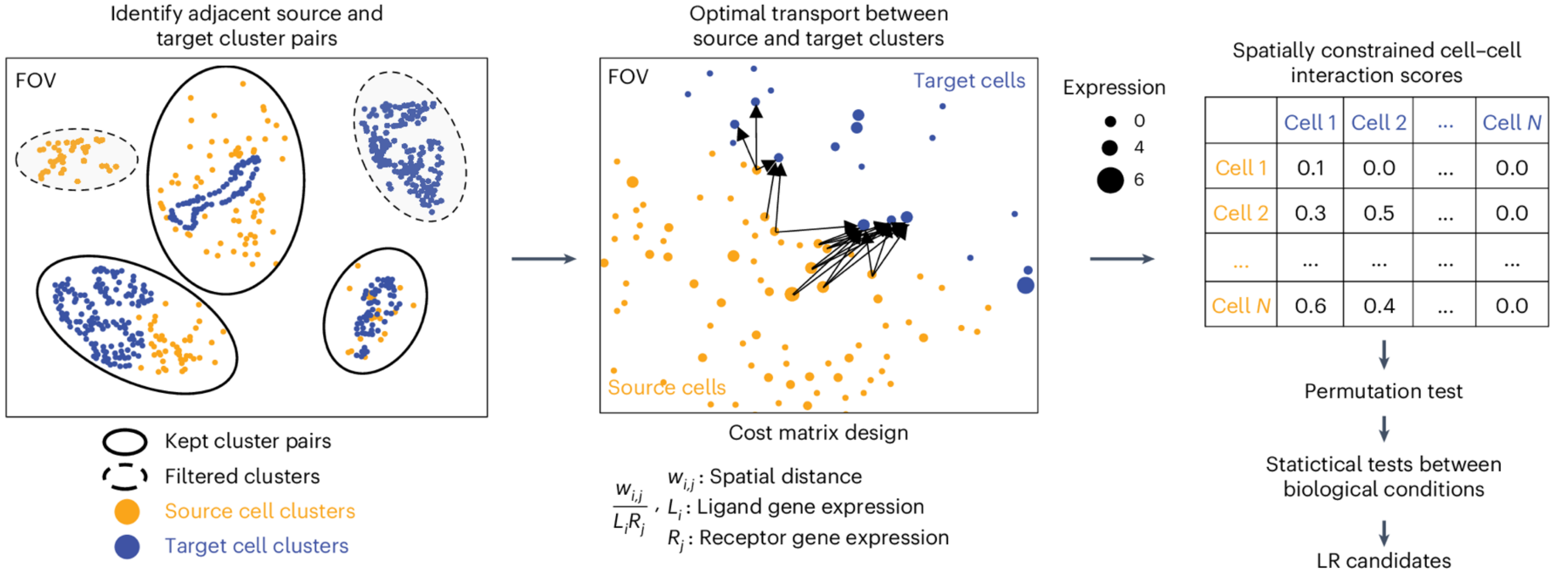
$$\underset{\gamma \in \mathbb{R}_{+}^{n\times m}}{\operatorname{argmin}}\Big[(1-\alpha) < \boldsymbol{\gamma}, \mathbf{M} >_{F} + \rho\left(\mathrm{KL}\left(\boldsymbol{\gamma}\mathbf{1}^{m}|\omega_1\right) + \mathrm{KL}\left(\boldsymbol{\gamma}^{T}\mathbf{1}^{n}|\omega_2\right)\right) + \alpha \sum_{i,j,k,l} L(\mathbf{D}_{\mathrm{sc}}(i,k), \mathbf{D}_{\mathrm{spa}}(j,l))\boldsymbol{\gamma}_{i,j}\boldsymbol{\gamma}_{k,l}\Big]$$

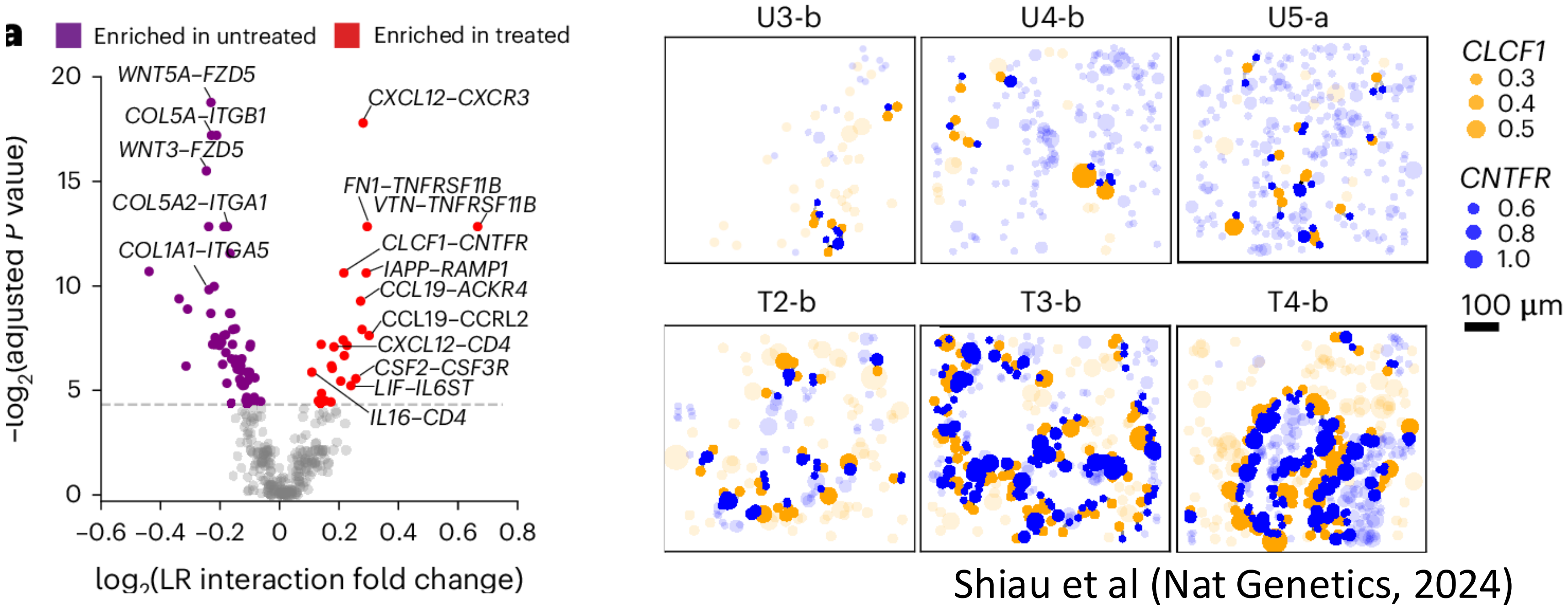


Cang and Nie, Nat Methods, 2020

SCOTIA

- Spatial clustering to identify nearby clusters
- OT with cost function combining physical distance and expression

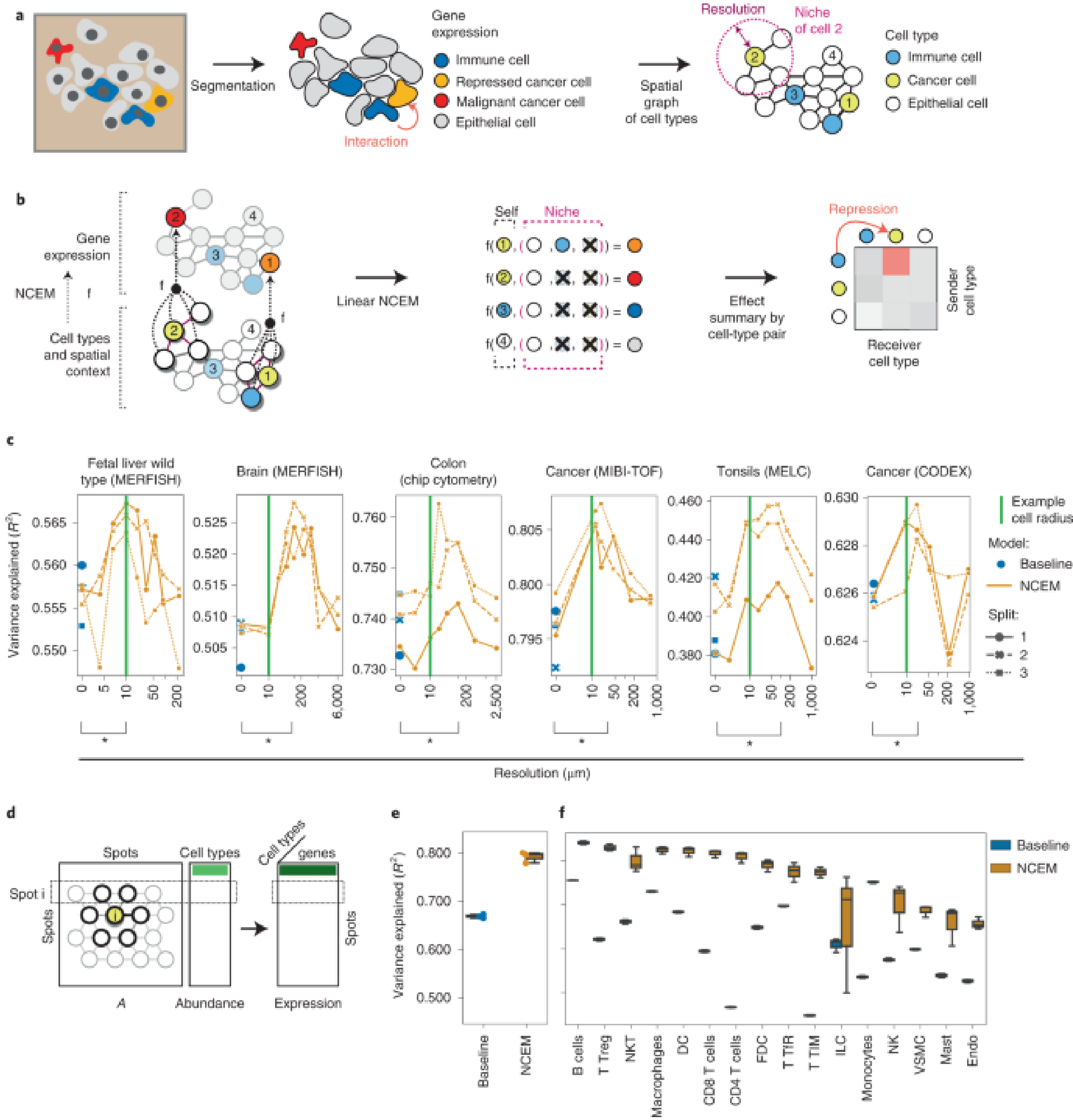




Shiau et al (Nat Genetics, 2024)

NCEM infers influence of neighbors

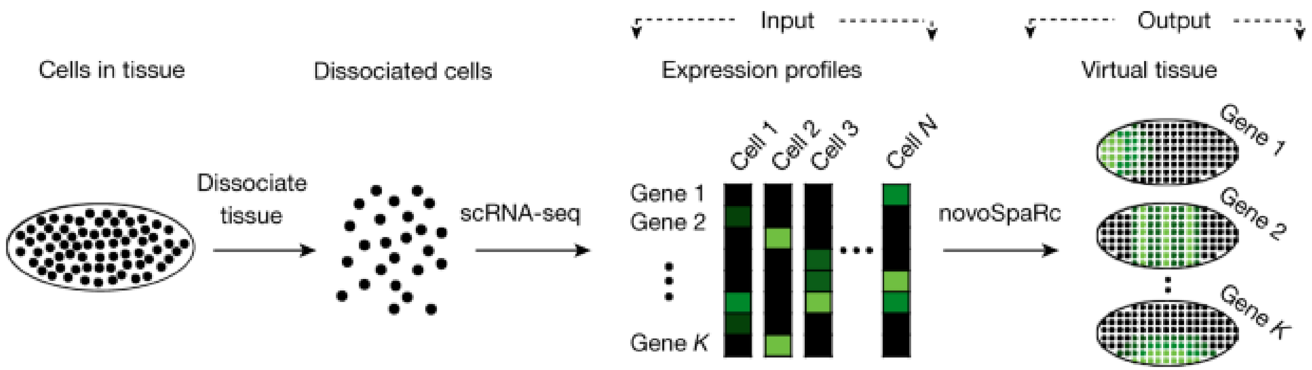
- Use graph neural network to predict expression as a function of cell type and neighbors

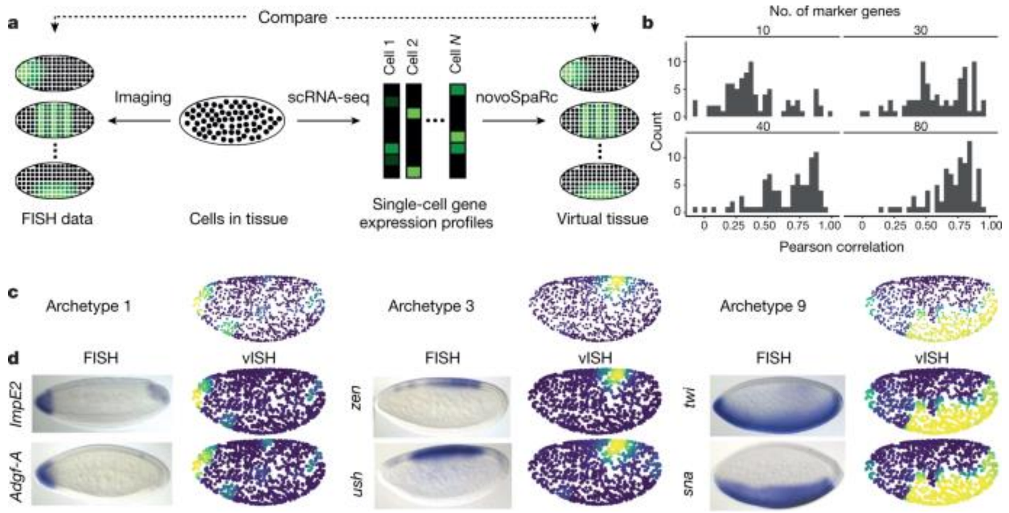


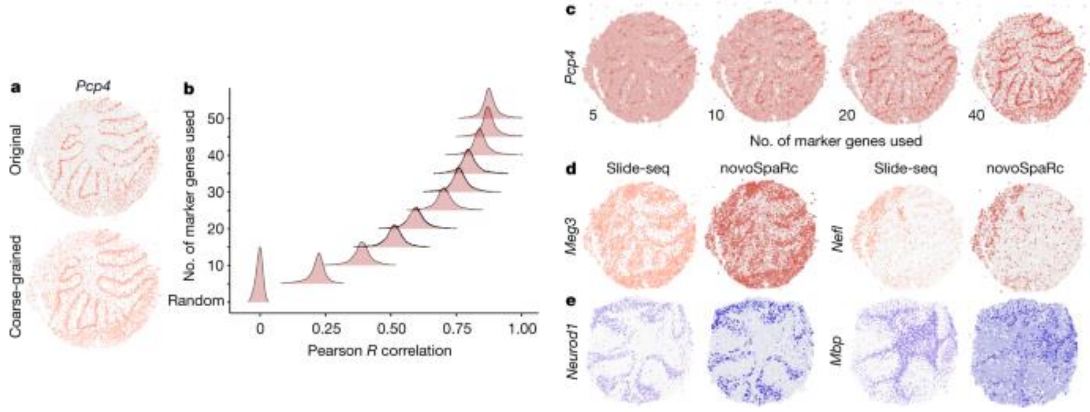
Fischer et al, Nat Biotech, 2022

novoSpaRc reconstructs tissues de novo

- Optimal transport framework
- Can incorporate reference info



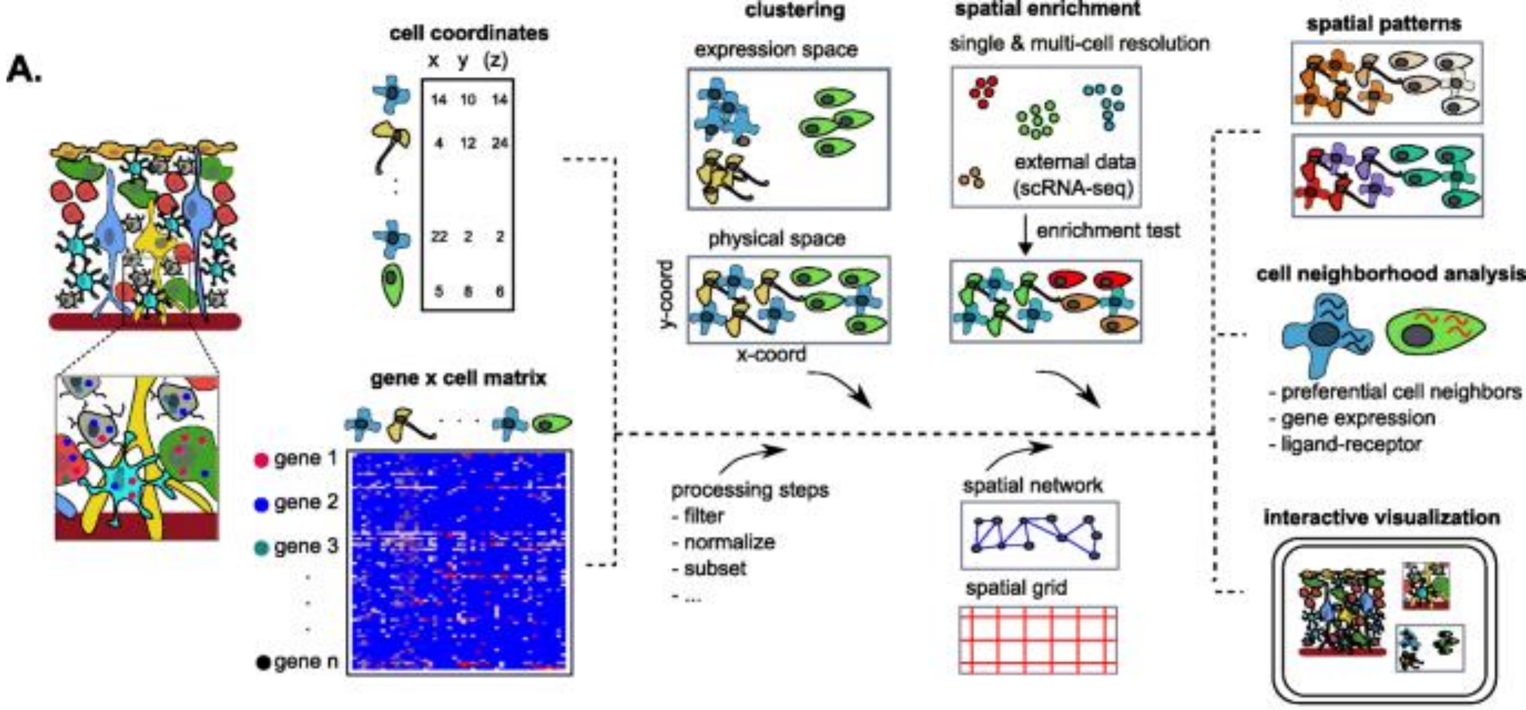


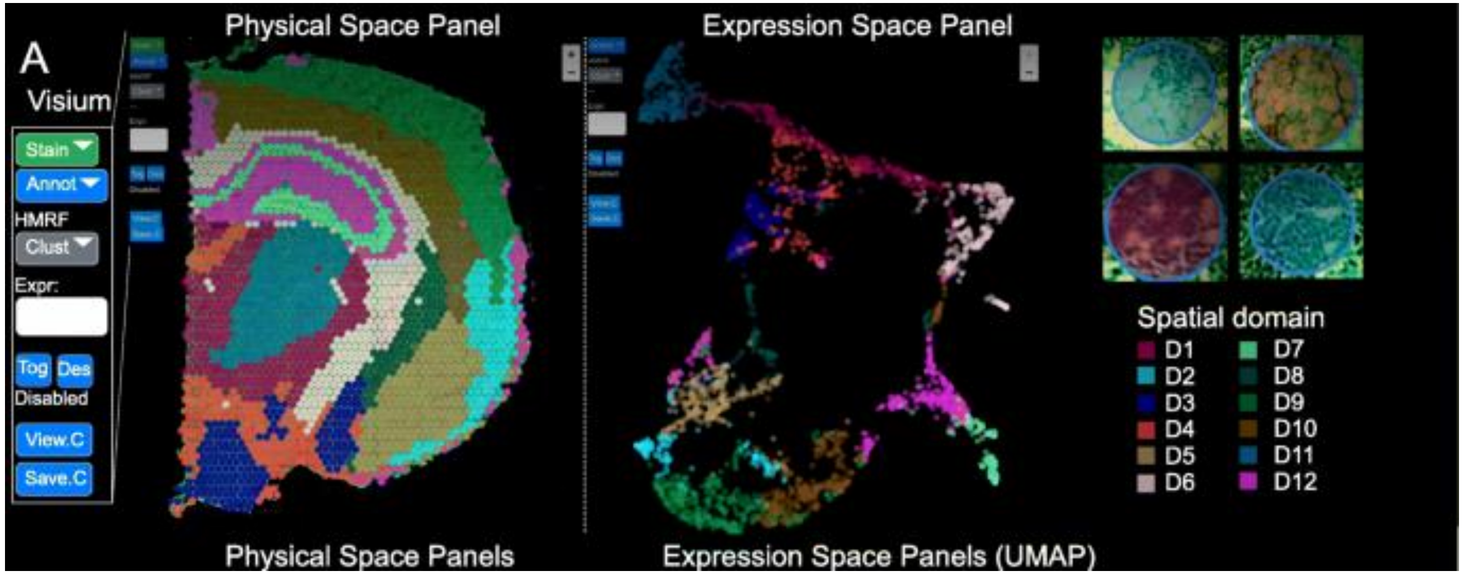


Nitzan et al, Nature, 2019

Giotto

- Toolbox for spatial data analysis
  - Visualization
  - Marker genes
  - Spatial domains
  - Cell-cell communication

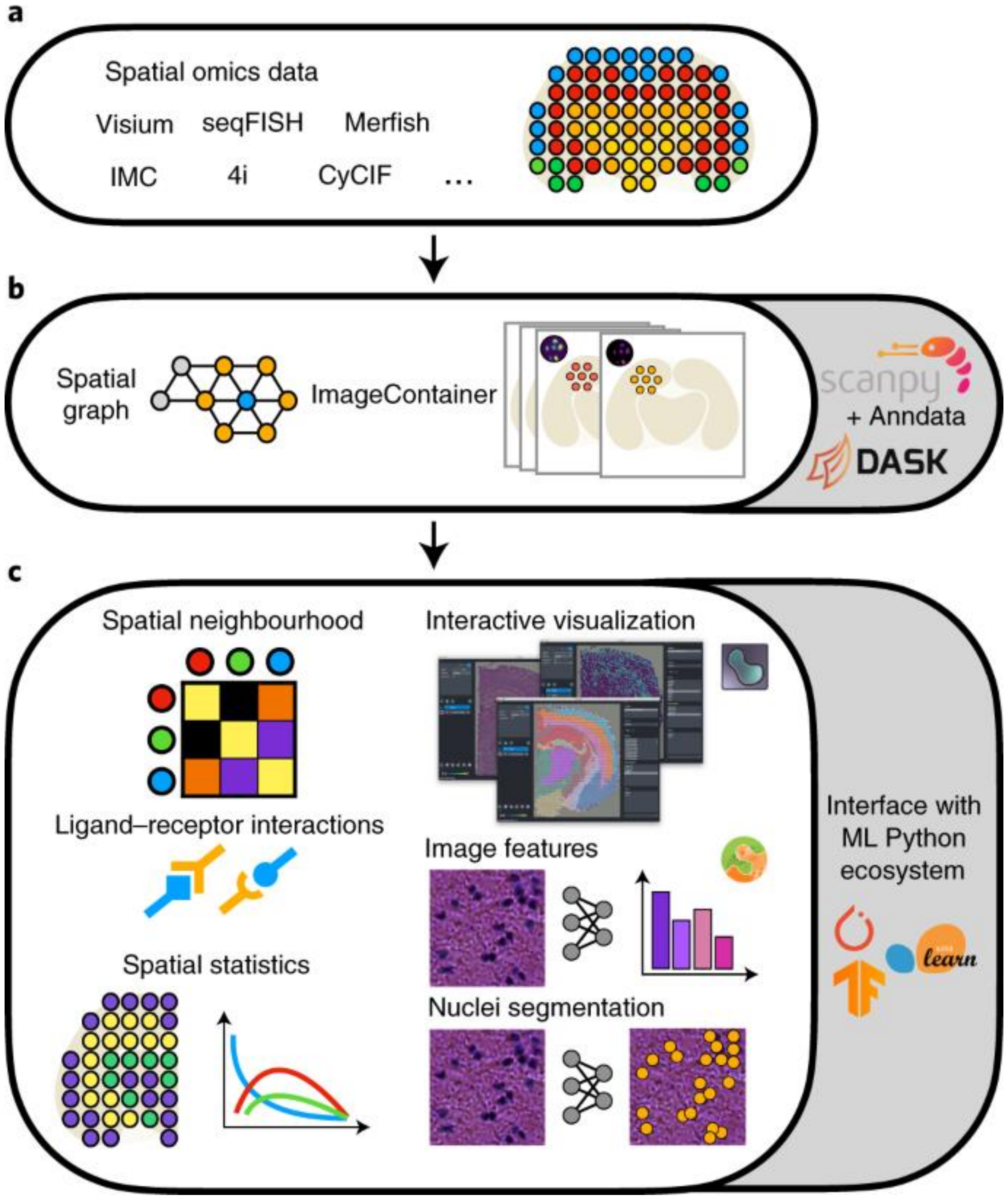




Dries et al, Genome Biology, 2021

Squidpy & Seurat

- Popular frameworks expanded to spatial transcriptomics



Palla et al, Nature Methods, 2024

Future outlook

- Interesting year(s) ahead for spatial transcriptomics technologies
- No standardized workflow
  - Need technologies to stabilize first
- Plenty of opportunities to identify new possibilities
  - Multimodal integration will be needed
- 3D tissue reconstruction will soon be here